

Supplementary Appendix for “Self-fulfilling fluctuations in HANK economies”

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A Derivation of IS curve

A.1 Household problem

Instead of directly solving the household problem in continuous time, we solve it in the discrete time limit in which each period is Δ units of time long. Then we derive the optimal decisions in continuous time by taking limits as $\Delta \rightarrow 0$. In discrete time, the problem of the household can be written as, where we have discarded the j subscript for convenience:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} e^{-\rho \Delta t} \left[\frac{c_{t\Delta}^{1-\gamma^{-1}}}{1-\gamma^{-1}} - \psi n_{t\Delta} \right] \Delta$$

s.t.

$$a_{t+\Delta} - a_t = (1 + r_t \Delta) (\xi_h w_t n_t + D_t - c_t) \Delta + r_{t+\Delta} \Delta a_t \quad \text{and} \quad a_{t+\Delta} \geq -\underline{a}$$

This problem can be formulated as a Bellman equation. The value function of a household with idiosyncratic productivity ξ_h and wealth a_t can be written as:

$$V(a_t, \xi_h) = \left[\frac{c_t^{1-\gamma^{-1}}}{1-\gamma^{-1}} - \psi n_t \right] \Delta + (1 - \rho \Delta) [(1 - \lambda_{l,t+\Delta} \Delta) V(a_{t+\Delta}, \xi_h) + \lambda_{l,t+\Delta} \Delta V(a_{t+\Delta}, \xi_l)], \text{(a.1)}$$

where $a_{t+\Delta}$ is given by:

$$a_{t+\Delta} = (1 + r_t \Delta) [(\xi_h w_t n_t + D_t - c_t) \Delta + a_t] \quad \text{and} \quad a_{t+\Delta} \geq -\underline{a}, \text{(a.2)}$$

and we have used the fact that for small Δ , $e^{-\rho \Delta} = 1 - \rho \Delta$. Similarly, the value function for a household with idiosyncratic productivity ξ_l and wealth a_t can be written as:

$$V(a_t, \xi_l) = \left[\frac{c_t^{1-\gamma^{-1}}}{1-\gamma^{-1}} - \psi n_t \right] \Delta + (1 - \rho \Delta) [(1 - \lambda_{h,t+\Delta} \Delta) V(a_{t+\Delta}, \xi_l) + \lambda_{h,t+\Delta} \Delta V(a_{t+\Delta}, \xi_h)],$$

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where $a_{t+\Delta}$ is given by:

$$a_{t+\Delta} = (1 + r_t \Delta) [(\xi_l w_t n_t + D_t - c_t) \Delta + a_t] \quad \text{and} \quad a_{t+\Delta} \geq -\underline{a},$$

The optimal choice of hours worked n_t by a household with idiosyncratic productivity ξ_h is given by:

$$\frac{\psi}{\xi_h w_t} = (1 + r_t \Delta) (1 - \rho \Delta) [(1 - \lambda_{l,t+\Delta} \Delta) V_a(a_{t+\Delta}, \xi_h) + \lambda_{l,t+\Delta} \Delta V_a(a_{t+\Delta}, \xi_l)], \quad (\text{a.3})$$

and by a household with idiosyncratic productivity ξ_l is given by:

$$\frac{\psi}{\xi_l w_t} = (1 + r_t \Delta) (1 - \rho \Delta) [(1 - \lambda_{h,t+\Delta} \Delta) V_a(a_{t+\Delta}, \xi_l) + \lambda_{h,t+\Delta} \Delta V_a(a_{t+\Delta}, \xi_h)], \quad (\text{a.4})$$

The optimal choice of consumption for a household with productivity ξ_h and ξ_l can be written as:

$$c_t(a, \xi_h) \leq \{(1 + r_t \Delta) (1 - \rho \Delta) [(1 - \lambda_{l,t+\Delta} \Delta) V_a(a_{t+\Delta}, \xi_h) + \lambda_{l,t+\Delta} \Delta V_a(a_{t+\Delta}, \xi_l)]\}^{-\gamma} \quad (\text{a.5})$$

$$c_t(a, \xi_l) \leq \{(1 + r_t \Delta) (1 - \rho \Delta) [(1 - \lambda_{h,t+\Delta} \Delta) V_a(a_{t+\Delta}, \xi_l) + \lambda_{h,t+\Delta} \Delta V_a(a_{t+\Delta}, \xi_h)]\}^{-\gamma}, \quad (\text{a.6})$$

where we have divided both sides of each equation by Δ . The inequality captures the fact that households may be borrowing constrained. Next, the envelope conditions for ξ_h and ξ_l households are

$$V_a(a_t, \xi_h) = (1 + r_t \Delta) (1 - \rho \Delta) [(1 - \lambda_{l,t+\Delta} \Delta) V_a(a_{t+\Delta}, \xi_h) + \lambda_{l,t+\Delta} \Delta V_a(a_{t+\Delta}, \xi_l)] \quad (\text{a.7})$$

$$V_a(a_t, \xi_l) = (1 + r_t \Delta) (1 - \rho \Delta) [(1 - \lambda_{h,t+\Delta} \Delta) V_a(a_{t+\Delta}, \xi_l) + \lambda_{h,t+\Delta} \Delta V_a(a_{t+\Delta}, \xi_h)] \quad (\text{a.8})$$

Using the envelope conditions along with (a.3)–(a.6), we have:

$$c_{h,t} \equiv c_t(a, \xi_h) = \left(\frac{\xi_h w_t}{\psi} \right)^\gamma \quad \text{and} \quad c_{l,t} \equiv c_t(a, \xi_l) = \left(\frac{\xi_l w_t}{\psi} \right)^\gamma, \quad (\text{a.9})$$

which shows that the consumption of all households with the same idiosyncratic productivity ξ_j is the same, regardless of their financial wealth.

Next, it is easy to see that for any real interest rate r_t , the expected consumption growth of ξ_l households is larger than that of ξ_h households. The expected consumption growth between dates $t + \Delta$ and t , for a household with productivity ξ_l at date t can be written as:

$$(1 - \lambda_{h,t+\Delta} \Delta) \frac{c_{l,t+\Delta}}{c_{l,t}} + \lambda_{h,t+\Delta} \Delta \frac{c_{h,t+\Delta}}{c_{l,t}} = \left\{ 1 + \lambda_{h,t+\Delta} \Delta \left[\left(\frac{\xi_h}{\xi_l} \right)^\gamma - 1 \right] \right\} \left(\frac{w_{t+\Delta}}{w_t} \right)^\gamma,$$

while the expected consumption growth of a household with productivity ξ_h at date t can be written as $\left\{ 1 - \lambda_{l,t+\Delta} \Delta \left[\frac{\xi_h^\gamma - \xi_l^\gamma}{\xi_h^\gamma} \right] \right\} \left(\frac{w_{t+\Delta}}{w_t} \right)^\gamma$. Thus, for any path of aggregate variables, the expected consumption growth is higher for ξ_l households. Hence, in equilibrium, at any date t , all ξ_l households must be borrowing constrained, and choose $a_{t+\Delta} = -\underline{a}$. In contrast, the ξ_h households are on their Euler equation, which can be derived by rearranging the envelope condition for ξ_h households as:

$$\frac{\rho - r_t (1 - \rho \Delta)}{(1 + r_t \Delta) (1 - \rho \Delta)} V_a(a_t, \xi_h) = \left[\frac{V_a(a_{t+\Delta}, \xi_h) - V_a(a_t, \xi_h)}{\Delta} + \lambda_{l,t+\Delta} \{V_a(a_{t+\Delta}, \xi_l) - V_a(a_{t+\Delta}, \xi_h)\} \right]$$

Taking the limit of this equation as $\Delta \rightarrow 0$, we get:

$$(\rho - r_t) V_a(a_t, \xi_h) = \dot{V}_a(a_t, \xi_h) + \lambda_{l,t} \{V_a(a_t, \xi_l) - V_a(a_t, \xi_h)\} \quad (\text{a.10})$$

Next, using (a.3)-(a.6) along with the envelope conditions, we have:

$$V_a(a_t, \xi_h) = c_{h,t}^{-\gamma^{-1}}, \quad V_a(a_t, \xi_l) = c_{l,t}^{-\gamma^{-1}} \quad \text{and} \quad \dot{V}_a(a_t, \xi_h) = -\gamma^{-1} c_{h,t}^{-\gamma^{-1}} \frac{\dot{c}_{h,t}}{c_{h,t}}$$

Using this in (a.10), we have the Euler equation:

$$\frac{\dot{c}_{h,t}}{c_{h,t}} = \gamma(r_t - \rho) + \gamma \lambda_{l,t} \left[\left(\frac{c_{l,t}}{c_{h,t}} \right)^{-\frac{1}{\gamma}} - 1 \right],$$

which is the same as (9) in the main text. □

A.2 Government debt and asset market clearing

In our baseline model, we assume that there is zero government debt issued $B_t = 0$ at each date. Since at any date t , all η of the ξ_l households are borrowing constrained and thus, in net, they borrow $\eta \underline{a}$, where η is the constant fraction of ξ_l households. Consequently, asset market clearing implies that ξ_h households at date t must be net savers and hold $\eta \underline{a}$ as a group.

Allowing for non-zero government debt does not affect the savings decision of ξ_l households and at any date t , they are still borrowing constrained and borrow $\eta \underline{a}$ as a group. However, now that there is a non-zero amount of government debt in the economy, asset market clearing implies that all ξ_h households as a whole must save $\eta \underline{a} + b_t$ at date t , where b_t denotes real government debt at date t (see below).

Notice that despite changing the equilibrium asset holdings of ξ_h households, adding non-zero government debt into our economy does not affect the consumption of each type of households, which is still given by (a.9). This is because quasi-linear preferences imply that consumption of each household is independent of their wealth. Consequently, because we always normalize government expenditures to 0, with or without government debt, the goods market clearing condition can be written as:

$$y_t = (1 - \eta)c_{h,t} + \eta c_{l,t}, \quad (\text{a.11})$$

Using the expressions for c_h and c_l in (a.9), (a.11) can be re-written as:

$$y_t = \left[(1 - \eta) \left(\frac{\xi_h}{\psi} \right)^\gamma + \eta \left(\frac{\xi_l}{\psi} \right)^\gamma \right] w_t^\gamma$$

Normalizing $\psi = [(1 - \eta)\xi_h^\gamma + \eta\xi_l^\gamma]^\frac{1}{\gamma}$, we can rewrite this as:

$$w_t = y_t^\frac{1}{\gamma}, \quad (\text{a.12})$$

which is the same as (11) in the main text. Furthermore, setting $\dot{\pi} = 0$ and $\pi = 0$ in the Phillips curve

(5) implies that real wages in the steady state with on-target inflation is $w = 1$. Consequently, (a.12) implies that output in the targeted steady state is $y = 1$. Finally, using this relationship in (a.9), we can express the per-capita consumption of households with skill ξ in terms of output at any date t :

$$c_{h,t} = \frac{\xi_h^\gamma}{(1-\eta)\xi_h^\gamma + \eta\bar{\xi}_l^\gamma} y_t > \frac{\xi_l^\gamma}{(1-\eta)\xi_h^\gamma + \eta\bar{\xi}_l^\gamma} y_t = c_{l,t} \quad (\text{a.13})$$

Government budget constraint The government budget constraint be written in nominal terms as:

$$\dot{B}_t = i_t B_t - P_t T_t$$

where B_t denotes the stock of nominal government debt at date t , T_t denote taxes net of transfers, and we have normalized government expenditures $g = 0$. Defining real debt as $b_t = B_t/P_t$, we can rewrite the government budget constraint in real terms as

$$\dot{b}_t = (i_t - \pi_t) b_t - T_t \quad (\text{a.14})$$

In our baseline model (Section III), we restrict attention to the case in which $T_t = b_t = 0$ at all dates. However, in Section IV.B, we assume that fiscal policy sets T_t according to the following rule:

$$T_t = \bar{T} + \bar{r}\phi_b b_t \quad (\text{a.15})$$

Combining (a.14) and (a.15), the evolution of real government debt can be written as:

$$\dot{b}_t = (i_t - \pi_t - \bar{r}\phi_b) b_t - \bar{T}, \quad (\text{a.16})$$

Defining b^* as the level of real government debt in the targeted steady state, imposing $\dot{b} = 0$ and $b = b^*$ in (a.16) implies that

$$\bar{T} = \bar{r}(1 - \phi_b) b^*$$

Using this, we can rewrite (a.16) as:

$$\dot{b}_t^g = (r_t - \bar{r}\phi_b) b_t - \bar{r}(1 - \phi_b) b^*, \quad (\text{a.17})$$

where $r_t = i_t - \pi_t$ is the real interest rate at date t . Rather than working with (a.17), it is more convenient to characterize the dynamics of $b_t^g = b_t - b^*$, i.e, the gap between the actual level of government debt and its level in the targeted steady state. We can then express (a.17) in terms of b_t^g as:

$$\dot{b}_t^g = (r_t - \bar{r}) b^* + (r_t - \bar{r}\phi_b) b_t^g \quad (\text{a.18})$$

A.3 Aggregate dynamics

The aggregate dynamics of y_t , π_t and b_t^g are described by the IS curve, the Phillips curve and the government budget constraint. We start by deriving the IS curve. Taking the time-derivative of expression

for $c_{h,t}$ in (a.13) yields $\frac{\dot{c}_{h,t}}{c_{h,t}} = \frac{\dot{y}_t}{y_t}$. We can then rewrite the Euler equation of the ξ_h household (9) as:

$$\frac{\dot{y}_t}{y_t} = \gamma(i_t - \pi_t - \rho) + \gamma\bar{\lambda}_l \left(\frac{\xi_h}{\xi_l} - 1 \right) y_t^{-\Theta}$$

Next, defining $x_t = \frac{1}{\gamma} \ln y_t$, we can rewrite the above as:

$$\dot{x}_t = i_t - \pi_t - \rho + \bar{\lambda}_l \left(\frac{\xi_h}{\xi_l} - 1 \right) e^{-\gamma\Theta x_t}$$

Substituting out $i_t = \bar{r} + \phi_\pi \pi_t$ using the monetary policy rule (6), we have

$$\dot{x}_t = \bar{r} + (\phi_\pi - 1)\pi_t - \rho + \sigma e^{-\gamma\Theta x_t}, \quad (\text{a.19})$$

where $\sigma = \bar{\lambda}_l \left(\frac{\xi_h}{\xi_l} - 1 \right)$, and $\bar{r} = \rho - \sigma$ is the real interest rate in the targeted steady state (and also the intercept in the monetary policy rule). Similarly, rewriting the Phillips curve (5) in terms of x_t , we have

$$\dot{\pi}_t = \rho\pi_t - \kappa(e^{x_t} - 1) \quad (\text{a.20})$$

Finally, by substituting $i_t = \bar{r} + \phi_\pi \pi_t$ using the monetary policy rule (6), into the government budget constraint (a.18), we can rewrite (a.18) as:

$$\dot{b}_t = (\phi_\pi - 1)b^*\pi_t + \bar{r}(1 - \phi_b)b_t^g + (\phi_\pi - 1)\pi_t b_t^g, \quad (\text{a.21})$$

Flexible-price limit Since we abstract from aggregate shocks, in the flexible-price limit of our economy ($\kappa \rightarrow \infty$), the Phillips curve implies that at any date t , $x_t = \pi_t = 0$. Using this in (a.19) and rearranging, the real interest rate in the flexible price limit $\bar{r}$ is given by $\bar{r} = \rho - \sigma$, where $\sigma = \bar{\lambda}_l \left(\frac{\xi_h}{\xi_l} - 1 \right) \geq 0$ captures the effect of consumption risk faced by households in steady state.

Using the fact that $\bar{r} = \rho - \sigma$, the equilibrium dynamics of x_t, π_t, b_t^g are given by the three dimensional system of non-linear ODEs:

$$\dot{x}_t = (\phi_\pi - 1)\pi_t + \sigma(e^{-\gamma\Theta x_t} - 1) \quad (\text{a.22})$$

$$\dot{\pi}_t = \rho\pi_t - \kappa(e^{x_t} - 1) \quad (\text{a.23})$$

$$\dot{b}_t^g = (\phi_\pi - 1)b^*\pi_t + \bar{r}(1 - \phi_b)b_t^g + (\phi_\pi - 1)\pi_t b_t^g \quad (\text{a.24})$$

A.3.1 Aggregate dynamics in the baseline model with zero government debt

In our baseline model with zero government debt, which implies that $b_t = b^* = b_t^g = \dot{b}_t^g = 0$ at all dates. Thus, (a.24) simply states that $0 = 0$, and so the aggregate dynamics of x_t, π_t in our baseline economy with zero government debt are fully described by the IS curve (a.22) and Phillips curve (a.23).

In the flexible price limit of the baseline economy $\kappa \rightarrow \infty$, as previously mentioned, the Phillips curve implies that $x_t = \pi_t = 0$ is the only bounded solution which satisfies both the IS and Phillips curves, and is thus the unique bounded equilibrium. $\square$

B Baseline model with inflation targeting rule

This section contains the proof of claims relating to the baseline model described by (16a)-(16b).

B.1 Global determinacy in RANK and in HANK with acyclical risk

In the RANK limit ($\sigma = 0$) / with acyclical risk ($\sigma > 0, \Theta = 0$), aggregate dynamics are given by:

$$\begin{aligned}\dot{x}_t &= (\phi_\pi - 1)\pi_t \\ \dot{\pi}_t &= \rho\pi_t - \kappa(e^{x_t} - 1)\end{aligned}$$

The Jacobian of the system evaluated at any (x, π) can be written as:

$$\begin{bmatrix} 0 & \phi_\pi - 1 \\ -\kappa e^x & \rho \end{bmatrix}$$

To show that $\phi_\pi > 1$ delivers global determinacy, we can invoke the Bendixson–Dulac theorem (Bendixson, 1901; Dulac, 1937),¹ which states if the trace does not change sign anywhere in the domain, then there are no non-constant periodic solutions lying entirely within $(x, \pi) \in (-\infty, \infty)^2$. This is true by inspection since the trace is given by $\rho > 0$. Thus, there are no non-constant periodic solutions.

Next, with $\phi_\pi > 1$, the determinant of the Jacobian evaluated at the targeted steady state $(x, \pi) = (0, 0)$ is given by $\kappa(\phi_\pi - 1) > 0$. Together with the fact that the trace is always positive $\rho > 0$, this implies that both eigenvalues have positive real parts. Thus, the targeted steady state $(0, 0)$ is unstable, and hence the only bounded equilibrium is given by the trajectory $(x_t, \pi_t) = (0, 0)$ for all t . $\square$

B.2 Proof of Proposition 2

Close to the targeted steady state $(0, 0)$, the dynamics of the system (16a)-(16b) are governed by:

$$\begin{bmatrix} \dot{x} \\ \dot{\pi}_t \end{bmatrix} = A \begin{bmatrix} x \\ \pi \end{bmatrix} + \mathcal{O}(x^2) \quad \text{for} \quad (x, \pi) \rightarrow (0, 0),$$

where A is given by

$$A = \begin{bmatrix} -\sigma\gamma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix},$$

Since both x and π are “jump” variables, local determinacy requires that both eigenvalues of A have a positive real part. As is well known, the sum of the two eigenvalues of A , denoted by z_1 and z_2 , is given by the trace of A , while their product is given by the determinant of A :

$$\begin{aligned}z_1 + z_2 &= \rho - \sigma\gamma\Theta, \\ z_1 \times z_2 &= \kappa(\phi_\pi - 1) - \rho\sigma\gamma\Theta\end{aligned}$$

¹See Theorem 4.1 on page 39 in Verhulst (1990) for a simple statement of the Bendixson–Dulac theorem in English.

Since this is a two dimensional system, either both z_1 and z_2 are real, or they are complex conjugates. Thus, for z_1 and z_2 to both have positive real parts, it is sufficient that both the sum and product of z_1, z_2 be positive. In other words, as long as $\Theta < \Theta^* \equiv \frac{\rho}{\sigma\gamma}$, a sufficient condition for local determinacy is that

$$\phi_\pi > 1 + \frac{\rho\sigma\gamma\Theta}{\kappa},$$

which is the same condition as in Proposition 2. Finally, for $\Theta > \Theta^*$, the sum of the two eigenvalues $z_1 + z_2 < 0$ regardless of the magnitude of ϕ_π , implying that at least one of the eigenvalues must have a negative real part, i.e., regardless of the magnitude of ϕ_π , the equilibrium is locally indeterminate. $\square$

B.3 Multiple Steady States

For any $\Theta > 0$, our baseline HANK economy has two steady states (except in a knife edge case). The $\dot{x} = 0$ and $\dot{\pi}_t = 0$ nullclines imply that in any steady state, (x, π) must satisfy:

$$\begin{aligned} 0 &= (\phi_\pi - 1)\pi + \sigma(e^{-\gamma\Theta x} - 1) \\ 0 &= \rho\pi - \kappa(e^x - 1) \end{aligned}$$

Clearly, $(0,0)$ always satisfies both equations. To see that there is generically another steady state, combine the two equations to eliminate π , to get an expression exclusively in terms of x :

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}(e^x - 1) + \sigma(e^{-\gamma\Theta x} - 1), \quad (\text{b.1})$$

and any x which satisfies $F(x) = 0$ constitutes a steady state. Again, clearly $x = 0$ solves this equation. The derivative of $F(x)$ is given by:

$$F'(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}e^x - \sigma\gamma\Theta e^{-\gamma\Theta x},$$

which, evaluated at $x = 0$ yields

$$F'(0) = \frac{\kappa}{\rho}(\phi_\pi - \varphi(\Theta)) \quad \text{where} \quad \varphi(\Theta) = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa},$$

If $\phi_\pi = \varphi(\Theta)$, then $F'(0) = 0$ and $F(x)$ is tangent to the x -axis at $x = 0$, implying that it is the only zero of $F(x)$ since $F(x)$ is declining in the region $x < 0$ and increasing in the region $x > 0$. This is the knife edge case in which there is a unique steady state. If instead, $\phi_\pi > \varphi(\Theta)$, then $F'(0) > 0$. Since $\lim_{x \rightarrow -\infty} F(x) \rightarrow \infty$, there must be at least one intersection with $\underline{x} < 0$ and $F'(\underline{x}) < 0$. Since $F(x)$ is strictly convex, this intersection is unique. Further, note that $dF(x)/d\phi_\pi < 0$ for $x < 0$ by inspection. Thus, by the implicit function theorem, we have $d\underline{x}/d\phi_\pi < 0$.

Instead if $1 < \phi_\pi < \varphi(\Theta)$, then $F(x)$ intersects the x axis twice, one of which is $x = 0$. We also know that in this case $F'(0) < 0$ and that $F(x) \rightarrow \infty$ as $x \rightarrow \infty$, implying that there is at least one intersection at $\bar{x} > 0$ with $F'(\bar{x}) > 0$. Since $F(x)$ is convex as long as $\phi_\pi > 1$, this intersection is the

only other intersection except $x = 0$. Furthermore, the implicit function theorem implies that a smaller ϕ_π implies a larger $\bar{x}$ as long as $\phi_\pi > 1$.

Case with $\phi_\pi = 1$ Finally, it is worth mentioning that with $\phi_\pi = 1$, only the targeted steady state exists. If we impose $\phi_\pi = 1$, (b.1) simplifies to:

$$F(x) = \sigma \left(e^{-\gamma\Theta x} - 1 \right),$$

which is clearly a monotonic function of x , and so $x = 0$ is the only solution to the equation $F(x) = 0$. Thus, it follows that with $\phi_\pi = 1$, only the targeted steady state exists.

However, there is still both local and global indeterminacy as there are multiple bounded trajectories $\{x_t, \pi_t\}$ which originate away from the targeted equilibrium $(0, 0)$ and still stay bounded. The easiest way to see this is to check the condition for local determinacy. This requires that that the eigenvalues of the matrix A , defined in equation (17) in Section 3 of the main paper has two explosive roots. Setting $\phi_\pi = 1$, we can write the matrix A as:

$$A = \begin{bmatrix} -\sigma\gamma\Theta & 0 \\ -\kappa & \rho \end{bmatrix}$$

The product of the two eigenvalues of A is given by the determinant of A , which is given by $-\rho\sigma\gamma\Theta$. This is clearly negative as long as risk is countercyclical $\Theta > 0$. Since the product of the eigenvalues is negative, it is automatic that the eigenvalues are real, and one is positive while the other is negative. This structure of eigenvalues implies that there is a 1-dimensional stable manifold around the targeted steady state, and any trajectory which begins on this manifold converges to the targeted steady state and hence remains bounded. Thus, with $\phi_\pi = 1$, as in RANK, when risk is countercyclical the targeted equilibrium is locally and hence globally indeterminate, and hence, it is without loss of generality that we focus on the case with $\phi_\pi > 1$ in our baseline. □

B.4 Local stability of the untargeted steady state

For any $\Theta > 0$, we focus of the case in which $\phi_\pi > \varphi(\Theta)$, which is a necessary (but not sufficient) condition for the targeted equilibrium to be locally determinate (see Proposition 2). We now show that whenever this condition is satisfied, the untargeted steady state, which features lower output $\underline{x}$, is locally indeterminate. At the untargeted steady state, the Jacobian of the system (16a)-(16b) can be written as:

$$A_{\underline{x}} = \begin{bmatrix} -\sigma\gamma\Theta e^{-\gamma\Theta \underline{x}} & \phi_\pi - 1 \\ -\kappa e^{\underline{x}} & \rho \end{bmatrix} \quad (\text{b.2})$$

Thus, up to first-order, the dynamics around the untargeted steady state are identical to the local dynamics around the *targeted* steady state of an alternate economy with more cyclical risk: $\Theta' = \Theta e^{-\gamma\Theta \underline{x}} > \Theta$ and a flatter Phillips curve with slope $\kappa' = \kappa e^{\underline{x}} < \kappa$ (since $\underline{x} < 0$, and so $e^{-\gamma\Theta \underline{x}} > 1$ and $e^{\underline{x}} < 1$). Consequently, we can apply Proposition 2 to conclude that for the untargeted steady state

$(\underline{x}, \underline{\pi})$ to be unstable (locally determinate), we need:

$$\phi_\pi > 1 + \frac{\rho\sigma\gamma\Theta'}{\kappa'} = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa} e^{-(\gamma\Theta+1)\underline{x}}$$

However, for a given Θ and ϕ_π , this can never be satisfied as long as $\phi_\pi > \phi(\Theta)$. To see why, recall from Appendix B.3 that we can use (b.1) to write $d\underline{x}/d\phi_\pi$ as:

$$\frac{d\underline{x}}{d\phi_\pi} = \left(\frac{1 - e^{\underline{x}}}{e^{\underline{x}}} \right) \left[\phi_\pi - 1 - \frac{\rho\sigma\gamma\Theta}{\kappa} e^{-(\gamma\Theta+1)\underline{x}} \right]^{-1}$$

We know that this expression is negative as long as $\phi_\pi > 1 + \frac{\rho\sigma\gamma\Theta}{\kappa}$. Since $\underline{x} < 0$, this implies that

$$\phi_\pi < 1 + \frac{\rho\sigma\gamma\Theta}{\kappa} e^{-(1+\gamma\Theta)\underline{x}}, \quad (\text{b.3})$$

i.e., the untargeted steady state $(\underline{x}, \underline{\pi})$ is stable (locally indeterminate), if $\phi_\pi > 1 + \frac{\rho\sigma\gamma\Theta}{\kappa}$. $\square$

B.5 Proof of Proposition 3

Given the interest rate rule $i_t = \bar{r} + \phi_\pi \pi_t$, the aggregate dynamics of the output-gap x_t and inflation π_t is given by the following 2 dimensional system of ordinary differential equations:

$$\begin{aligned} \dot{x} &= (\phi_\pi - 1) \pi + \sigma (e^{-\gamma\Theta x} - 1) \\ \dot{\pi} &= \rho \pi - \kappa (e^x - 1) \end{aligned}$$

We can rewrite this system in matrix form as:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \underbrace{\begin{bmatrix} -\sigma\gamma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix}}_A \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \begin{bmatrix} \sigma (e^{-\gamma\Theta x_t} - 1 + \gamma\Theta x_t) \\ -\kappa (e^x - 1 - x_t) \end{bmatrix} \quad (\text{b.4})$$

Next, for a given $\Theta > 0$ and assuming that (18) is satisfied, we show that the global dynamics of (x_t, π_t) can be split into 3 broad regions depending on the magnitude of Θ . This result is formally presented in Proposition B.1 below.

Proposition B.1 (Global dynamics). *Consider the economy described in Proposition 1 for a given $\Theta > 0$ and assume that (18) is satisfied. Then the global dynamics depend on the magnitude of the cyclical risk Θ , and can be split into 3 broad regions.*

1. **Mildly countercyclical risk** $\Theta \in (0, \Theta^\diamond)$: $\exists \Theta^\diamond > 0$, such that for any $\Theta \in (0, \Theta^\diamond)$, there exists a saddle-connection, along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state. In this region, the targeted equilibrium is locally determinate, but there is global indeterminacy, as any trajectory starting on the saddle connection also remains bounded forever, while satisfying all equilibrium conditions. Figure B.1a depicts this case.
2. **Moderately countercyclical risk** $\Theta \in [\Theta^\diamond, \Theta^\star]$: For any $\Theta \in [\Theta^\diamond, \Theta^\star]$, trajectories originating in the

neighborhood of the targeted steady state $(0,0)$ initially diverge away from the steady state but eventually converge to a **super-critical limit-cycle** surrounding $(0,0)$, and thus remain bounded and imply the equilibrium is globally indeterminate, even though the targeted steady state is locally determinate. Moreover, the periodicity of the periodic solutions is a decreasing function of Θ in the range $(\Theta^\diamond, \Theta^\star)$. Figure B.1c depicts the stable cycle.

In the knife edge case with $\Theta = \Theta^\diamond$, there exists a homoclinic orbit, which is a stable trajectory which connects the untargeted steady state $(\underline{x}, \underline{\pi})$ to itself, and lies on the boundary of the periodic solutions described above. Again, this implies global indeterminacy, even though $(0,0)$ is locally determinate. This case is depicted by Figure B.1b.

Finally, if $\Theta = \Theta^\star$, the limit-cycles collapse onto the steady state $(0,0)$. The equilibrium is globally indeterminate since the higher-order terms ensure that any trajectory starting in the neighborhood of the targeted steady state converges back to $(0,0)$.

3. **Highly countercyclical risk** $\Theta > \Theta^\star$: For any $\Theta > \Theta^\star$, the targeted steady state $(0,0)$ is locally indeterminate even if (18) is satisfied, i.e., there exists multiple bounded trajectories in the neighborhood of $(0,0)$, which converge to the targeted steady state. Since the equilibrium is locally indeterminate, it is also globally indeterminate. In addition to the multiple bounded trajectories which start near the targeted steady state, there also exists a saddle-connection, along which, the economy converges to the targeted steady state even if it starts near the untargeted steady state. This is depicted by Figure B.1d

Overall, for any $\Theta > 0$, i.e, if risk is even mildly countercyclical, there is global indeterminacy. For any finite ϕ_π , the inflation targeting rule (6) fails at eliminating the existence of multiple bounded equilibria.

We will prove Proposition B.1 by using Theorem 7.1 in Kopell and Howard (1975), and the Hopf bifurcation theorem (Marsden and McCracken, 1976). We present these theorems here for convenience.

Theorem 1 (Hopf Bifurcation Theorem). *Consider a two-dimensional system*

$$\dot{\mathbf{x}} = F_\mu(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^2, \mu \in \mathbb{R}$$

with smooth F , which for all sufficiently small $|\mu|$ has the equilibrium $\mathbf{x} = (0,0)$, and the Jacobian $D_{\mathbf{x}}F_\mu(0,0)$ has eigenvalues

$$\lambda_{1,2}(\mu) = \Omega(\mu) \pm i\omega(\mu) \quad \text{where} \quad i = \sqrt{-1},$$

Then, if the following conditions are satisfied:

1. At $\mu = 0$, there exists a purely imaginary set of eigenvalues: $\Omega(0) = 0$ and $\omega(0) > 0$
2. The eigenvalues cross the imaginary axis with non-zero speed: $\frac{d\Omega(0)}{d\mu} \neq 0$
3. The first Lyapunov coefficient of the system $\ell_1(0) \neq 0$,

there exists a family of real periodic solutions $\mathbf{x} = \mathbf{x}(t, \epsilon)$, $\mu(\epsilon)$ which has properties $\mu(0) = 0$, $\mathbf{x}(t, 0) = (0,0)$, but $\mathbf{x}(t, \epsilon) \neq (0,0)$ for sufficiently small ϵ . The same holds for the period $T(\epsilon)$ and $T(0) = 2\pi / |\omega(0)|$.

Theorem 2 (Theorem 7.1 in Kopell and Howard (1975)). *Let $\dot{\mathbf{x}} = F_{\mu,v}(\mathbf{x})$ be a two parameter family of ODEs on $\mathbb{R}^2$, F smooth in all of its four arguments, such that $F_{\mu,v}(0,0) = \mathbf{0}$. Also assume:*

1. $dF_{0,0}(0,0)$ has a double zero eigenvalue and a single eigenvector $\mathbf{v}$.
2. The mapping $(\mu, \nu) \rightarrow (\det(dF_{\mu,\nu}(0,0)), \text{tr}(dF_{\mu,\nu}(0,0)))$ has a nonzero Jacobian at $(\mu, \nu) = (0, 0)$.
3. Let $Q(\mathbf{x}, \mathbf{x})$ be the 2×1 vector containing the terms quadratic in $\mathbf{x}$ and independent of (μ, ν) in a Taylor series expansion of $F_{\mu,\nu}(\mathbf{x})$ around 0. Then $[dF_{0,0}(0,0), Q(\mathbf{v}, \mathbf{v})]$ has rank 2.

Then there is a curve $f(\mu, \nu) = 0$ such that if $f(\mu_0, \nu_0) = 0$, then $\dot{\mathbf{x}} = F_{\mu_0, \nu_0}(\mathbf{x})$ has a homoclinic orbit. This one-parameter family of homoclinic orbits (in (X, μ, ν) space) is on the boundary of a two-parameter family of periodic solutions. For all $|\mu|, |\nu|$ sufficiently small, if $\dot{\mathbf{x}} = F_{\mu,\nu}(\mathbf{x})$ has neither a homoclinic orbit nor a periodic solution, there is a unique trajectory joining the critical points.

We first use Theorem 1 to prove point 2 of Proposition B.1. We use the cyclicity of risk Θ as the bifurcation parameter (which plays the role of μ in Theorem 1 above). Define $\Theta^* = \frac{\rho}{\sigma\gamma}$. Imposing $\Theta = \Theta^*$ in (b.4) yields

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \underbrace{\begin{bmatrix} -\rho & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix}}_{A^*} \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \begin{bmatrix} \sigma \left(e^{-\frac{\rho}{\sigma} x_t} - 1 + \frac{\rho}{\sigma} x_t \right) \\ -\kappa (e^{x_t} - 1 - x_t) \end{bmatrix}, \quad (\text{b.5})$$

It is clear by inspection that the trace of the matrix A^* is equal to zero, which implies that the two eigenvalues sum to 0. Also, the determinant of A^* is given by

$$\text{Det}(A^*) = \kappa(\phi_\pi - 1) - \rho^2$$

Next, recall that the augmented Taylor principle (18) requires that $\phi_\pi > 1 + \frac{\rho\sigma\gamma\Theta}{\kappa}$ for local determinacy. Evaluating this expression at $\Theta = \Theta^*$, we have $\phi_\pi > 1 + \frac{\rho^2}{\kappa}$, which in turn implies that $\text{Det}(A^*) > 0$. Consequently, at $\Theta = \Theta^*$ the eigenvalues of A^* are purely imaginary and given by $\pm\omega i$, where $i = \sqrt{-1}$ and $\omega = \sqrt{\kappa(\phi_\pi - 1) - \rho^2}$. Thus, requirement 1 of Theorem 1 is satisfied at $\Theta = \Theta^*$ (this corresponds to the $\mu = 0$ in the statement of the theorem).

Next, it is clear by inspection that the eigenvalues of the matrix A in (b.4) change smoothly in Θ , which implies that condition 2 of Theorem 1 is also satisfied. Thus, the only other condition we need to check is that the first Lyapunov coefficient (evaluated at the bifurcation point) is not 0. To check this, we first need to transform the system (b.5) into normal-form, for which we diagonalize A^* as:²

$$A^* = PDP^{-1},$$

where

$$D = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} \rho & \omega \\ \kappa & 0 \end{bmatrix}$$

²The normal form involves a change of variables so that the first-order accurate system can be written in the well known “decoupled” system. Diagonalizing a matrix with real eigenvalues leads to a decoupled system with entries only on the main diagonal. However, since we are diagonalizing a 2x2 matrix with purely complex roots, the diagonalized matrix features non-zero entries only on the anti-diagonal. See the matrix in equation (b.6).

Next, we can pre-multiply both sides of (b.5) by P^{-1} to express the system in normal form:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}, \quad (\text{b.6})$$

where

$$\begin{bmatrix} u \\ v \end{bmatrix} = P^{-1} \begin{bmatrix} x \\ \pi \end{bmatrix},$$

$$\begin{aligned} f(u, v) &= -e^{\rho u + \omega v} + 1 + \rho u + \omega v \\ g(u, v) &= \frac{\rho e^{\rho u + \omega v} + \sigma e^{-\frac{\rho}{\sigma}(\rho u + \omega v)} - (\rho + \sigma)}{\omega} \end{aligned}$$

Finally, the first Lyapunov coefficient at the bifurcation point $\Theta = \Theta^*$ is given by:

$$\begin{aligned} \ell_1(0) &= f_{uuu}(0, 0) + f_{uvv}(0, 0) + g_{uuv}(0, 0) + g_{vvv}(0, 0) \\ &\quad + \frac{1}{\omega} \left[f_{uv}(0, 0) (f_{uu}(0, 0) + f_{vv}(0, 0)) - g_{uv}(0, 0) (g_{uu}(0, 0) + g_{vv}(0, 0)) - f_{uu}(0, 0) g_{uu}(0, 0) \right. \\ &\quad \left. + f_{vv}(0, 0) g_{vv}(0, 0) \right] \\ &= -(\rho + \sigma) \frac{\rho^2 \kappa^2 (\phi_\pi - 1)^2}{\sigma^2 \omega^2} < 0, \end{aligned}$$

which is non-zero, for any $\phi_\pi > 0$. Thus, condition 3 of Theorem 1 is also satisfied, and the system (b.4) undergoes a Hopf bifurcation at $\Theta = \Theta^*$. Furthermore, since $\ell_1(0) < 0$ regardless of the value of $\phi_\pi > 1$, the Hopf bifurcation is always *supercritical*, i.e. the higher-order terms of the system (b.4), push x in towards the equilibrium $(0, 0)$.

In terms of our HANK model, this means that there exists $\Theta^\diamond \in (0, \Theta^*)$ such that for any Θ in the neighborhood $\Theta \in (\Theta^\diamond, \Theta^*)$, starting from any initial condition in the neighborhood $(x, \pi) = (0, 0)$, the system converges to a stable cycle around the targeted steady state. In the main text, if cyclicity of risk is such that $\Theta^\diamond < \Theta < \Theta^*$, we say that risk is “moderately countercyclical”. Thus, for any Θ in the moderately countercyclical region, even though the targeted equilibrium is locally determinate, the equilibrium is globally indeterminate since all trajectories starting near the targeted steady state initially diverge but then converge to a cycle, remaining bounded.

Theorem 1 also implies that the amplitude and periodicity of the stable cycles are decreasing functions of Θ in the moderately countercyclical range, which in turn depends on the inverse of ω (the imaginary part of the eigenvalue). In particular, at $\Theta = \Theta^\diamond$, Proposition B.1 shows that there exists a homoclinic orbit (we prove this using Theorem 2 below). The homoclinic orbit has infinite periodicity, which means that $\Theta^\diamond$ is implicitly defined by the Θ for which the cycle has infinite periodicity. While this cannot be characterized analytically, we can numerically compute the value of $\Theta^\diamond$, and this value is presented in the main text. Thus, as we increase Θ from $\Theta^\diamond$ towards Θ^* , the amplitude and periodicity of the cycles decrease. Finally, at $\Theta = \Theta^*$ the periodicity of the cycle is 0 and the cycle collapses on to the targeted steady state itself. However, as we show next, the equilibrium is still indeterminate. Since the eigenvalues of the Jacobian at $\Theta = \Theta^*$ are purely imaginary, the first-order terms do not move the

system away from or towards $(0,0)$. However, since the Hopf bifurcation is supercritical, the higher-order terms push in towards the origin, implying that all trajectories which start in the neighborhood of the targeted steady state $(0,0)$ remain bounded, implying global indeterminacy.

As mentioned above, to prove the existence of the homoclinic orbit at $\Theta = \Theta^\diamond$, we need to use Theorem 2, which states that as long the conditions 1,2,3 are satisfied, then (b.4) has a homoclinic orbit on the boundary of the stable cycles we described above. While Theorem 1 only required conditions on one parameter Θ , Theorem 2 requires imposing some additional conditions on a second parameter. For us, it is most convenient to choose ϕ_π as the second parameter. First, notice that for $\Theta = \Theta^*$ and $\phi_\pi = \varphi(\Theta^*) = 1 + \frac{\rho^2}{\kappa}$, the Jacobian of (b.4) is:

$$A^\diamond = \begin{bmatrix} -\rho & \frac{\rho^2}{\kappa} \\ -\kappa & \rho \end{bmatrix}, \quad (\text{b.7})$$

which has both trace and determinant equal to 0, implying that both eigenvalues are 0, satisfying condition 1 of Theorem 2. Also, since the eigenvalues repeat, it is easy to check that the matrix has eigenvector $\mathbf{v} = \begin{bmatrix} \rho \\ \kappa \end{bmatrix}$, and a generalized eigenvector $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$.

Next, we show that condition 2 is also satisfied. Recall that the Jacobian of (b.4) is given by the matrix A

$$A(\Theta, \phi_\pi) = \begin{bmatrix} -\sigma\gamma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix},$$

and the trace of $A(\Theta, \phi_\pi)$ is $Tr_A = \rho - \sigma\gamma\Theta$, while the determinant is $Det_A = \kappa(\phi_\pi - 1) - \rho\sigma\gamma\Theta$. Then the Jacobian of $[Tr_A, Det_A]$ is given by:

$$J = \begin{bmatrix} -\sigma\gamma & -\rho\sigma\gamma \\ 0 & \kappa \end{bmatrix}$$

The determinant of this matrix is non-zero as long as $\kappa\sigma\gamma \neq 0$. This confirms that condition 2 of Theorem 2 is satisfied. Next, to check condition 3, we need to construct the matrix $Q(\mathbf{x}, \mathbf{x})$, which is a 2×1 vector that contains terms quadratic in (x, π) and independent of (Θ, ϕ_π) in a Taylor series expansion of $F(x, \pi, \Theta^*, \varphi(\Theta^*))$ around $(x, \pi) = (0,0)$. Since (b.4) has no higher-order terms in π , it is easy to see that $Q(\mathbf{x}, \mathbf{x})$ can be written as:

$$Q = \begin{bmatrix} 0 \\ -0.5\kappa x^2 \end{bmatrix}$$

Evaluating this at the eigenvector $\mathbf{v}$, we have:

$$Q(\mathbf{v}, \mathbf{v}) = \begin{bmatrix} 0 \\ -0.5\kappa\rho^2 \end{bmatrix}$$

Then, clearly the condition 3 of Theorem 2 is satisfied since

$$\text{rank} \begin{bmatrix} -\rho & \kappa^{-1}\rho^2 & 0 \\ -\kappa & \rho & -0.5\kappa\rho^2 \end{bmatrix} = 2,$$

as long as $\rho \neq 0$. Thus, the conditions for Theorem 2 are satisfied. Consequently, Theorem 2 states that at $\Theta = \Theta^\diamond$, a homoclinic orbit emerges, which completes the proof of point 2 of Proposition B.1.

Next, we prove point 1 of Proposition B.1. Point 2 of Proposition B.1 established that stable cycles exist for $\Theta \in (\Theta^\diamond, \Theta^*)$ and a homoclinic orbit exists at $\Theta = \Theta^\diamond$. Then Theorem 2 implies that since there is no cycle or homoclinic orbit, there exists a saddle connection along which the economy moves from the targeted to the untargeted steady state for $0 < \Theta < \Theta^\diamond$. Any trajectory which originates on this saddle connection remain bounded. Hence, there is global indeterminacy even when $0 < \Theta < \Theta^\diamond$. Finally, point 3 of Proposition B.1 is also true because of similar reasons. When $\Theta > \Theta^*$, the Hopf bifurcation theorem implies that there are no cycles in this part of the parameter space. Consequently, Theorem 2 implies that there must be a saddle connection from the untargeted to targeted steady state. Since the targeted equilibrium is already locally indeterminate for $\Theta > \Theta^*$, there is also global indeterminacy. □

C Proof of Proposition 4

The monetary policy rule which implements a unique equilibrium is given by

$$i_t = \bar{r} + \phi_\pi \pi_t + \phi_r (r^*(x_t) - \bar{r}) \quad \text{with} \quad \phi_\pi > 1 \text{ and } \phi_r \geq 1 \quad (\text{c.1})$$

Such a rule ensures the existence of a unique bounded equilibrium in which the economy remains at the targeted steady state $(x, \pi) = (0, 0)$ at all dates. To see this, we can plug in the policy rule (c.1) into the IS equation (14). Then, the global dynamics of (x_t, π_t) are described by the following system of ODEs:

$$\dot{x}_t = (\phi_\pi - 1)\pi_t + (1 - \phi_r)\sigma \left(e^{-\gamma\Theta x_t} - 1 \right) \quad (\text{c.2})$$

$$\dot{\pi}_t = \rho\pi_t - \kappa(e^{x_t} - 1) \quad (\text{c.3})$$

This system nests our baseline economy if we set $\phi_r = 0$. First, note that $x_t = \pi_t = 0$ clearly satisfies the system of equations above, thus proving that the targeted steady state $x = \pi = 0$ is a valid equilibrium with the policy rule (c.1).

First, note that the Jacobian of the system at any (x, π) can be written as:

$$J(x, \pi) = \begin{bmatrix} (\phi_r - 1)\sigma\gamma\Theta e^{-\gamma\Theta x} & \phi_\pi - 1 \\ -\kappa e^x & \rho \end{bmatrix}$$

The local determinacy property of the targeted equilibrium $x = \pi = 0$ is determined by the eigenvalues of the matrix $J(0, 0)$. The sum of the eigenvalues of $J(0, 0)$ is given by $\rho + (\phi_r - 1)\sigma\gamma\Theta$ and the product

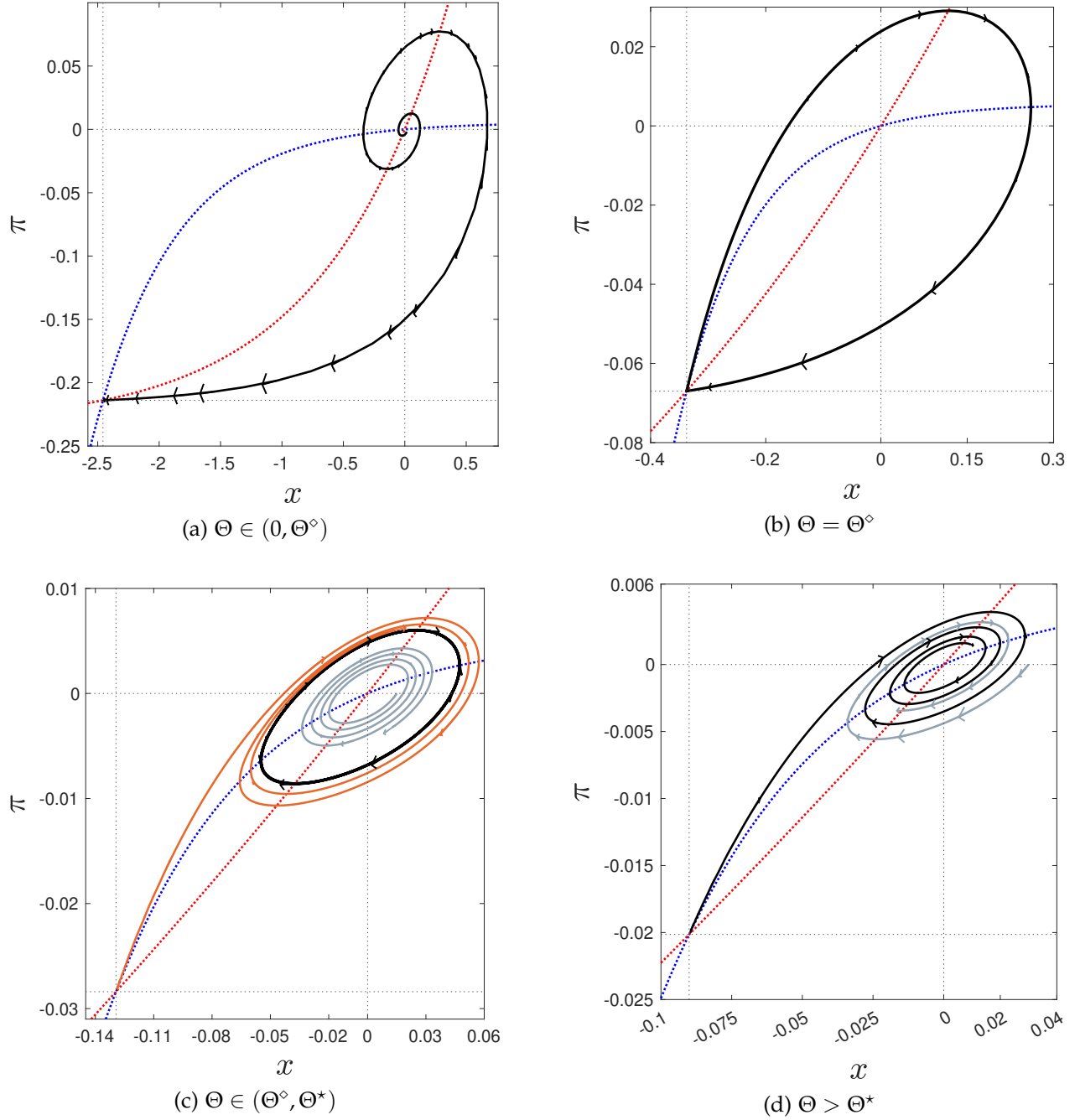


Figure B.1: Global dynamics depending on magnitude of Θ

of the eigenvalues is given by the determinant of $\rho(\phi_r - 1)\sigma\gamma\Theta + \kappa(\phi_\pi - 1)$. A sufficient condition for both the sum and product of roots to be positive is that $\phi_\pi > 1$ and $\phi_r \geq 1$. If this condition is satisfied, then for any $\Theta > 0$, $J(0, 0)$ has two eigenvalues with positive real parts, ensuring local determinacy of the targeted equilibrium.

Second, notice that the trace of $J(x, \pi)$ is given by $\rho + (\phi_r - 1)\sigma\gamma\Theta e^{-\gamma\Theta x}$, which is positive for any $(x, \pi) \in (-\infty, \infty)^2$ as long as $\phi_r \geq 1$. Since the trace of the Jacobian does not change sign in the entire domain, the Bendixson–Dulac theorem (Bendixson, 1901; Dulac, 1937) implies that there are no non-constant periodic solutions lying entirely within $(x, \pi) \in (-\infty, \infty) \times (-\infty, \infty)$. Thus, we have global

determinacy as long as there is a unique constant solution.

We already know that $x = \pi = 0$ is a constant solution. Next, we show that this is the only constant solution. To characterize all constant solutions, we set $\dot{x} = \dot{\pi} = 0$ in (c.2) and (c.3). This yields:

$$0 = (\phi_\pi - 1)\pi + (1 - \phi_r)\sigma \left(e^{-\gamma\Theta x} - 1 \right) \quad \text{and} \quad \pi = \frac{\kappa}{\rho}(e^x - 1)$$

We can use the second expression to substitute out π from the first. Doing so yields a single equation in x , which describes all the steady state values of x :

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}(e^x - 1) + (1 - \phi_r)\sigma \left(e^{-\gamma\Theta x} - 1 \right),$$

and $x = 0$ is a solution to $F(x) = 0$. To see that $F(x)$ has no other zeros, note that the first derivative of $F(x)$ can be written as:

$$F'(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}e^x + (\phi_r - 1)\gamma\sigma\Theta e^{-\gamma\Theta x}$$

A sufficient condition for this expression to be strictly positive for any $x \in (-\infty, \infty)$ is that (i) $\phi_\pi > 1$, and (ii) $\phi_r \geq 1$. Thus, as long as $\phi_\pi > 1$ and $\phi_r \geq 1$, the only zero of $F(x)$ is at $x = 0$, i.e., the targeted steady state is the unique steady state. Consequently, $x = \pi = 0$ is the only bounded solution with the policy rule (c.1). $\square$

C.1 Escape clause

In this section, we provide details of the escape clause policy described in Section IV.A. In particular, we assume that the monetary policy is described by the simple inflation targeting rule $i_t = \bar{r} + \phi_\pi \pi_t$ as long as output is above some threshold level $\tilde{x}$. However, when output is below this threshold $x_t < \tilde{x}$, monetary policy switches to a strict-inflation targeting rule $\pi_t = 0$ from that date on. In what follows, we will assume that $\tilde{x} < 0$, i.e., the threshold level of output lies below output in the targeted equilibrium ($x = 0$). We will also assume that for a given Θ and ϕ_π , $\tilde{x}$ is marginally higher than the output in the untargeted steady state that would exist if monetary policy was described as in our baseline model.

The strict-inflation targeting rule $\pi_t = 0$ can be interpreted as a whatever-it-takes stance which rules out any equilibria along which at any date the economy passes through a point (x_t, π_t) such that $x_t < \tilde{x}$ and $\pi_t \neq 0$. The first thing to notice is that $\pi_t = 0$ is not an instrument rule like the policy rule (6); it is instead a *targeting rule* and can be interpreted as a commitment by the central bank that it will set the path of nominal interest rates to whatever level is required to ensure that $\pi_t = 0$ at all dates from then on. In terms of an instrument rule, this targeting rule is often expressed as the limit of a standard inflation targeting Taylor rule $i_t = \bar{r} + \phi_\pi \pi_t$ with ϕ_π set to ∞ .

To see how such a rule can rule out any equilibria along which at any date the economy passes through a point (x_t, π_t) such that $x_t < \tilde{x}$ and $\pi_t \neq 0$, suppose that at date t , the economy features a level of output with some finite x_t , which satisfies $x_t < \tilde{x}$. Notice that the IS curve (with the inflation

targeting rule plugged in) and the Phillips curve in the context of our baseline model is given by:

$$\dot{x}_t = (\phi_\pi - 1)\pi_t + \sigma(e^{-\gamma\Theta x_t} - 1)$$

With $\phi_\pi \rightarrow \infty$, then for $\dot{x}_t$ to be finite, we need that $\pi_t = 0$. Otherwise, if $\pi_t > 0$, we would have $\dot{x}_t = +\infty$ and if $\pi_t < 0$, we would have $\dot{x}_t = -\infty$. In other words, if $\pi_t \neq 0$ at all dates, x_t will grow or shrink infinitely fast and become unbounded. Thus, the only bounded solution which satisfies the IS and the Phillips curve is $x_t = \pi_t = 0$ at all dates. Thus, the combination of the IS curve and the policy rule imply that $\pi_t = \dot{\pi}_t$ must equal 0 at all dates from then on. However, since $x_t < \tilde{x}$, with π_t set to 0, the date t Phillips curve implies that $\dot{\pi}_t = -\kappa x_t > 0$, which contradicts the fact that $\pi_t = \dot{\pi}_t = 0$. This shows that the strict inflation targeting rule ensures that there exists no combination of (x, π) with $x < \tilde{x}$ which satisfies both the IS curve and the Phillips curve, and thus does not constitute an equilibrium. Thus, as long as $\tilde{x} > \underline{x}$, it follows that the untargeted steady state which features an output-gap of $\underline{x} < 0$ and inflation $\underline{\pi} < 0$ cannot emerge in equilibrium, given monetary policy which switches to a strict-inflation targeting rule $\pi_t = 0$ if output falls below $\tilde{x}$.

However, for $\tilde{x}$ sufficiently close to $\underline{x}$, such a policy stance need not rule out the stable cycle. To see this, suppose that risk is moderately countercyclical $\Theta \in (\Theta^\circ, \Theta^*)$. Then since monetary policy still follows the same rule as in our baseline as long as the economy is near the untargeted steady state, we can still use the same proof strategy as in Appendix B.5 to prove the existence of the stable cycle. The reason behind this is that the proof of the existence of the stable cycle in Appendix B.5 only uses the properties of the ODEs describing the economy *local* to the targeted steady state. Since the escape clause only kicks in once the economy is far away from the targeted steady state, the behavior of the economy local to the targeted steady state is unchanged. Consequently, as long as $\tilde{x}$ is less than the smallest value of output observed on the stable cycle, the escape clause cannot rule out the economy from converging to a stable cycle surrounding the targeted steady state. Figure 3a in the paper presents an example of such a situation in our calibrated economy when the escape clause is triggered if output declines more than 5% below its targeted value. Since the untargeted steady state features a level of output about 6% below its targeted level, but the maximum amplitude of output on the stable cycle is about $\pm 2.5\%$, the policy rules out the untargeted steady state but fails to eliminate the stable cycle.

D Monetary-fiscal interaction

As described in Appendix A.3, the dynamics of x_t, π_t, b_t^g are given by the three dimensional system of non-linear ODEs:

$$\dot{x}_t = (\phi_\pi - 1)\pi_t + \sigma(e^{-\gamma\Theta x_t} - 1) \quad (\text{d.1})$$

$$\dot{\pi}_t = \rho\pi_t - \kappa(e^{x_t} - 1) \quad (\text{d.2})$$

$$\dot{b}_t^g = (\phi_\pi - 1)b^* \pi_t + \bar{r}(1 - \phi_b)b_t^g + (\phi_\pi - 1)\pi_t b_t^g \quad (\text{d.3})$$

Separating the first-order terms which govern dynamics local to the steady state $(x, \pi, b^g) = (0, 0, 0)$ and higher-order terms which dominate dynamics away from $(0, 0, 0)$, we can equivalently rewrite the

3 dimensional system above as:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \\ \dot{b}_t^g \end{bmatrix} = A \begin{bmatrix} x_t \\ \pi_t \\ b_t^g \end{bmatrix} + \begin{bmatrix} \sigma (e^{-\gamma\Theta x_t} - 1 + \gamma\Theta x_t) \\ -\kappa (e^{x_t} - 1 - x_t) \\ (\phi_\pi - 1) \pi_t b_t^g \end{bmatrix}, \quad (\text{d.4})$$

where the matrix A is given by:

$$A = \begin{bmatrix} -\gamma\sigma\Theta & \phi_\pi - 1 & 0 \\ -\kappa & \rho & 0 \\ 0 & (\phi_\pi - 1) b^* & \bar{r}(1 - \phi_b) \end{bmatrix} \quad (\text{d.5})$$

The determinacy properties of the economy with government debt depend on the eigenvalues of the matrix A in (d.5). Since we have one predetermined variable b^g , and two jump variables x, π , we need 2 eigenvalues with positive real parts and one negative eigenvalue for local determinacy. The characteristic polynomial associated with the matrix A can be written as:

$$\mathcal{P}(z) = [\bar{r}(1 - \phi_b) - z] \cdot \text{Det}(A^\dagger - zI) = 0, \quad (\text{d.6})$$

where $A^\dagger$ is the leading principal minor of order 2 of the matrix A , and is given by

$$A^\dagger = \begin{bmatrix} -\gamma\sigma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix} \quad (\text{d.7})$$

Then, we know that one of the eigenvalues of the system is given by $z_1 = \bar{r}(1 - \phi_b)$, while the two remaining eigenvalues are the two roots of the quadratic equation $\text{Det}(A^\dagger - zI) = 0$. Furthermore, we know that

$$\text{Tr}(A^\dagger) = \gamma\sigma(\Theta^* - \Theta) \quad \text{and} \quad \text{Det}(A^\dagger) = \kappa[\phi_\pi - \varphi(\Theta)], \quad (\text{d.8})$$

where

$$\Theta^* = \frac{\rho}{\sigma\gamma} \quad \text{and} \quad \varphi(\Theta) = 1 + \frac{\rho\gamma\sigma\Theta}{\kappa}$$

are the same as in our baseline model. In what follows, we follow the nomenclature of Leeper (1991) and consider two regimes based on the magnitudes of ϕ_π and ϕ_b : (i) Active-Monetary, Passive-Fiscal in which $\phi_\pi > 1$ and $\phi_b > 1$, and (ii) Passive-Monetary, Active-Fiscal in which $\phi_\pi \in [0, 1]$ and $\phi_b \in [0, 1]$.³

D.1 Active-Monetary, Passive-Fiscal (AMPF)

It is worth noting that our baseline model is just a special case of this AMPF regime with $\phi_b \rightarrow \infty$ and $\bar{b} = 0$. As such, the determinacy properties of the AMPF regime with $\phi_b < \infty$ and $\bar{b} \neq 0$ are identical to that of our baseline economy without government debt. We show this next.

³It is easy to see that with $\phi_b = 1$, the eigenvalue $z_1 = 0$ and thus, we cannot have local determinacy in the knife-edge case of $\phi_b = 1$. Thus, we define the regions to exclude the case with $\phi_b = 1$.

D.1.1 Local Determinacy

Passive fiscal policy ($\phi_b > 1$) implies that the eigenvalue $z_1 = \bar{r}(1 - \phi_b) < 0$. So local determinacy requires that the other two eigenvalues have positive real parts. For this, we need the trace and determinant of the matrix A^+ be positive. We divide our analysis into three cases depending on how countercyclical risk is and how aggressively monetary policy responds to inflation.

Case I: Mildly or moderately countercyclical risk $0 < \Theta < \Theta^*$ and $1 < \phi_\pi \leq \varphi(\Theta)$ First, consider the case in which risk is not highly countercyclical $\Theta < \Theta^*$, and monetary policy is active but not too aggressive $1 < \phi_\pi < \varphi(\Theta)$. Then, from (d.8), we know that even though the trace of A^+ is positive, the determinant of A^+ is negative, implying that we have 2 negative and one positive root. Thus, the targeted steady state has a 2-dimensional stable manifold around it, and hence, is *locally indeterminate* in this configuration under the AMPF regime.

Case II: Mildly or moderately countercyclical risk $0 < \Theta < \Theta^*$, and $\phi_\pi > \varphi(\Theta)$ Next, consider the case in which risk is mildly or moderately countercyclical $0 < \Theta < \Theta^*$ and that monetary policy is sufficiently active $\phi_\pi > \varphi(\Theta)$. With $\phi_\pi > \varphi(\Theta)$, (d.8) implies that both the determinant and trace of A^+ are positive, implying that there are two positive and one negative root. Thus, the AMPF regime delivers local determinacy if risk is not highly countercyclical and monetary policy responds sufficiently aggressively to changes in inflation.

Case III: Highly countercyclical risk $\Theta > \Theta^*$ Finally, when risk is highly countercyclical ($\Theta > \Theta^*$), equation (d.8) shows that the trace of A^+ is negative. Consequently, as in our baseline model, local determinacy is not possible under the AMPF regime if risk is highly countercyclical $\Theta > \Theta^*$.

Thus, as in our baseline model, in the AMPF regime, a large enough ϕ_π can deliver local determinacy as long as risk is not highly countercyclical. Next, we show that as in our baseline, no matter how large ϕ_π is, the AMPF regime cannot deliver global determinacy.

D.1.2 Global indeterminacy

Since the AMPF regime does not deliver local determinacy if risk is highly countercyclical ($\Theta > \Theta^*$) (Case III above), it follows that the equilibrium also features global indeterminacy. Similarly, if $\Theta < \Theta^*$, but $1 < \phi_\pi < \varphi(\Theta)$ (Case I), the AMPF regime also does not deliver global determinacy. Thus, the remaining case to consider is when risk is not highly countercyclical, $\Theta < \Theta^*$, and monetary policy is sufficiently aggressive, $\phi_\pi > \varphi(\Theta)$ (Case II). In this case the target equilibrium is locally determinate. However, as in our baseline model, this does not translate into global determinacy. We start with the case of moderately countercyclical risk.

Stable cycle (Moderately countercyclical risk) Similar to Appendix B.5, in the AMPF regime, the economy undergoes a Hopf bifurcation at $\Theta = \Theta^* = \rho/\sigma\gamma$, i.e. if $\Theta = \Theta^*$, the eigenvalues of the matrix A described in (d.5) has two purely imaginary roots. Then, the three dimensional generalization of Theorem 1 guarantees that there exists $\Theta^\diamond < \Theta^*$ such that for any $\Theta \in (\Theta^\diamond, \Theta^*)$, any trajectory

(except on the one dimensional stable manifold) which starts in the neighborhood of $(x, \pi, b^g) = (0, 0, 0)$ converge to a stable cycle along which the economy oscillates around the targeted steady state, remaining bounded forever. We present this Theorem next for convenience.

Theorem 3 (Hopf Bifurcation in n-dimensions). *Consider a $n \geq 2$ dimensional system*

$$\dot{\mathbf{x}} = F_\mu(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n, \mu \in \mathbb{R}$$

with smooth F , which for all sufficiently small $|\mu|$ has the equilibrium $\mathbf{x} = \mathbf{0}$, and the Jacobian $D_{\mathbf{x}}F_\mu(\mathbf{0})$ has eigenvalues given by $\Omega(\mu) + \omega(\mu)\mathbf{i}$ where $\mathbf{i} = \sqrt{-1}$. Then, if the following conditions are satisfied:

1. *At $\mu = 0$, $D_{\mathbf{x}}F_{\mu=0}(\mathbf{0})$ has two purely imaginary eigenvalues $\pm\bar{\omega}_0$ and no other eigenvalue of $D_{\mathbf{x}}F_{\mu=0}(\mathbf{0})$ is an integral multiple of $\bar{\omega}_0\mathbf{i}$.*
2. *Without loss of generality, for ease of exposition, assume that the first two out of the n eigenvalues are the purely imaginary ones at $\mu = 0$, and their continuation is given by:*

$$\lambda_{1,2}(\mu) = \Omega(\mu) \pm \omega(\mu)\mathbf{i}$$

Then, these eigenvalues cross the imaginary axis with non-zero speed: $d\Omega(0)/d\mu \neq 0$.

Under the above conditions there exist a family of real periodic solutions $\mathbf{x} = \mathbf{x}(t, \epsilon)$, $\mu(\epsilon)$ which has the properties $\mu(0) = 0$, $\mathbf{x}(t, 0) = \mathbf{0}$, but $\mathbf{x}(t, \epsilon) \neq \mathbf{0}$ for sufficiently small ϵ . The same holds for the period $T(\epsilon)$ and $T(0) = 2\pi/|\omega(0)|$.

To see that Theorem 3 applies, imposing $\Theta = \Theta^* = \rho/\sigma\gamma$ in (d.5), we can write A as:

$$A(\Theta^*) = \begin{bmatrix} -\rho & \phi_\pi - 1 & 0 \\ -\kappa & \rho & 0 \\ 0 & (\phi_\pi - 1)\bar{b} & \bar{r}(1 - \phi_b) \end{bmatrix}$$

We know from the characteristic polynomial of A (equation (d.6)) that one of the eigenvalues of $A(\Theta^*)$ is given by $z_1 = \bar{r}(1 - \phi_b)$, which is negative in the AMPF regime since $\phi_b > 1$. Then the sum of the other two roots is the trace of the principal minor of order 2 associated with $A(\Theta^*)$, which is given by $-\rho + \rho = 0$. The zero trace means that the eigenvalues z_2 and z_3 cancel each other out. Furthermore, the determinant of the principal minor of order 2 associated with $A(\Theta^*)$ is given by

$$\kappa [\phi_\pi - \varphi(\Theta^*)] \quad \text{where} \quad \varphi(\Theta^*) = 1 + \frac{\rho^2}{\kappa},$$

which is positive since we are considering the case with $\phi_\pi > \varphi(\Theta^*)$. Together these properties of the trace and determinant imply that the other two eigenvalues are purely imaginary, and thus condition 1 is satisfied. Furthermore, it is clear from inspection that the eigenvalues change smoothly in Θ , and thus condition 2 of the Theorem is satisfied. This shows that, as in our baseline model, the economy with non-zero government debt undergoes a Hopf bifurcation at $\Theta = \Theta^*$ in the AMPF regime.

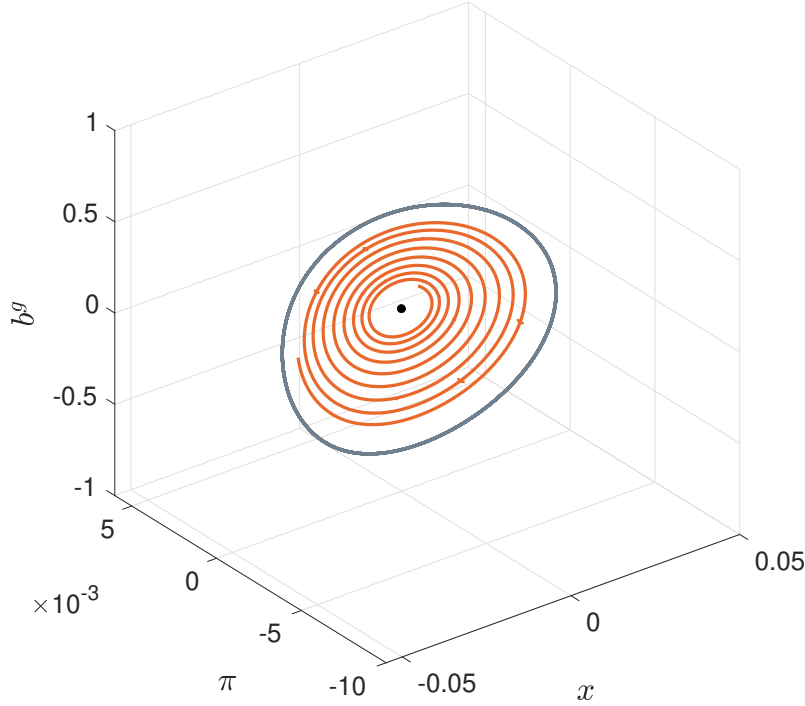


Figure D.1: Stable cycle with moderately countercyclical risk in AMPF

Finally, to show that the cycle is stable in the economy with government debt, as in our baseline model, we need to show that the first Lyapunov coefficient is negative. While it is analytically cumbersome to compute the first Lyapunov coefficient and show that it is negative, Figure D.1 shows that under our calibrated value of $\Theta = 28.1$ (which corresponds to the moderately countercyclical risk case, i.e., $\Theta^\diamond < \Theta < \Theta^*$), any trajectory originating in the neighborhood of $(x, \pi, b^g) = (0, 0, 0)$ (depicted by the orange trajectory) eventually converges to a stable limit-cycle (depicted by the gray trajectory) around the targeted steady state when risk is moderately countercyclical $\Theta \in (\Theta^\diamond, \Theta^*)$. Thus, if $\phi_\pi > \phi(\Theta)$, and risk is moderately countercyclical $\Theta^\diamond < \Theta < \Theta^*$, the AMPF regime delivers local but not global determinacy, since along with the targeted equilibrium $(x, \pi, b^g) = (0, 0, 0)$, the stable cycle and all the trajectories converging to the stable cycle also constitute bounded sequences which satisfy all equilibrium conditions.

Saddle Connection (Mildly countercyclical risk) Next, we consider the case in which risk is mildly countercyclical $\Theta \in (0, \Theta^\diamond)$. First, notice that for any $\Theta > 0$, as in our baseline model, there still exists a untargeted steady state alongside the targeted steady state $(x, \pi, b^g) = (0, 0, 0)$. This can be seen by setting $\dot{x} = \dot{\pi} = \dot{b}^g = 0$ in (a.22)-(a.24), which yields:

$$\begin{aligned} 0 &= (\phi_\pi - 1)\pi + \sigma(e^{-\gamma\Theta x} - 1) \\ \pi &= \frac{\kappa}{\rho}(e^x - 1) \\ b^g &= \frac{(\phi_\pi - 1)\bar{b}}{\bar{r}(\phi_b - 1) - (\phi_\pi - 1)\pi}\pi \end{aligned}$$

As in Appendix B.3, combining the first two of these equations allows us to characterize the output levels in all steady states as a solution to one equation in one unknown:

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho} (e^x - 1) + \sigma (e^{-\gamma\Theta x} - 1)$$

Since AMPF implies that $\phi_\pi > 1$, the same argument as in Appendix B.3 applies, and hence $F(x)$ has two zeros as long as $\Theta > 0$: one targeted with $x = \pi = b^g = 0$ and one untargeted steady state. Since we are looking at the case with $\phi_\pi > \varphi(\Theta)$, as Appendix B.3 shows in the context of our baseline model, the untargeted steady state has lower output and inflation than the targeted steady state. However, unlike in our baseline model (which featured only 2 jump variables), the existence of a second steady state by itself does not automatically imply indeterminacy because now we have a predetermined variable b_t^g , and to prove global indeterminacy, we need to show that for at least some initial value of b_0^g , there exist at least two bounded sequences $\{x_t, \pi_t, b_t^g\}_{t=0}^\infty$ and $\{\tilde{x}_t, \tilde{\pi}_t, \tilde{b}_t^g\}_{t=0}^\infty$ where $\tilde{b}_0^g = b_0^g$.

To show this, we can invoke the 3-dimensional version of the Theorem 2 (which corresponds to Corollary 7.1 in Kopell and Howard (1975)) to prove the existence of a saddle connection along which the economy can transition from near the targeted steady state to the untargeted steady state. We present this Theorem next:

Theorem 4 (Corollary 7.1 in Kopell and Howard (1975)). *Let $\dot{\mathbf{x}} = F_{\mu,\nu}(\mathbf{x})$ be a two parameter family of ODEs on $\mathbb{R}^n$, $n \geq 2$, with F smooth in all of its arguments, such that $F_{\mu,\nu}(0) = \mathbf{0}$. Also assume:*

1. *$dF_{0,0}(0)$ has rank $n - 1$ and a zero eigenvalue with multiplicity 2 with $\mathbf{v}$ being the right-eigenvector associated with the zero eigenvalue.*
2. *The mapping $(\mu, \nu) \rightarrow (\det(dF_{\mu,\nu}(0), \sigma_{n-1}dF_{\mu,\nu}(0)))$ has a nonzero Jacobian at $(\mu, \nu) = (0, 0)$, where $\sigma_{n-1}dF_{\mu,\nu}(0)$ is the coefficient on the linear term in the characteristic polynomial of $dF_{\mu,\nu}(0)$.*
3. *Let $Q(\mathbf{x}, \mathbf{x})$ be the $n \times 1$ vector containing the terms quadratic in $\mathbf{x}$ and independent of (μ, ν) in a Taylor series expansion of $F_{\mu,\nu}(\mathbf{x})$ around 0. Then $[dF_{0,0}(0), Q(\mathbf{v}, \mathbf{v})]$ has rank $n - 1$.*

Then there is a curve $f(\mu, \nu) = 0$ such that if $f(\mu_0, \nu_0) = 0$, then $\dot{\mathbf{x}} = F_{\mu_0, \nu_0}(\mathbf{x})$ has a homoclinic orbit. This one-parameter family of homoclinic orbits (in (X, μ, ν) space) is on the boundary of a two-parameter family of periodic solutions. For all $|\mu|, |\nu|$ sufficiently small, if $\dot{\mathbf{x}} = F_{\mu,\nu}(\mathbf{x})$ has neither a homoclinic orbit nor a periodic solution, there is a unique trajectory joining the critical points.

As in our baseline model, we use Θ and ϕ_π as the bifurcation parameters. Condition 1 is clearly satisfied since $dF(0)$ is simply the A matrix in d.5. To see that point 1 is satisfied, imposing $\Theta = \Theta^*$ and $\phi_\pi = \varphi(\Theta^*)$ we can rewrite A in (d.5) as:

$$\begin{bmatrix} -\rho & \frac{\rho^2}{\kappa} & 0 \\ -\kappa & \rho & 0 \\ 0 & \frac{\rho^2}{\kappa} \bar{b} & \bar{r}(1 - \phi_b) \end{bmatrix},$$

which clearly has rank 2, since the first 2 rows are linearly dependent. Also, the principal minor of order two has trace and determinant zero, implying that the roots of the system are $\bar{r}(1 - \phi_b), 0, 0$. Furthermore, one can compute the right-eigenvector associated with 0 to be:

$$\mathbf{v} = \begin{bmatrix} 1 \\ \frac{\kappa}{\rho} \\ \frac{\rho \bar{b}}{\bar{r}(1 - \phi_b)} \end{bmatrix} \quad (\text{d.9})$$

Next, the regularity condition 2 is satisfied since

$$\begin{vmatrix} -\bar{r}(1 - \phi_b) \rho \gamma \sigma & [\bar{r}(1 - \phi_b) + \rho] \gamma \sigma \\ \kappa \bar{r}(1 - \phi_b) & -\kappa \end{vmatrix} = -\kappa \gamma \sigma \bar{r}^2 (1 - \phi_b)^2 \neq 0$$

Finally, taking a second order Taylor approximation of the 3 ODEs (a.22)-(a.24) and imposing $\Theta = \Theta^*$ and $\phi_\pi = \varphi(\Theta^*)$, we have

$$Q(\mathbf{x}, \mathbf{x}) = \begin{bmatrix} \frac{\rho^2}{2\sigma} x^2 \\ -\kappa x^2 \\ \frac{\rho^2}{\kappa} \pi b^g \end{bmatrix}$$

Evaluating at $\mathbf{x} = \mathbf{v}$ (which is defined in (d.9)), we get

$$Q(\mathbf{v}, \mathbf{v}) = \begin{bmatrix} \frac{\rho^2}{2\sigma} \\ -\kappa \\ -\frac{\rho^2 \bar{b}}{\bar{r}(1 - \phi_b)} \end{bmatrix}$$

and we can finally verify that

$$\text{rank} \begin{bmatrix} -\rho & \frac{\rho^2}{\kappa} & 0 & \frac{\rho^2}{2\sigma} \\ -\kappa & \rho & 0 & -\kappa \\ 0 & \frac{\rho^2}{\kappa} \bar{b} & \bar{r}(1 - \phi_b) & -\frac{\rho^2 \bar{b}}{\bar{r}(1 - \phi_b)} \end{bmatrix} = 3,$$

which implies that all three conditions for Theorem 4 are satisfied. Since we know that a stable cycle only exists if risk is moderately countercyclical ($\Theta^\diamond < \Theta < \Theta^*$), Theorem 4 implies that when risk is mildly countercyclical ($0 < \Theta < \Theta^\diamond$), then there must be a saddle connection along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state. As in the baseline economy, any trajectory originating on this saddle connection remains bounded and hence there are multiple bounded sequences which satisfy equilibrium. Thus, even when risk is mildly countercyclical, there is global indeterminacy in the AMPF regime.

D.2 Passive-Monetary, Active-Fiscal (PMAF)

Active fiscal policy ($0 \leq \phi_b < 1$) implies that the eigenvalue $z_1 = \bar{r}(1 - \phi_b) > 0$. Then local determinacy requires that one of the eigenvalues of A^+ be positive and the other negative.

D.2.1 Local Determinacy

Equation (d.8) shows that since $\phi_\pi < 1 < \varphi(\Theta)$, the determinant of A^+ is negative for any $\Theta > 0$, implying that the product of the two remaining roots is negative, which means that one of the roots of A^+ is negative and one is positive. Thus, the PMAF regime delivers local determinacy of the targeted equilibrium $(x, \pi, b^g) = (0, 0, 0)$ for any Θ , i.e., regardless of whether risk is mildly, moderately or highly countercyclical.

D.2.2 Global Determinacy

Next, we show that the PMAF regime eliminates the ways in which global determinacy manifests in the AMPF regime. We start by explicitly showing that the PMAF regime eliminates the untargeted steady state as well as the stable cycle. Following this demonstration, we then prove that the PMAF regime delivers global determinacy: for any b_0^g , there is a unique x_0, π_0 such that the trajectory starting at (x_0, π_0, b_0^g) is the only trajectory which remains bounded. Moreover, this trajectory converges to the targeted steady state $(0, 0, 0)$ asymptotically.

Eliminating the stable cycle At $\Theta = \Theta^* = \rho/\sigma\gamma$, the trace of A^+ equals zero (see equation (d.7)), so the two eigenvalues $z_2 = -z_3$. Additionally, since the determinant of A^+ is negative, it means that all three eigenvalues are real, and that the absolute value of z_2 and z_3 is identical, but the signs are opposite. Thus, unlike in the AMPF regime, the system does not undergo a Hopf bifurcation since the roots are not complex. Thus, the PMAF regime eliminates the stable cycle, which was one of the manifestations of the global indeterminacy in the baseline model and the AMPF regime.

Eliminating the second steady state and saddle connection The PMAF steady state also prevents the untargeted steady state from existing; only the targeted steady state survives. To see this, we can set $\dot{x} = \dot{\pi} = \dot{b}^g = 0$ in (a.22)-(a.24), to get:

$$\begin{aligned} 0 &= (\phi_\pi - 1)\pi + \sigma(e^{-\gamma\Theta x} - 1) \\ \pi &= \frac{\kappa}{\rho}(e^x - 1) \\ b^g &= \frac{(\phi_\pi - 1)\bar{b}}{\bar{r}(\phi_b - 1) - (\phi_\pi - 1)\pi}\pi \end{aligned}$$

As in Appendix B.3, combining the first two of these equations allows us to characterize the output levels in all steady states as a solution to one equation in one unknown:

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}(e^x - 1) + \sigma(e^{-\gamma\Theta x} - 1)$$

Taking the first derivative of this function:

$$F'(x) = -\frac{\kappa(1 - \phi_\pi)}{\rho}e^x - \gamma\sigma\Theta e^{-\gamma\Theta x} < 0$$

Since $\phi_\pi \leq 1$ in the PMAF regime, $F'(x) < 0$ for all $x \in (-\infty, \infty)$. This means that the only zero of $F(x)$ is at $x = 0$, i.e., the targeted steady state. Hence, the PMAF regime also eliminates the untargeted steady state, which was a manifestation of the global indeterminacy. Furthermore, since the untargeted steady state does not exist in the PMAF regime, neither can the saddle connection along which the economy can travel from the targeted to the untargeted steady state, as in the AMPF regime. Thus, the PMAF regime eliminates all three manifestations of global indeterminacy which appear in our baseline model and AMPF regime.

Next, we can now prove that the PMAF regime delivers global determinacy. Notice that in the system of 3 ODEs (d.4), the subsystem which describes the dynamics of x_t, π_t does not explicitly depend on b^g . This is easily seen by computing the Jacobian of (d.4) at any (x, π, b^g) and noticing that the entries in the first two rows of the 3rd column are zero:

$$J(x, \pi, b^g) = \begin{bmatrix} -\sigma\gamma\Theta e^{-\gamma\Theta x} & \phi_\pi - 1 & 0 \\ -\kappa e^x & \rho & 0 \\ 0 & (\phi_\pi - 1)(b^* + b^g) & \bar{r}(1 - \phi_b) + (\phi_\pi - 1)\pi \end{bmatrix}$$

Thus, we can study the dynamics of (x, π) separately by focusing on the principal co-minor of the Jacobian above, which we will refer to as $J^\diamond(x, \pi)$:

$$J^\diamond(x, \pi) = \begin{bmatrix} -\sigma\gamma\Theta e^{-\gamma\Theta x} & \phi_\pi - 1 \\ -\kappa e^x & \rho \end{bmatrix}$$

We have already established that the determinant of the Jacobian evaluated at the unique steady state $x, \pi = 0, 0$ is negative in the AMPF regime, which implies that $A^\diamond = J^\diamond(0, 0)$ has one negative and one positive eigenvalue. It follows that local to $(x, \pi) = (0, 0)$, there exists a one dimensional stable manifold which can be written as $x = \Psi(\pi)$ such that only trajectories for any π close to 0, only trajectories beginning at $((\Psi(\pi), \pi))$ converge to $(0, 0)$, while others diverge locally. However, we can extend this argument for the existence of a 1-dimensional stable manifold $x = \Psi(\pi)$ because in the PMAF regime, the off-diagonal elements of $J^\diamond(x, \pi)$ are always strictly negative for any $(x, \pi) \in \mathbb{R}^2$, which implies that the subsystem which describes the dynamics of (x, π) is a *competitive ODE system*. The existence of the 1-dimensional stable manifold $x = \Psi(\pi)$, along which $\{x_t, \pi_t\} \rightarrow 0, 0$, follows from Hirsch's theorem for competitive ODE systems (Hirsch, 1982).⁴ Furthermore, Hirsch (1982) also shows that $\frac{d\Psi(\pi)}{d\pi} < 0$, which implies that the stable manifold is strictly monotone and that all trajectories which start off this stable manifold become unbounded.

Next, notice that the dynamics of b^g , which are described by (d.3) only depend on b_t^g and π_t and can be written as:

$$\dot{b}_t^g = (\phi_\pi - 1) b^* \pi_t + \bar{r}(1 - \phi_b) b_t^g + (\phi_\pi - 1) \pi_t b_t^g$$

Since the dynamics of π_t are determined separately from b_t^g , this ODE is simply a linear non homoge-

⁴We thank Greg Kaplan for suggesting the use of Hirsch's theorem.

neous ordinary differential equation in b^g , and for a given initial condition b_0^g admits the solution:

$$b_t^g = b_0^g e^{\int_0^t \mu_s ds} + \int_0^t v_s e^{\int_s^t \mu_\tau d\tau} ds$$

where $\mu_t = \bar{r}(1 - \phi_b) + (\phi_\pi - 1)\pi_t$ and $v_t = (\phi_\pi - 1)b^* \pi_t$. Next, as we take the limit $t \rightarrow \infty$, we know from the x, π subsystem that along the stable manifold $(\Psi(\pi), \pi)$, we have $\lim_{t \rightarrow \infty} \pi_t \rightarrow 0$. Then to ensure that $\lim_{t \rightarrow \infty} b_t^g \rightarrow 0$, we have:

$$b_0^g = -(\phi_\pi - 1)b^* \int_0^\infty \pi_s e^{-\int_0^s [\bar{r}(1 - \phi_b) + (\phi_\pi - 1)\pi_\tau] d\tau} ds$$

This expression above can be interpreted as a mapping from π_0 to b_0^g . Also, since the x, π subsystem is competitive, it follows from Hirsch's theorem that a higher π_0 increases π_t for all $t > 0$ and hence the condition above implies a strictly positive relationship between b_0^g and the π_0 which is needed to ensure that b_0^g converges to 0 asymptotically. Thus, for a given b_0^g there exists a unique π_0 for which b_0^g does not grow unbounded and instead converges to 0 asymptotically. We can describe this relation as $\pi_0 = \Lambda(b_0^g)$. Together, the characterization above implies that for any given b_0^g , only the trajectory originating at $(\Psi(\Lambda(b_0^g)), \Lambda(b_0^g), b_0^g)$ converges to $(0, 0, 0)$ asymptotically, while all other trajectories grow unbounded. Thus, in the PMAF regime, we have global determinacy. $\square$

E Extensions of the baseline model

E.1 Monetary policy rule with output-gap stabilization

We first consider the case in which monetary policy also responds to output-gap fluctuations. The augmented rule can be written as:

$$i_t = \bar{r} + \phi_\pi \pi_t + \phi_x x_t, \quad (\text{e.1})$$

which nests the inflation targeting rule (6) if we set $\phi_x = 0$. Given the interest rate rule (e.1), dynamics of the output-gap x_t , and inflation π_t are described by:

$$\dot{x}_t = (\phi_\pi - 1)\pi_t + \phi_x x_t - (r^*(x_t) - \bar{r}) \quad (\text{e.2a})$$

$$\dot{\pi}_t = \rho \pi_t - \kappa (e^{x_t} - 1) \quad (\text{e.2b})$$

How does ϕ_x affect local determinacy? As is well known in the RANK literature, allowing the nominal rate to respond to output-gap fluctuations in addition to inflation reduces the burden on ϕ_π to ensure that the targeted equilibrium is locally determinate (see, e.g. Bullard and Mitra (2002)). The same is true in the context of our HANK economy with countercyclical risk. Recall that in the inflation targeting rule (6) (with $\phi_x = 0$), a higher ϕ_π could only guarantee local determinacy only if risk was mildly or moderately countercyclical ($0 < \Theta < \Theta^*$). However, if risk is highly countercyclical ($\Theta \geq \Theta^*$), no finite ϕ_π can deliver local determinacy. However, as Proposition E.1 shows, a monetary policy rule which responds to output-gap fluctuations can *always* guarantee local determinacy, no

matter how countercyclical risk is.

Proposition E.1 (Output-gap stabilization). *For any $\Theta \geq 0$, the combination of a large enough ϕ_x and ϕ_π is sufficient for local determinacy of the targeted steady state. In particular, local determinacy requires:*

$$\phi_\pi > \varphi(\Theta) = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa} \quad \text{and} \quad \phi_x > \sigma\gamma(\Theta - \Theta^*) \quad (\text{e.3})$$

Proof. With $\phi_x \neq 0$, for (x, π) close to the targeted steady state $(0, 0)$, the dynamics of the system (16a)-(16b) are governed by the following system:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = A \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \mathcal{O}(x^2) \quad \text{for } (x, \pi) \rightarrow (0, 0),$$

where A is given by

$$A = \begin{bmatrix} \phi_x - \sigma\gamma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix},$$

Since both x and π are jump-variables, local determinacy requires that both eigenvalues of A have a positive real part. As is well known, the sum of the two eigenvalues of A , denoted by z_1 and z_2 , is given by the trace of A , while their product is given by the determinant of A :

$$\begin{aligned} z_1 + z_2 &= \rho + \phi_x - \sigma\gamma\Theta, \\ z_1 \times z_2 &= \kappa(\phi_\pi - 1) + \rho\phi_x - \rho\sigma\gamma\Theta \end{aligned}$$

Since this is a two dimensional system, either both z_1 and z_2 are real, or they are complex conjugates. Thus, for z_1 and z_2 to both have positive real parts, it is sufficient that both the sum and product of z_1, z_2 be positive. In other words, a sufficient condition for local determinacy is

$$\phi_\pi + \frac{\rho}{\kappa}\phi_x > \varphi(\Theta) = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa} \quad \text{and} \quad \phi_x > \sigma\gamma(\Theta - \Theta^*) \quad (\text{e.4})$$

This condition is satisfied if (ϕ_π, ϕ_x) jointly satisfy the following condition

$$\phi_\pi > \varphi(\Theta) \quad \text{and} \quad \phi_x > \sigma\gamma(\Theta - \Theta^*),$$

which is the same condition as in Proposition E.1. For mildly or moderately countercyclical risk $0 < \Theta < \Theta^*$, the above expression shows that setting $\phi_x = 0$ and $\phi_\pi > \varphi(\Theta)$ is sufficient for local determinacy. This is the same condition as in Proposition 2. However, when risk is highly countercyclical $\Theta \geq \Theta^*$, setting $\phi_x = 0$ is no longer enough for local determinacy. Local determinacy now requires a large enough $\phi_x > \sigma\gamma(\Theta - \Theta^*)$ alongside $\phi_\pi > \varphi(\Theta)$. $\square$

Consistent with Proposition 2, (e.3) shows that if risk is for mild or moderately countercyclical, $\Theta \in (0, \Theta^*]$, then $\phi_x = 0$ is sufficient for local determinacy as long as ϕ_π is large enough: $\phi_\pi > \varphi(\Theta)$. However, if risk is highly countercyclical ($\Theta > \Theta^*$), setting ϕ_x slightly above 0 is not sufficient, and local

determinacy requires $\phi_x > \sigma\gamma(\Theta - \Theta^*) > 0$, alongside a large enough ϕ_π . Overall, (e.3) shows that the more countercyclical risk, the higher ϕ_π and ϕ_x need to be to ensure local determinacy; raising only one of the two is not sufficient. To understand why a high enough ϕ_x can guarantee local determinacy, it is again useful to concentrate on the IS curve. With $\phi_x \neq 0$, the IS curve (e.2a) can be written as:

$$\dot{x}_t = \underbrace{(\phi_\pi - 1)\pi_t + \phi_x x_t}_{\text{destabilizing}} - \sigma\gamma\Theta x_t + \mathcal{O}(x^2) \quad \text{for} \quad (x, \pi) \rightarrow (0, 0)$$

Relative to the case with $\phi_x = 0$, the expression above shows that in addition to the $(\phi_\pi - 1)\pi$ term which supplies the destabilizing forces, now an additional term, $\phi_x x$, also generates destabilizing dynamics. When x is away from its steady state value, say $x > 0$, this term makes $\dot{x}$ more positive, thus pushing x further away from 0. The stabilizing force, provided by the term, $-\sigma\gamma\Theta x$ is the same as in the case with $\phi_x = 0$. Thus, as (e.3) shows, a large enough ϕ_π and ϕ_x make it easier for the destabilizing forces to overwhelm the stabilizing force, generating local determinacy.

While the combination of a high enough ϕ_π and ϕ_x helps resolve the problem of *local* determinacy, it does little to resolve the problem of global indeterminacy. All the forces which contributed to the existence of multiple equilibria in the baseline still continue to operate regardless of how large ϕ_x is.

First, with $\phi_x > 0$, no matter how large, the untargeted steady state $(\underline{x}, \underline{\pi})$ continues to exist, and features lower output than the targeted steady state as well as below target inflation. In fact, the larger the $\phi_x > 0$, the lower are output and inflation in the untargeted steady state. To see this, note that the $\dot{x} = 0$ and $\dot{\pi}_t = 0$ nullclines with $\phi_x \neq 0$, imply that in any steady state, (x, π) must satisfy:

$$\begin{aligned} 0 &= (\phi_\pi - 1)\pi + \phi_x x + \sigma(e^{-\gamma\Theta x} - 1) \\ 0 &= \rho\pi - \kappa(e^x - 1) \end{aligned}$$

Clearly, $(0, 0)$ still satisfies both equations. To see that the untargeted steady state still exists, combine the two equations to eliminate π , to get an expression exclusively in terms of x :

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}(e^x - 1) + \phi_x x + \sigma(e^{-\gamma\Theta x} - 1),$$

and any x which solves $F(x) = 0$ constitutes a steady state. Again, clearly $x = 0$ solves this equation. The derivative of $F(x)$ is given by:

$$F'(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}e^x + \phi_x - \sigma\gamma\Theta e^{-\gamma\Theta x},$$

which, evaluated at $x = 0$ yields

$$F'(0) = \frac{\kappa}{\rho}\left(\phi_\pi + \frac{\rho}{\kappa}\phi_x - \varphi(\Theta)\right) \quad \text{where} \quad \varphi(\Theta) = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa},$$

If $\phi_\pi + \frac{\rho}{\kappa}\phi_x = \varphi(\Theta)$, then $F'(0) = 0$ and $F(x)$ is tangent to the x -axis at $x = 0$, implying that it is the only zero of $F(x)$ since $F(x)$ is declining in the region $x < 0$ and increasing in the region $x > 0$. This is the knife edge case in which there is a unique steady state. If instead, we impose the condition for local

determinacy of the targeted equilibrium, $\phi_\pi + \frac{\rho}{\kappa}\phi_x > \varphi(\Theta)$, then $F'(0) > 0$. Since $\lim_{x \rightarrow -\infty} F(x) \rightarrow \infty$, there must be at least one intersection with $\underline{x} < 0$ and $F'(\underline{x}) < 0$. Since $F(x)$ is strictly convex, this intersection is unique. Further, note that $dF(x)/d\phi_\pi < 0$ for $x < 0$ by inspection. Thus, by the implicit function theorem, we have $d\underline{x}/d\phi_\pi < 0$.

The existence of the second steady state again implies that the equilibrium is *globally* indeterminate, because the two steady states imply that there are at least two bounded trajectories which satisfy all equilibrium conditions: $(x_t, \pi_t) = (0, 0)$ and $(x_t, \pi_t) = (\underline{x}, \underline{\pi})$. However, separate from the second steady state, as in the baseline, there are other equilibria as well, and as with $\phi_x = 0$, the untargeted steady state is not the only other bounded trajectory. Proposition E.2 provides an exhaustive characterization of global dynamics for a given cyclicality Θ as a function of ϕ_x .

Proposition E.2. *For a given $\Theta > 0$, assume that $\phi_\pi > \varphi(\Theta)$. Then, defining $\phi_x^* = \sigma\gamma(\Theta - \Theta^*)$, the global dynamics of the economy with monetary policy rule (e.1) can be split into 3 regions:*

1. $\phi_x < \phi_x^*$ (**small ϕ_x**): If $\phi_x < \phi_x^*$, trajectories which start in the neighborhood of $(0, 0)$ converge to the targeted steady state, and thus, the targeted equilibrium is both locally and globally indeterminate. In addition to the multiple trajectories originating near the targeted steady state which remain bounded, there also exists a **saddle-connection** along which the economy can transition from the neighborhood of the untargeted steady state to the targeted steady state. All trajectories which originate at any point on this saddle connection also remain bounded.
2. $\phi_x \in [\phi_x^*, \bar{\phi}_x]$ (**medium ϕ_x**): $\exists \bar{\phi}_x > \phi_x^*$, such that for any $\phi_x \in [\phi_x^*, \bar{\phi}_x]$, the targeted equilibrium is locally determinate, but globally indeterminate. Trajectories which start in the neighborhood of the targeted steady state initially diverge away from it, but eventually remain bounded and converge to a **stable limit-cycle** surrounding the targeted steady state. The amplitude and periodicity of the cycles is increasing in ϕ_x in this range. At the upper limit of this interval, $\phi_x = \bar{\phi}_x$, the limit-cycles are absorbed into a **homoclinic orbit**. At the other end of this range, $\phi_x = \phi_x^*$, the limit-cycles collapse onto the targeted steady state. The equilibrium is still globally indeterminate since the higher-order terms ensure that any trajectory starting in the neighborhood of $(0, 0)$ converge back to $(0, 0)$.
3. $\phi_x > \bar{\phi}_x$ (**large ϕ_x**): For $\phi_x > \bar{\phi}_x$, the targeted equilibrium is locally determinate but still globally indeterminate. There are no stable limit-cycles in this range of ϕ_x , but the equilibrium is still globally indeterminate, owing to the existence of a saddle-connection, along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state. Any trajectory which originates on this saddle connection also remains bounded.

No matter how large ϕ_π and ϕ_x are, the equilibrium is still globally indeterminate if risk is even mildly countercyclical.

Proof. Given the interest rate rule $i_t = \bar{r} + \phi_\pi \pi_t + \phi_x x_t$, the dynamics of x_t and π_t are given by the ODEs:

$$\begin{aligned}\dot{x} &= (\phi_\pi - 1)\pi + \phi_x x_t + \sigma(e^{-\gamma\Theta x} - 1) \\ \dot{\pi} &= \rho\pi - \kappa(e^x - 1)\end{aligned}$$

We can rewrite this system in matrix form as:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \underbrace{\begin{bmatrix} \phi_x - \sigma\gamma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix}}_A \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \begin{bmatrix} \sigma(e^{-\gamma\Theta x_t} - 1 + \gamma\Theta x_t) \\ -\kappa(e^x - 1 - x_t) \end{bmatrix} \quad (\text{e.5})$$

Notice that the matrix A has trace equal to zero at $\phi_x = \phi_x^* = \sigma\gamma\Theta - \rho = \sigma\gamma(\Theta - \Theta^*)$, where $\Theta^* = \frac{\rho}{\sigma\gamma}$. Evaluating (e.5) at $\phi_x = \phi_x^*$, we have:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \underbrace{\begin{bmatrix} -\rho & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix}}_{A^*} \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \begin{bmatrix} \sigma(e^{-\gamma\Theta x_t} - 1 + \gamma\Theta x_t) \\ -\kappa(e^x - 1 - x_t) \end{bmatrix}, \quad (\text{e.6})$$

Since the matrix A^* is identical to the one in (b.5) in the model with $\phi_x = 0$ (See Appendix B.5 for details), it has the same eigenvalues and can be diagonalized as:

$$A^* = PDP^{-1},$$

where

$$D = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} \rho & \omega \\ \kappa & 0 \end{bmatrix}$$

Thus, as in Appendix B.5, we can transform the system into normal form, which can be written as:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix},$$

where

$$\begin{aligned} f(u, v) &= -e^{\rho u + \omega v} + 1 + \rho u + \omega v \\ g(u, v) &= \frac{\sigma}{\omega} (e^{-\gamma\Theta x} - 1 + \gamma\Theta x) + \frac{\rho}{\omega} (e^x - 1 - x) \end{aligned}$$

Computing the derivatives of $f()$ and $g()$ at $u = v = 0$, the first Lyapunov coefficient can be written as:

$$\ell_1(0) = -\frac{\sigma\gamma^2\Theta^2(\rho^2 + \omega^2)}{16\omega^2} [\rho\sigma\gamma^2\Theta^2 + \omega^2\gamma\Theta + \omega^2 + \rho^2],$$

which is negative for all $\Theta > 0$, i.e., if risk is countercyclical. Since all the conditions for Theorem 1 are satisfied, and the system undergoes a Hopf bifurcation at $\phi_x = \phi_x^*$. Consequently, $\exists \bar{\phi}_x > \phi_x^*$ such that for $\phi_x \in (\phi_x^*, \bar{\phi}_x)$, trajectories starting in the neighborhood of $(0,0)$ initially diverge, but then converge to a stable cycle which surrounds the targeted steady state. Since all these trajectories remain bounded, in this region $\phi_x \in (\phi_x^*, \bar{\phi}_x)$, the equilibrium is globally indeterminate, even though the targeted steady state is unstable (locally determinate). For $\phi_x = \phi_x^*$, the stable cycles collapse onto $(0,0)$. The equilibrium in this case is still globally indeterminate because the higher-order terms of the system push any trajectory originating near the targeted steady state towards $(0,0)$. Next, evaluating

A^* in (e.6) at $\phi_\pi = \varphi(\Theta)$ yields the same matrix as $A^\diamond$ in (b.7) in Appendix B.5. Furthermore, since setting $\phi_x \neq 0$ does not change the higher-order terms of the system, by the same reasoning as in Appendix B.5, all the conditions of Theorem 2 are also satisfied with $\phi_x \neq 0$. Thus, it follows that for $\phi_x = \bar{\phi}_x$, the stable cycles get absorbed into a homoclinic orbit. This proves point 2 of Proposition E.2.

For $\phi_x > \bar{\phi}_x$, while the stable cycles disappear but the equilibrium is still globally indeterminate. This is because Theorem 2 guarantees the existence of a saddle connection along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state. All trajectories beginning from any point on this saddle connection always remain bounded, thus proving the existence of multiple bounded trajectories which satisfy all equilibrium conditions. Thus even though for a large ϕ_x , the targeted equilibrium is locally determinate, there is global indeterminacy. This proves point 3 of Proposition E.2.

For $\phi_x < \phi_x^*$, Proposition E.1 proves that the targeted equilibrium is locally indeterminate. Thus, the equilibrium is also globally indeterminate. In addition, Theorem 2 ensures the existence of a saddle connection along which the economy can converge from the neighborhood of the untargeted steady state to the targeted steady state. This proves point 1 of Proposition E.2. $\square$

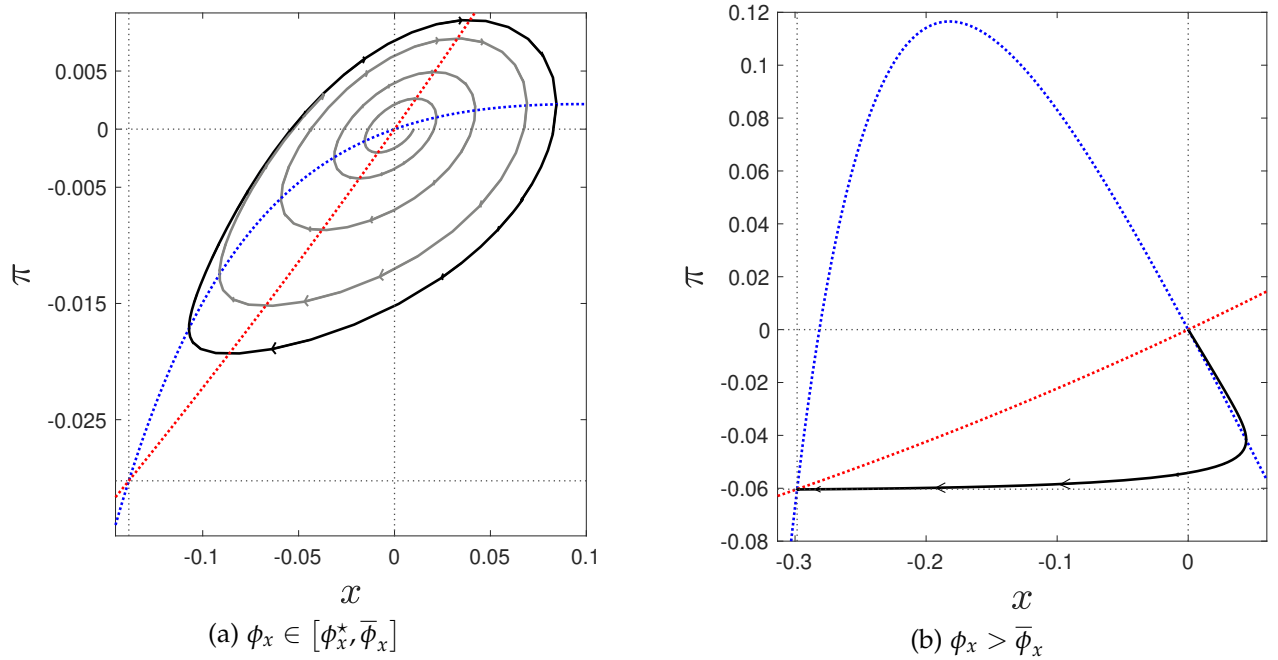


Figure E.1: Global dynamics with $\phi_x > 0$ when $\Theta < \Theta^*$

Proposition E.2 shows that no combination of ϕ_π and ϕ_x can ensure global determinacy. Figure E.1 depicts the non-fundamental fluctuations that can emerge in or HANK economy. To plot this Figure, we set $\Theta = 28.1$, which is the model estimate of Bilbiie, Primiceri and Tambalotti (2023).⁵ Figure E.1 plots global dynamics as a function of ϕ_x when risk is moderately countercyclical $\Theta \in (\Theta^\diamond, \Theta^*)$. The dotted-red curve depicts the $\dot{\pi} = 0$ -nullcline and the dotted-blue curve depicts the $\dot{x} = 0$ -nullcline.

⁵Since $\Theta = 28.1 < 31.1 = \Theta^*$, we have $\phi_x^* = \sigma\gamma(\Theta - \Theta^*) = -0.0041$. In other words, the *small* ϕ_x region corresponds to the interval $\phi_x \in (-\infty, -0.0041)$, which we ignore, because the sensible range of ϕ_x is in the interval $[0, \infty)$. Consequently, Figure E.1 only plots dynamics in the *medium* ϕ_x (i.e., $\phi_x \in [\phi_x^*, \bar{\phi}_x]$) and *large* ϕ_x (i.e., $\phi_x > \bar{\phi}_x$) cases.

Figure E.1a plots global dynamics with $\phi_x = 0.01$, which lies in the range $\phi_x \in (\phi_x^*, \bar{\phi}_x)$: in this range, trajectories starting in the neighborhood of $(0,0)$ (dark gray trajectory) converges to the stable cycle (black trajectory).⁶ In contrast, Figure E.1b features a larger $\phi_x = 0.5$, which is the standard calibration of the Taylor rule (Taylor, 1993) and shows that there exists a saddle connection from the targeted to the untargeted steady state. A trajectory beginning at any point on this saddle-connection remains bounded, and constitutes a bounded sequence which satisfies equilibrium. The same is true for more countercyclical risk, $\Theta > \Theta^*$.

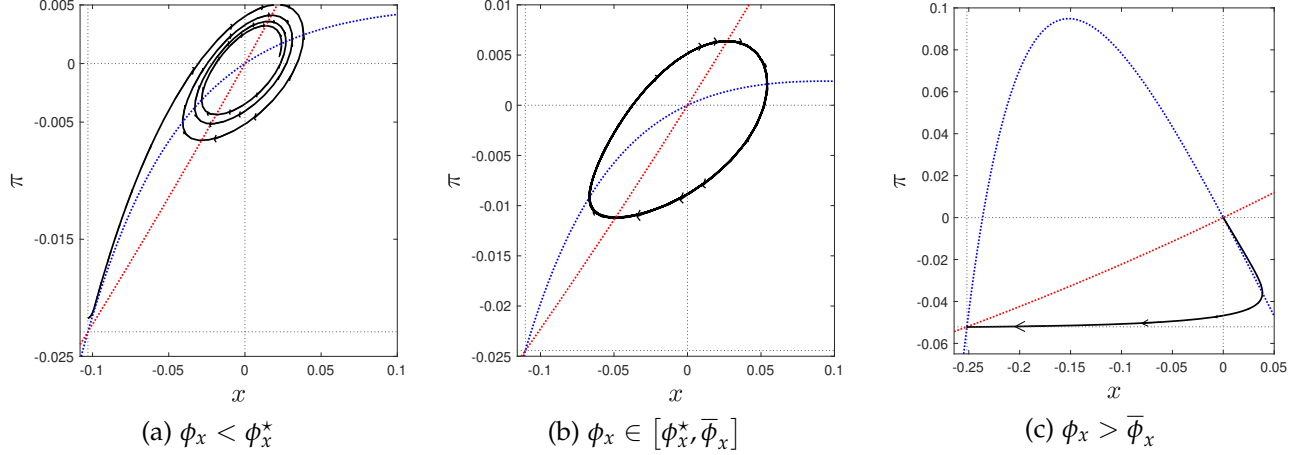


Figure E.2: Global dynamics as a function of ϕ_x when $\Theta > \Theta^*$

Figure E.2 depicts global dynamics when risk is highly countercyclical risk ($\Theta > \Theta^*$). In particular, we set $\Theta = 32$, which given our calibration is larger than $\Theta^* = 31.1$. We impose $\phi_\pi > \varphi(\Theta)$ in all plots. Figure E.2a plots dynamics when $\phi_x < \sigma\gamma(\Theta - \Theta^*)$. As stated in Propositions E.1 and E.2, in this region of ϕ_x , the equilibrium is both locally and globally indeterminate. The Figure shows a saddle connection along which the economy moves from the neighborhood of the untargeted steady state to the targeted steady state. Figure E.2b plots dynamics in the region where $\phi_x \in [\phi_x^*, \bar{\phi}_x]$. The black trajectory depicts a stable cycle. All trajectories originating near the targeted steady state initially diverge away from it but then converge to the stable cycle and remain bounded. Thus, while the targeted equilibrium is locally determinate, it is globally indeterminate. Finally, Figure E.2c plots global dynamics in the case with $\phi_x > \bar{\phi}_x$. Propositions E.1 and E.2 show that in this region the targeted equilibrium is locally determinate but globally indeterminate. The black trajectory depicts a saddle connection along which a trajectory starting near the targeted steady state diverges away from it, only to converge to the untargeted steady state. Thus, Figures E.1 and E.2 show that for any $\Theta > 0$, the equilibrium is always globally indeterminate, regardless of the combination of ϕ_π, ϕ_x .

E.2 Inertial policy rule

The monetary policy rule we have studied so far only react to changes in current economic conditions. However, empirically, many central banks have been noted to have a tendency to only adjust the policy rate gradually in response to changes in economic conditions. Such inertial rules have been shown to

⁶In addition, trajectories originating near the untargeted steady state also converge to the cycle as in Figure B.1c, but we omit these trajectories from Figure E.1a to avoid clutter.

be desirable from the point of view of delivering local determinacy (Bullard and Mitra, 2007), and for various other reasons (Woodford, 2003). In continuous time, such a rule can be written as:⁷

$$di_t = -\alpha \left[i_t - \bar{r} - \phi_\pi \pi_t \right] dt, \quad (\text{e.7})$$

where $\alpha > 0$ controls the relative weight on inflation in the past relative to current inflation. A smaller α in (e.8) implies a larger weight on past inflation relative to current inflation, and that the rule is *more backward-looking*. In fact, the limit $\alpha \rightarrow 0$ corresponds to the *price-level targeting* (PLT) limit. In the limit $\alpha \rightarrow \infty$ converges to the policy rule (6), which only responded to changes in current inflation, and is not backward-looking at all. Instead of working with (e.7), we transform it into the equivalent *average-inflation targeting* (AIT) rule:

$$i_t = \bar{r} + \phi_\pi \pi_t^b, \quad \text{where} \quad \pi_t^b = \alpha \int_{-\infty}^t e^{-\alpha(t-\tau)} \pi_\tau d\tau \quad \text{for} \quad \alpha \in (0, \infty), \quad (\text{e.8})$$

(e.8) shows that the inertial behavior is equivalent to the central bank responding to changes in the weighted-average of past and current inflation (denoted π_t^b), instead of just changes in current inflation. The dynamics of the economy under an AIT rule are described by a 3-dimensional system of ODEs:

$$\dot{x}_t = \phi_\pi \pi_t^b - \pi_t - (r^*(x_t) - \bar{r}) \quad (\text{e.9})$$

$$\dot{\pi}_t = \rho \pi_t - \kappa (e^{x_t} - 1) \quad (\text{e.10})$$

$$\dot{\pi}_t^b = \alpha (\pi_t - \pi_t^b), \quad (\text{e.11})$$

where (e.11) is derived by taking a time-derivative of π_t^b described in (e.8). Imposing steady state in (e.9)-(e.11), it is easy to see that $x = 0$ and $\pi = \pi^b = 0$ is still a steady state (the targeted steady state).

Compared to the purely inflation targeting rule (6), a sufficiently backward-looking inertial rule (e.8) can guarantee local-determinacy of the targeted steady state, even if risk is highly countercyclical ($\Theta > \Theta^*$). This is formalized in Proposition E.3.⁸

Proposition E.3 (Local determinacy with an inertial rule). *For a given $\Theta > 0$, and any π_0^b in the small neighborhood of the targeted steady state $x = 0$, $\pi = \pi^b = 0$, there exists a unique $\{x_0, \pi_0\}$, such that the trajectory $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ remains bounded inside this neighborhood, as long as α is sufficiently close to 0 and*

⁷This rule is analogous to the more familiar discrete-time specification for such rules:

$$1 + i_t = \varrho (1 + i_{t-1}) + (1 - \varrho) [1 + \bar{r} + \phi_\pi \pi_t],$$

where $\varrho \in (0, 1)$ captures the idea that the policy rate at date t displays inertia: $1 + i_t$ depends not just on the deviation of current inflation from target, but also depends on what the policy rate was set to in the past $1 + i_{t-1}$.

⁸With an inertial rule, the definition of local determinacy is slightly different relative to our baseline model. This is because our baseline model featured two forward looking variables x, π and so local determinacy required two eigenvalues with positive real parts. In contrast, the inertial rule adds as *predetermined* variable π^b , and so local determinacy requires two eigenvalues with positive real parts, and one negative eigenvalue. In other words, local determinacy of the targeted equilibrium now requires that for a given value of the predetermined variable π_0^b in a small neighborhood of the targeted steady state, there exist a unique (x_0, π_0) starting from which the trajectory $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ satisfies all equilibrium conditions and remains bounded. Global determinacy, then, requires that from *any* given value of the predetermined variable π_0^b (not necessarily in the neighborhood of the targeted steady state), there exists a unique (x_0, π_0) , starting from which the trajectory $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ remains bounded while satisfying all equilibrium conditions.

$\phi_\pi > \phi(\Theta)$. In other words, the targeted equilibrium is locally determinate if

$$\phi_\pi > \phi(\Theta) \quad \text{and} \quad \alpha \in \left[0, \alpha^*(\theta)\right),$$

If risk is mildly or moderately countercyclical $\alpha^*(\Theta) = \infty$, but if risk is highly countercyclical, then $\alpha^*(\Theta) < \infty$ (the exact expression is available in proof below).

Proof. With the inertial rule $\alpha > 0$, for (x, π, π^b) close to the targeted steady state $(0, 0, 0)$, the dynamics of the system (e.9), (e.10) and (e.11) are governed by the following system:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \\ \dot{\pi}_t^b \end{bmatrix} = A \begin{bmatrix} x_t \\ \pi_t \\ \pi_t^b \end{bmatrix} + \mathcal{O}(x^2) \quad \text{for } (x, \pi, \pi^b) \rightarrow (0, 0, 0),$$

where A is given by

$$A = \begin{bmatrix} -\sigma\gamma\Theta & -1 & \phi_\pi \\ -\kappa & \rho & 0 \\ 0 & \alpha & -\alpha \end{bmatrix},$$

Since we have one predetermined-variable π^b and two jump-variables x, π , for the targeted steady state to be locally determinate, we need one negative root and two roots with positive real parts. This would ensure that for a given π_0^b in the neighborhood of the targeted steady state, there exists a unique (x_0, π_0) such that the trajectory $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ remains bounded.

To see that this is the case when ϕ_π is large enough and α is small enough, we need to characterize the eigenvalues of A . The characteristic polynomial associated with A can be written as:

$$\mathcal{P}(z) = a_0 z^3 + a_1 z^2 + a_2 z + a_3$$

where

$$\begin{aligned} a_0 &= -1 \\ a_1 &= -\sigma\gamma(\Theta - \Theta^*) - \alpha \\ a_2 &= \kappa\phi(\Theta) - \alpha\sigma\gamma(\Theta - \Theta^*) \\ a_3 &= -\alpha\kappa(\phi_\pi - \phi(\Theta)), \end{aligned}$$

where

$$\phi(\Theta) = 1 + \frac{\rho^2}{\kappa} \left(\frac{\Theta}{\Theta^*} \right)$$

The stability of the system is governed by the pattern of sign changes in the sequence:

$$a_0, \quad a_1, \quad -\frac{a_0 a_3 - a_1 a_2}{a_1}, \quad a_3$$

For the Jacobian to have one negative root and two roots with positive real parts, we need the sequence

to have 2 sign changes. The first term in this sequence is always $-$. Imposing $\phi_\pi > \varphi(\Theta)$ guarantees that the fourth term in the sequence is also $-$. To determine the sign of the other two terms in the sequence, we need to consider two cases:

- (i) **Highly countercyclical risk** ($\Theta > \Theta^*$) : In this case, for any $\alpha > 0$, the second term in the sequence a_1 is negative. So the sequence is $-,-,?, -$. Local determinacy then requires that the third term in the sequence be positive (two sign changes) for local determinacy, i.e.,

$$\frac{\sigma\gamma(\Theta - \Theta^*) [\kappa\varphi(\Theta) - \alpha\sigma\gamma(\Theta - \Theta^*) - \alpha^2] + \alpha\kappa\phi_\pi}{\sigma\gamma(\Theta - \Theta^*) + \alpha} > 0,$$

which can be reformulated as:

$$\psi(\alpha) < \phi_\pi,$$

where

$$\psi(\alpha) = -\frac{\sigma\gamma(\Theta - \Theta^*)}{\alpha\kappa} [\kappa\varphi(\Theta) - \alpha\sigma\gamma(\Theta - \Theta^*) - \alpha^2] \quad (\text{e.12})$$

We know that

$$\psi'(\alpha) = \frac{\sigma\gamma(\Theta - \Theta^*)}{\kappa} \left[1 + \frac{\kappa\varphi(\Theta)}{\alpha^2} \right] > 0 \quad \psi(0) \rightarrow -\infty \quad \psi(\infty) \rightarrow \infty,$$

which, by the intermediate-value theorem implies that $\exists \alpha^*(\Theta) \in (0, \infty)$, such that $\psi(\alpha^*(\Theta)) = \phi_\pi$ and $\psi(\alpha) < \phi_\pi$ for all $\alpha \in (0, \alpha^*(\Theta))$. In fact, we can write $\alpha^*(\Theta)$ as:

$$\alpha^*(\Theta) = \frac{1}{2} \left\{ \frac{\kappa\phi_\pi - \sigma^2\gamma^2(\Theta - \Theta^*)^2}{\sigma\gamma(\Theta - \Theta^*)} + \sqrt{\frac{[\kappa\phi_\pi - \sigma^2\gamma^2(\Theta - \Theta^*)^2]^2}{\sigma^2\gamma^2(\Theta - \Theta^*)^2} + 4\kappa\varphi(\Theta)} \right\}$$

Thus, even when risk is highly countercyclical $\Theta > \Theta^*$ local determinacy is ensured as long as α is small enough, i.e., the rule is backward-looking enough.

- (ii) **Mildly or moderately countercyclical risk** ($\Theta \leq \Theta^*$) : In this region, we need to check two cases. First consider the case in which α is large: $\alpha \geq \sigma\gamma(\Theta^* - \Theta)$. In this case, the second term is still negative, and so the sequence is $-,-,?, -$. Thus, we need the third term in the sequence to be positive. For this to be the case, we need

$$\phi_\pi > \psi(\alpha),$$

where $\psi(\alpha)$ is the same as in (e.12). However, now with $\Theta < \Theta^*$, we have:

$$\psi'(\alpha) = \frac{\sigma\gamma(\Theta - \Theta^*)}{\kappa} \left[1 + \frac{\kappa\varphi(\Theta)}{\alpha^2} \right] < 0 \quad \psi(\sigma\gamma(\Theta^* - \Theta)) = \varphi(\Theta) \quad \psi(\infty) \rightarrow -\infty,$$

Thus, the third term is always positive in this case. $\alpha \geq \sigma\gamma(\Theta^* - \Theta)$. Thus, we have local

determinacy for any α in this region.

Finally, the remaining case is when α is small: $0 < \alpha < \sigma\gamma(\Theta^* - \Theta)$. In this case, the second term of the sequence is positive. So the sequence is $-, +, ?, -$. Thus, there are two sign changes regardless of the sign of the third term, and we have local determinacy for any α in this region.

Overall, if $\Theta \in (0, \Theta^*]$, the targeted equilibrium is locally determinate for any $\alpha \in (0, \infty)$, as long as $\phi_\pi > \varphi(\Theta)$. In other words, for $\Theta \in (0, \Theta^*]$, we have $\alpha^*(\Theta) = \infty$. However, if $\Theta > \Theta^*$, $\phi_\pi > \varphi(\Theta)$ is no longer sufficient for local determinacy. Local determinacy requires $\phi_\pi > \varphi(\Theta)$ alongside a small enough α : $\alpha < \alpha^*(\Theta)$, i.e., a rule which is also backward-looking enough. $\square$

Proposition E.3 shows that when risk is mildly or moderately countercyclical ($\Theta < \Theta^*$), any $\alpha \geq 0$ delivers local determinacy, as long as $\phi_\pi > \varphi(\Theta)$. This follows directly from the fact that in the limit as $\alpha \rightarrow \infty$, the inertial rule (e.8) converges to the policy rule (6), and we know from Proposition 2 that if $0 < \Theta < \Theta^*$, then $\phi_\pi > \varphi(\Theta)$ is sufficient for local determinacy. However, even when $\Theta \geq \Theta^*$, a small enough α ensures that the targeted equilibrium is locally determinate. In fact, a corollary of this result is that in the PLT limit, $\alpha \rightarrow 0$, the targeted equilibrium is always locally determinate regardless of how countercyclical risk may be.⁹ Thus, a sufficiently backward-looking inertial rule ensures local determinacy of the target equilibrium, no matter how countercyclical risk is.

However, this improved performance in terms of ensuring local determinacy does not translate into global determinacy. In fact, Proposition E.5 shows that as long as risk is countercyclical, the equilibrium is always globally indeterminate, no matter how backward-looking the rule. First, as the Proposition below states, inertial rules do not eliminate the second steady state; in fact for a given ϕ_π , the second steady state is the same as in our baseline economy and is unaffected by α (which measures how backward looking the rule is).

Proposition E.4 (Multiple Steady states). *For any $\Theta > 0$, there exist two steady states in which the output-gap and inflation are unaffected by the value of α . The targeted steady state is always one of the steady states. Furthermore, if the targeted steady state is locally determinate, then the untargeted steady state is locally indeterminate and has a negative output gap and inflation below target.*

Proof. The stationary points of the economy are not affected by changing the policy rule from (6) to the AIT policy rule (e.8). With the AIT policy rule, the steady state is represented by three nullclines, which can be written as:

$$\begin{aligned} 0 &= \phi_\pi \pi^b - \pi + \sigma(e^{-\gamma\Theta x} - 1) \\ 0 &= \rho\pi - \kappa(e^x - 1) \\ 0 &= \alpha(\pi - \pi^b) \end{aligned}$$

Since the third nullcline implies that $\pi = \pi^b$, the first two nullclines are the same as in our baseline. Thus, the exact value of α does not affect the level of output and inflation in the untargeted steady

⁹Bilbiie (2025) also shows that a PLT rule guarantees local-determinacy in this THANK model. However, that paper does not study whether such a rule ensures global determinacy.

state, but it does affect the stability properties. The steady state value of x in the untargeted steady state is still defined by the same equation as in the baseline model:

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}(e^x - 1) + \sigma(e^{-\gamma\Theta x} - 1),$$

and the same argument as in Appendix B.3 establishes the existence of the untargeted steady state. Next, we show that if $\phi_\pi > \varphi(\Theta)$, then the untargeted steady state is locally indeterminate.

Suppose that $\phi_\pi > \varphi(\Theta)$. Then, the Jacobian of the system (e.9)-(e.11) evaluated at the untargeted steady state can be written as:

$$A_{\underline{x}} = \begin{bmatrix} -\sigma\gamma\Theta e^{-\gamma\Theta \underline{x}} & -1 & \phi_\pi \\ -\kappa e^{\underline{x}} & \rho & 0 \\ 0 & \alpha & -\alpha \end{bmatrix}$$

Since we have one predetermined-variable π^b and two jump-variables x, π , for the untargeted steady state to be locally determinate, we need one negative root, and two roots with positive real parts. This would then ensure that for a given π_0^b in the neighborhood of the untargeted steady state, there exists a unique (x_0, π_0) such that the trajectory $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ remains bounded. If, for a given π_0^b , there exist multiple (x_0, π_0) for which the trajectory $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ remains bounded, then the untargeted steady state is locally indeterminate.

The characteristic polynomial of the $A_{\underline{x}}$ is given by:

$$\mathcal{P}(z) = a_0 z^3 + a_1 z^2 + a_2 z + a_3$$

where

$$\begin{aligned} a_0 &= -1 \\ a_1 &= -\sigma\gamma(\Theta e^{-\gamma\Theta \underline{x}} - \Theta^*) - \alpha \\ a_2 &= \kappa e^{\underline{x}} \left(1 + \frac{\rho\sigma\gamma\Theta}{\kappa} e^{-(1+\gamma\Theta)\underline{x}} \right) - \alpha\sigma\gamma(\Theta e^{-\gamma\Theta \underline{x}} - \Theta^*) \\ a_3 &= -\alpha\kappa e^{\underline{x}} \left[\phi_\pi - 1 - \frac{\rho\sigma\gamma\Theta}{\kappa} e^{-(1+\gamma\Theta)\underline{x}} \right] \end{aligned}$$

The stability of the system is governed by the pattern of sign changes in the sequence:

$$a_0, \quad a_1, \quad -\frac{a_0 a_3 - a_1 a_2}{a_1}, \quad a_3$$

For the Jacobian to have one negative root and two roots with positive real parts, we need the sequence to have 2 sign changes. Recall from (b.3) in Appendix B.4, that if $\phi_\pi > \varphi(\Theta)$, then we have:

$$\phi_\pi - 1 - \frac{\rho\sigma\gamma\Theta}{\kappa} e^{-(1+\gamma\Theta)\underline{x}} < 0$$

This implies that the fourth term on the sequence is +. Clearly, the sign of the first term in the sequence

is $-$. If $\alpha > \sigma\gamma (\Theta^* - \Theta e^{-\gamma\Theta x})$, then the sign of the second term in the sequence is $-$. So we have: $-$, $-$, $?$, $+$, and no matter what the sign of the third term is, we cannot have two sign changes. Thus, with large α , we have 2 negative and 1 positive root, which implies that for a given π^b , there are multiple bounded trajectories in the neighborhood of the untargeted steady state which converge to it.

Now consider the case in which $\alpha \leq \sigma\gamma (\Theta^* - \Theta e^{-\gamma\Theta x})$. With small α , the sequence of signs is now $-$, $+$, $?$, $+$. So if the third term is negative, then a small α can ensure that the untargeted steady state is locally determinate. However, this is not the case, and the third term is positive:

$$\frac{\alpha \kappa e^x (\phi^\diamond - \phi_\pi) + \sigma\gamma \left[\Theta^* - \frac{\alpha}{\sigma\gamma} - \Theta e^{-\gamma\Theta x} \right] \{ \kappa e^x \phi^\diamond + \alpha \sigma\gamma (\Theta^* - \Theta e^{-\gamma\Theta x}) \}}{\sigma\gamma \left(\Theta^* - \frac{\alpha}{\sigma\gamma} - \Theta e^{-\gamma\Theta x} \right)} > 0,$$

where $\phi^\diamond = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa} e^{-(1+\gamma\Theta)x}$, and we have used the fact that if $\phi_\pi > \varphi(\Theta)$, then $\phi_\pi < \phi^\diamond$. Furthermore, we know that $\Theta^* \geq \Theta e^{-\gamma\Theta x} + \frac{\alpha}{\sigma\gamma} > \Theta e^{-\gamma\Theta x}$ in this case. So even with small α , the sequence is $-$, $+$, $+$, $+$, which only has one sign change. Thus, regardless of the magnitude of α , the untargeted steady state is locally indeterminate as long as $\Theta > 0$ and $\phi_\pi > \varphi(\Theta)$. $\square$

Proposition E.4 showed that as long as $\Theta > 0$, the untargeted steady state always exists, and is locally indeterminate as long as $\phi_\pi > \varphi(\Theta)$, regardless of the magnitude of α . Consequently, there is always global indeterminacy. This is because local determinacy of the untargeted steady state implies that for a given π_0^b in the neighborhood of the untargeted steady state there exists multiple (x_0, π_0) such that there are at least two trajectories $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ which remain bounded forever. In fact, since the untargeted steady state has two negative and one positive eigenvalue, there exists a 2 dimensional stable manifold containing the untargeted steady state. Any trajectory which originates in this stable manifold remains bounded, in fact it converges to the untargeted steady state. Consequently, there is at least one π_0^b , for which there exists multiple (x_0, π_0) such that the trajectories $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ satisfy equilibrium and always remain bounded. There are even more bounded trajectories which start close to the targeted steady state, as Proposition E.5 shows.

Proposition E.5. *Suppose that risk is countercyclical ($\Theta > 0$) and monetary policy is described by the inertial rule (e.8) satisfying $\phi_\pi > \varphi(\Theta)$. Then the equilibrium is always globally indeterminate, regardless of the magnitude of $\alpha \in [0, \infty)$. The global dynamics under the inertial rule (e.8) can be divided into three regions based on the magnitude of $\alpha \geq 0$:*

1. **Mildly backward-looking** ($\alpha > \alpha^*(\Theta)$): *For large α , the targeted equilibrium is both locally and globally indeterminate. Not only do trajectories which originate in the neighborhood of the targeted steady state converge to it, there also exists a saddle connection along which trajectories which originate near the untargeted steady state converge to the targeted steady state.*
2. **Moderately backward-looking** ($\alpha \in [\underline{\alpha}(\Theta), \alpha^*(\Theta)]$): *$\exists \underline{\alpha}(\Theta) < \alpha^*(\Theta)$ such that for any $\alpha \in (\underline{\alpha}(\Theta), \alpha^*(\Theta))$, any trajectory which originates near the targeted steady state initially diverges but then converges to a stable cycle which surrounds the targeted steady state. The amplitude of these cycles is a decreasing function of α in this region. Overall, for $\alpha \in (\underline{\alpha}(\Theta), \alpha^*(\Theta))$, the targeted equilibrium is locally determinate but there is global indeterminacy. At the lower boundary $\alpha = \underline{\alpha}(\Theta)$, the stable cycles are*

absorbed into a homoclinic orbit, and at the upper boundary $\alpha = \alpha^*(\Theta)$, the limit cycles are degenerate, but there is still global indeterminacy since the higher-order terms push any trajectory starting near the targeted steady state back towards it.

3. **Strongly backward-looking** ($\alpha < \underline{\alpha}(\Theta)$): For small enough α , there are no stable cycles. However, there exists a saddle connection along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state. Thus, a small α ensures local determinacy but not global determinacy.

Overall, if risk is countercyclical, there is global indeterminacy, no matter how backward-looking the policy rule.

Proof. For a given $\Theta > 0$, the Jacobian of the system (e.9), (e.10), (e.11), evaluated at the targeted steady state, can be written as:

$$A = \begin{bmatrix} -\sigma\gamma\Theta & -1 & \phi_\pi \\ -\kappa & \rho & 0 \\ 0 & \alpha & -\alpha \end{bmatrix}$$

The trace of A can be written as:

$$\text{tr}(A) = -\sigma\gamma\Theta + \rho - \alpha = -\sigma\gamma(\Theta - \Theta^*) - \alpha,$$

where we have used the definition $\Theta^* = \frac{\rho}{\sigma\gamma}$. Next, the determinant of A is given by:

$$\det(A) = -\alpha\kappa(\phi_\pi - \varphi(\Theta)),$$

Since we maintain that $\phi_\pi > \varphi(\Theta)$, we know that $\det(A) < 0$ for any $\alpha \in (0, \infty)$. We need to show that $\exists \alpha$ for which the two complex roots have zero real parts. To prove this, we use Orlando's formula, which can be written as:¹⁰

$$H = -\det(A) + \text{tr}(A) \times G(A),$$

where $G(A)$ denotes the pairwise product of the eigenvalues of the matrix A and is given by:¹¹

$$G(A) = \begin{vmatrix} -\sigma\gamma\Theta & -1 \\ -\kappa & \rho \end{vmatrix} + \begin{vmatrix} -\sigma\gamma\Theta & \phi_\pi \\ 0 & -\alpha \end{vmatrix} + \begin{vmatrix} \rho & 0 \\ \alpha & -\alpha \end{vmatrix} = \alpha\sigma\gamma(\Theta - \Theta^*) - \kappa\varphi(\Theta)$$

Using these expressions, we can write H as:

$$H = \alpha\kappa(\phi_\pi - \varphi(\Theta)) + [\sigma\gamma(\Theta - \Theta^*) + \alpha][\kappa\varphi(\Theta) - \alpha\sigma\gamma(\Theta - \Theta^*)],$$

which can be further simplified to

$$H = \kappa \times \alpha(\phi_\pi - \psi(\alpha)),$$

¹⁰See pp. 196-198 in Chapter XV of Gantmacher (1960).

¹¹Let z_1, z_2 and z_3 denote the three eigenvalues of the matrix A . Then $\text{tr}(A) = z_1 + z_2 + z_3$, $\det(A) = z_1 z_2 z_3$ and $G(A) = z_1 z_2 + z_2 z_3 + z_3 z_1$.

where $\psi(\alpha)$ is defined in (e.12). Notice that $H = 0$ describes values of α (if one exists) for which two of the eigenvalues cancel each other out. $H = 0$ requires that either (i) two of the eigenvalues are purely imaginary or (ii) two real eigenvalues have the same magnitude but opposite sign. Since Proposition E.3 showed that for $\Theta > \Theta^*$, as long as $\alpha < \alpha^*(\Theta)$, there are two complex roots with positive real parts and one real negative root. Thus, it follows that at $\alpha = \alpha^*(\Theta)$, the two complex roots are purely imaginary and thus cancel each other out, leaving the trace to equal the remaining negative root. Thus, a Hopf bifurcation occurs at $\alpha = \alpha^*(\Theta)$. While verifying that the first Lyapunov coefficient of this 3 dimensional system is possible, it is extremely cumbersome and we verify numerically that it is negative, implying that the Hopf bifurcation is supercritical. Consequently, the Hopf bifurcation theorem implies that for any $\Theta > 0$, $\exists \underline{\alpha}(\Theta) < \alpha^*(\Theta)$, such that for $\alpha \in (\underline{\alpha}(\Theta), \alpha^*(\Theta)]$, all trajectories (except those which begin on the one dimensional stable manifold around the targeted steady state), which originate near the targeted steady state converge to a stable cycle. In this region, since $\alpha < \alpha^*(\Theta)$, Proposition E.3 implies that the targeted steady state is locally determinate. However, since for a given π_0^b in the neighborhood of the targeted steady state, there exists multiple (x_0, π_0) such that the trajectories $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ remain bounded forever, there is global indeterminacy. The convergence to a stable cycle is depicted graphically in Figure E.3a, where we have set $\Theta = 28.1$ and $\alpha = 9$. Since $\Theta = 28.1 < \Theta^* = 31.1$ under our calibration, $\alpha^*(28.1) = \infty$ and $\underline{\alpha}(28.1) \approx 1.03$ under our calibration. Thus, for any $\alpha > 1.03$, any trajectory originating near the targeted steady state converges to a stable cycle.

Next, Theorem 7.2 and Corollary 7.1 of Kopell and Howard (1975) (which are generalizations of Theorem 2 to $n > 2$ dimensions) imply that for α close to $\underline{\alpha}(\Theta)$, there exists a homoclinic orbit at the boundary of the stable cycles mentioned above. Thus, for $\alpha = \underline{\alpha}(\Theta)$, there exists a homoclinic orbit which cycles around the targeted steady state, while passing through the untargeted steady state. The homoclinic orbit is depicted graphically in Figure E.3b. Even in this case, since $\underline{\alpha}(\Theta) < \alpha^*(\Theta)$, Proposition E.3 shows that the targeted equilibrium is locally determinate. However, there is still global indeterminacy, since for any π_0^b in the neighborhood of the targeted steady state, all combinations of (x_0, π_0) in the neighborhood of the targeted steady state are such that the trajectories $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ remain bounded forever. There is a unique one dimensional manifold around the targeted steady state along which the trajectories converge to the targeted steady state and remain bounded, while all trajectories originating off this stable manifold converge to the homoclinic orbit. At the bifurcation point, $\alpha = \alpha^*(\Theta)$, for a given π_0^b in the neighborhood of the targeted steady state, while the first-order terms do not move the system towards or away from the targeted steady state, the higher-order terms ensure that all trajectories $\{x_t, \pi_t, \pi_t^b\}_{t=0}^\infty$ converge to the targeted steady state, and hence remain bounded, implying global indeterminacy in this case as well.

Finally, for $\alpha < \underline{\alpha}(\Theta)$, there are no stable cycles and Proposition E.3 implies that the targeted steady state is locally determinate. However, Theorem 7.2 and Corollary 7.1 of Kopell and Howard (1975) also guarantee that for $\alpha \in (0, \alpha^*(\Theta))$, there exists a saddle connection along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state. Furthermore, any trajectory starting on this saddle connection constitutes a bounded equilibrium since it converges to the untargeted steady state and remains bounded forever. Thus, even in the range $\alpha \in [0, \underline{\alpha}(\Theta))$, the equilibrium is globally indeterminate. $\square$

Proposition E.5 characterizes the global dynamics of the economy under the average inflation targeting rule for all degrees of backward-lookingness, and shows that no matter how backward looking a policy rule, it cannot eliminate global indeterminacy. Figure E.3 plots global dynamics for $\Theta < \Theta^*$.

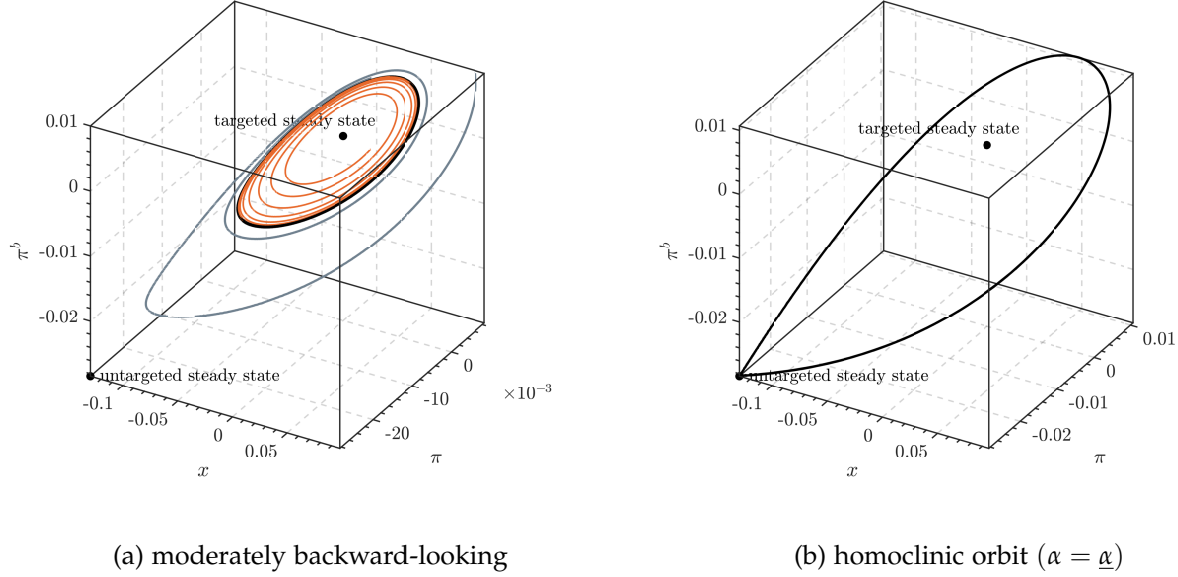


Figure E.3: Global dynamics with an inertial rule

When $\Theta = 28.1 < \Theta^*$, $\alpha^*(\Theta) = \infty$ and set $\alpha \geq \alpha^*(\Theta)$ is empty. Thus, the global dynamics in this case are described by points 2 and 3 of Proposition E.5. Figure E.3a depicts the dynamics for a moderately backward-looking rule, where we have set $\alpha = 9$, which is larger than $\underline{\alpha}(\Theta) = 1.03$ and smaller than $\alpha^*(\Theta) = \infty$. The figure shows that trajectories which originate near the targeted steady state (orange curves) or even further away from the steady state (grey trajectory), both converge to a stable cycle (black trajectory). Figure E.3b depicts the homoclinic orbit which occurs if $\alpha = \underline{\alpha}(\Theta) = 1.03$.¹²

Trajectories which originate inside the homoclinic orbit converge to it and remain bounded. While point 3 of Proposition E.5 guarantees that for $\alpha \in [0, \underline{\alpha})$, there exists a saddle connection (along which the economy can move from the neighborhood of the targeted steady state to the untargeted steady state), it is hard to numerically plot this trajectory because it is hard to numerically compute the 1 dimensional stable manifold. Hence we are unable to plot the dynamics described in point 3. Overall, the discussion above shows that as long as risk is countercyclical ($\Theta > 0$), there is global indeterminacy, no matter how backward looking the policy rule is.

E.3 Analysis with a Rotemberg Phillips curve

E.3.1 Deriving the micro-founded Rotemberg Phillips curve

As in Rotemberg (1982), we assume that there each firm k faces a quadratic cost of changing its price at each date, which in our continuous time setting can be written as $\frac{\psi}{2} \left(\frac{\dot{p}_{k,t}}{p_{k,t}} \right)^2 P_t y_t$, where ψ is a constant

¹²The existence of homoclinic orbits in 3 dimensional systems can also give rise chaotic dynamics, as studied by Shilnikov (See Chapter 6 of Kuznetsov (1998)). Barnett et al. (2022) study an application of Shilnikov chaos to a New Keynesian model.

which scales the cost. We still maintain the assumption that the production function is linear and that the only factor of production is labor. Then, at any date t , firm k 's nominal profit net of the price-adjustment cost can be written as:

$$D_{k,t} = \left[P_{j,t} \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} - (1 - \varepsilon^{-1}) P_t w_t \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} - \frac{\psi}{2} \left(\frac{\dot{P}_{j,t}}{P_{j,t}} \right)^2 P_t \right] y_t,$$

where $(1 - \varepsilon^{-1})w_t$ denotes the post-subsidy real wage, where the subsidy is designed to eliminate the monopolistic markup on average. The pricing problem of a firm can be expressed as a Hamiltonian:

$$\mathcal{H} = \left[P_{j,t} \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} - (1 - \varepsilon^{-1}) P_t w_t \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} - \frac{\psi}{2} \left(\frac{\dot{P}_{j,t}}{P_{j,t}} \right)^2 P_t \right] y_t + \eta_t \dot{P}_{j,t},$$

where η_t denotes the co-state variable. The optimal choice of $\dot{P}_{j,t}$ satisfies

$$\eta_t = \psi \left(\frac{\dot{P}_{j,t}}{P_{j,t}} \right) \left(\frac{P_t}{P_{j,t}} \right) y_t, \quad (\text{e.13})$$

while the evolution of the co-state under the optimal plan can be written as:

$$\dot{\eta}_t = i_t \eta_t - \left[(1 - \varepsilon) \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} + \varepsilon (1 - \varepsilon^{-1}) \left(\frac{P_t}{P_{j,t}} \right) w_t \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} + \psi \left(\frac{\dot{P}_{j,t}}{P_{j,t}} \right)^2 \left(\frac{P_t}{P_{j,t}} \right) \right] \quad (\text{e.14})$$

In a symmetric equilibrium, all firms choose the same $P_{j,t} = P_t$ and we can rewrite (e.13) and (e.14) as:

$$\eta_t = \psi \pi_t y_t \quad (\text{e.15})$$

$$\dot{\eta}_t = i_t \eta_t - \left[(1 - \varepsilon) + \varepsilon (1 - \varepsilon^{-1}) w_t + \psi \pi_t^2 \right] y_t, \quad (\text{e.16})$$

where $\pi_t = \dot{P}_t / P_t$. Next, taking the time derivative of (e.15), we have:

$$\dot{\eta}_t = \psi \dot{\pi}_t y_t + \psi \pi_t \dot{y}_t \quad (\text{e.17})$$

Combining (e.15), (e.16) and (e.17), we get the Phillips curve:

$$\dot{\pi}_t = \left(r_t - \frac{\dot{y}_t}{y_t} \right) \pi_t - \kappa (w_t - 1), \quad (\text{e.18})$$

where $\kappa = \frac{\varepsilon-1}{\psi}$.

E.3.2 Global indeterminacy with the Rotemberg Phillips curve

As in our baseline, we consider the inflation targeting rule:

$$i_t = \bar{r} + \phi_\pi \pi_t$$

Then, the dynamics of (x_t, π_t) can be written as:

$$\dot{x}_t = (\phi_\pi - 1) \pi_t + \sigma (e^{-\gamma \Theta x_t} - 1) \quad (\text{e.19})$$

$$\dot{\pi}_t = (r_t - \gamma \dot{x}_t) \pi_t - \kappa (e^{x_t} - 1) \quad (\text{e.20})$$

As before, we can rewrite this system by separating the first-order and higher-order terms:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \underbrace{\begin{bmatrix} -\gamma \sigma \Theta & \phi_\pi - 1 \\ -\kappa & \bar{r} \end{bmatrix}}_A \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \begin{bmatrix} \sigma (e^{-\gamma \Theta x_t} - 1 + \gamma \Theta x_t) \\ -\kappa (e^{x_t} - 1 - x_t) - (\gamma - 1) (\phi_\pi - 1) \pi_t^2 - \gamma \sigma (e^{-\gamma \Theta x_t} - 1) \pi_t \end{bmatrix} \quad (\text{e.21})$$

where $\bar{r} = \rho - \sigma$ denotes the real interest rate in the targeted steady state. In what follows, we assume that $\bar{r} > 0$.¹³ As in Appendix B, local determinacy in this economy requires that the matrix A have two eigenvalues with positive real parts.

Stable cycle via a Hopf bifurcation Similar to Appendix B, it is clear that the trace of A is 0 at $\Theta = \Theta_R^* \equiv \frac{\bar{r}}{\sigma \gamma}$. Imposing $\Theta = \Theta_R^*$ in A , we have:

$$A_R^* = \begin{bmatrix} -\bar{r} & \phi_\pi - 1 \\ -\kappa & \bar{r} \end{bmatrix},$$

and the eigenvalues of A_R^* are given by $\pm \omega i$ where $\omega = \sqrt{\kappa (\phi_\pi - 1) - \bar{r}^2}$ and $i = \sqrt{-1}$. Thus, the system undergoes a Hopf-bifurcation at $\Theta = \Theta_R^*$.

Next, we can diagonalize the matrix A_R^* as:

$$A_R^* = P D P^{-1},$$

where

$$D = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} \bar{r} & \omega \\ \kappa & 0 \end{bmatrix}, P^{-1} = \begin{bmatrix} 0 & \frac{1}{\kappa} \\ \frac{1}{\omega} & -\frac{\bar{r}}{\omega \kappa} \end{bmatrix}$$

Pre-multiplying both sides of (e.21) by P^{-1} , we get:

$$\begin{bmatrix} \dot{u}_t \\ \dot{v}_t \end{bmatrix} = D \begin{bmatrix} u_t \\ v_t \end{bmatrix} + \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix},$$

where

$$f(u, v) = -e^{\bar{r}u + \omega v} + 1 + \bar{r}u + \omega v - \kappa(\gamma - 1)(\phi_\pi - 1)u^2 - \gamma \sigma \left(e^{-\frac{\bar{r}^2}{\sigma}u - \frac{\bar{r}\omega}{\sigma}v} - 1 \right) u$$

and

$$g(u, v) = \frac{\sigma}{\omega} \left(e^{-\frac{\bar{r}^2}{\sigma}u - \frac{\bar{r}\omega}{\sigma}v} - 1 \right) (1 + \bar{r}\gamma u) + \frac{\bar{r}}{\omega} (e^{\bar{r}u + \omega v} - 1) + \frac{\bar{r}\kappa}{\omega} (\gamma - 1)(\phi_\pi - 1)u^2,$$

¹³We make this assumption since with $\bar{r} < 0$, the Phillips curve slopes the wrong way and becomes downward sloping.

$$\begin{bmatrix} u_t \\ v_t \end{bmatrix} = P^{-1} \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} = \begin{bmatrix} \bar{r}u_t + \omega v_t \\ \kappa u_t \end{bmatrix}$$

Finally, using the same steps as in the baseline model, the first Lyapunov coefficient at the bifurcation point $\Theta = \Theta_R^*$ can be written as:

$$\ell_1(0) = - \frac{\kappa \bar{r} (\phi_\pi - 1) [\rho + \sigma (1 - \gamma)] \left\{ \bar{r} \omega^2 + 2 \left(\gamma - \frac{1}{2} \right) \sigma [\kappa (\phi_\pi - 1) - \rho \bar{r} + \bar{r} \sigma] \right\}}{16 \sigma^2 \omega^2}$$

First, notice that since the denominator is always positive, the sign of $\ell_1(0)$ depends on the sign of the numerator $\mathbb{N}$, which can be written as:

$$\mathbb{N}(\sigma) = \kappa \bar{r} (\phi_\pi - 1) [\rho + \sigma (1 - \gamma)] \left\{ \bar{r} \omega^2 + 2 \left(\gamma - \frac{1}{2} \right) \sigma [\kappa (\phi_\pi - 1) - \rho \bar{r} + \bar{r} \sigma] \right\}$$

In particular, for $\ell_1(0) < 0$, we need that $\mathbb{N} > 0$.

Since we are maintaining the assumption that $\phi_\pi > 1 + \frac{\rho \sigma \gamma \Theta}{\kappa}$ (which is a sufficient condition for local determinacy of the targeted equilibrium), evaluating this condition at $\Theta = \Theta_R^* = \bar{r} / \sigma \gamma$, we have:

$$\phi_\pi > 1 + \frac{\rho \bar{r}}{\kappa} \quad \Rightarrow \quad \kappa (\phi_\pi - 1) - \rho \bar{r} > 0,$$

which implies that the term inside the last square brackets of $\mathbb{N}$ is positive. Next, note that for any $\gamma > 0$, a sufficient condition for $\ell_1(0) < 0$ is that the steady state level of consumption risk σ be sufficiently small. The easiest way to see this is to note that the numerator is positive if we evaluate it at $\sigma = 0$:

$$\mathbb{N}|_{\sigma=0} = \kappa \bar{r} (\phi_\pi - 1) \rho \bar{r} \omega^2 > 0$$

Therefore, by continuity, the numerator remains positive for any γ as long as σ is sufficiently small. This shows that as long as the steady state consumption risk σ is small enough, we have $\ell_1(0) < 0$ for any $\gamma > 0$. In other words, even with the Rotemberg Phillips curve, the Hopf bifurcation is *supercritical*: there exists an interval $(\Theta_R^\circ, \Theta_R^*)$ for which any trajectories originating in the neighborhood of $(x, \pi) = (0, 0)$ initially diverge away from the targeted steady state but then converge to a stable cycle staying bounded forever, implying that there is global indeterminacy. The Hopf bifurcation theorem also ensures that for $\Theta > \Theta_R^*$, as in the baseline, there are multiple bounded trajectories and hence there is global indeterminacy. Given our calibration of the baseline model, the first Lyapunov coefficient is equal to -0.004, which implies that even though the economy features local determinacy under our baseline calibration, the equilibrium is globally indeterminate. Trajectories which originate in the neighborhood of the targeted steady state initially diverge away, but eventually converge to a stable cycle which surrounds the targeted steady state, thus remaining bounded. These dynamics are qualitatively the same as in the calibrated version of the baseline model (which did not feature a non-linear Phillips curve).

E.4 Robustness to functional forms

In our baseline model, we assumed a particular functional-form for how the transition rate of switching from ξ_h to ξ_l depends on y_t . In particular, we assumed the functional form:

$$\lambda_{l,t} = \bar{\lambda}_l y_t^{-\Theta} = \bar{\lambda}_l e^{-\gamma \Theta x_t}, \quad \Theta > 0,$$

where $x_t = \gamma^{-1} \ln y_t$. In this Appendix, we show that our characterization of global indeterminacy does not depend $\lambda_{l,t}$ having this exact functional form. To do so, we allow for the transition rate $\lambda_{l,t}$ to be described by:

$$\lambda_{l,t} = \bar{\lambda}_l \cdot \Lambda(\ln y_t) = \bar{\lambda}_l \cdot \Lambda(\gamma x_t), \quad (\text{e.22})$$

where $\Lambda(\cdot)$ is some analytic function which takes non-negative values $\Lambda(\gamma x) \geq 0$ for any $x \in (-\infty, \infty)$. In addition, we normalize $\Lambda(0) = 1$ so that the transition rate in the targeted steady state with $x = 0$ is still given by $\bar{\lambda}_l$ as in the baseline model. Finally, to make the analysis comparable to the baseline model, we assume that

$$\Lambda'(0) = -\Theta < 0, \quad (\text{e.23})$$

which ensures that risk is countercyclical, with the parameter Θ controlling how countercyclical risk is as in the baseline model. However, it is important to note that (e.23) only makes an assumption about how the transition rate changes *local* to the targeted steady state and *does not* restrict the behavior of how the transition rate responds to changes in x away from $x = 0$.

Keeping everything else the same as in our baseline model except the transition rate, which is now given by (e.22), the IS curve and the Phillips curve can be written as:

$$\dot{x}_t = i_t - \pi_t - \rho + \sigma \Lambda(\gamma x_t) \quad (\text{e.24})$$

$$\dot{\pi}_t = \rho \pi_t - \kappa (e^{x_t} - 1), \quad (\text{e.25})$$

where, as in the baseline model, $\sigma = \bar{\lambda}_l \left(\frac{\xi_h}{\xi_l} - 1 \right) > 0$ still captures the average consumption risk faced by ξ_h households in the targeted steady state. Finally, assuming that monetary policy is still described by the policy rule $i_t = \rho - \sigma + \phi_\pi \pi_t$ (which is the same as (6)), the aggregate dynamics of x_t, π_t are fully characterized by the following system of ODEs:

$$\dot{x}_t = (\phi_\pi - 1) \pi_t + \sigma \left(\Lambda(\gamma x_t) - 1 \right) \quad (\text{e.26})$$

$$\dot{\pi}_t = \rho \pi_t - \kappa (e^{x_t} - 1) \quad (\text{e.27})$$

As in the main text, we can rewrite the system of ODEs as:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = A \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \underbrace{\begin{bmatrix} \sigma \left\{ \Lambda(\gamma x_t) - 1 + \gamma \Theta x_t \right\} \\ -\kappa (e^{x_t} - 1 - x_t) \end{bmatrix}}_{\text{higher-order terms}}, \quad (\text{e.28})$$

where, as in the baseline model, the properties of matrix A dominate the dynamics local to the targeted steady state $(x, \pi) = (0, 0)$, and the higher-order terms dominate the dynamics. With a general $\Lambda(\cdot)$ function, the matrix A can be written as:

$$A = \begin{bmatrix} \sigma\gamma\Lambda'(0) & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix} = \begin{bmatrix} -\sigma\gamma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix},$$

where the second equality uses (e.23), i.e., local to the steady state, the derivative of $\Lambda(\cdot)$ with respect to x is given by $-\Theta$. Consequently, the A matrix above is the same as in equation (17) in our baseline model, implying that the first-order dynamics of the two economies are identical local to the targeted steady state.

Local Determinacy Since the first-order dynamics of our baseline economy and the economy with a general $\Lambda(\cdot)$ function are identical local to the targeted steady state, it follows that the conditions which guarantee local determinacy of the targeted equilibrium in this economy with the general $\Lambda(\cdot)$ function are the same as in our baseline economy and summarized by Proposition 2, which states that as long as, *local to the targeted steady state*, risk is not too countercyclical: $0 < \Theta < \frac{\rho}{\sigma\gamma}$, local determinacy requires that $\phi_\pi > \varphi(\Theta)$, where $\varphi(\Theta)$ is the same as in (18). Furthermore, as in the baseline model, if *local to the targeted steady state*, risk is too countercyclical $\Theta > \frac{\rho}{\sigma\gamma}$, the targeted equilibrium is locally indeterminate no matter how large ϕ_π , as long as it is finite. Throughout the rest of Appendix E.4, we will maintain the assumption that $\phi_\pi > \varphi(\Theta)$, which is a necessary condition for local determinacy of the targeted equilibrium.

E.4.1 Global Indeterminacy: Existence of a stable cycle

In this section, we show that the existence of the stable cycle *only* depends on the properties of $\Lambda(\cdot)$ local to $x = 0$. As in our baseline model (see Appendix), it is easy to see that $\Theta = \Theta^* = \frac{\rho}{\sigma\gamma}$ is still the bifurcation point of this general system. To see this, we can evaluate (e.28) at $\Theta = \Theta^* = \rho/(\sigma\gamma)$ to get:

$$\begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \underbrace{\begin{bmatrix} -\rho & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix}}_{A^*} \begin{bmatrix} x_t \\ \pi_t \end{bmatrix} + \begin{bmatrix} \sigma \{ \Lambda(\gamma x_t) - 1 + \frac{\rho}{\sigma} x_t \} \\ -\kappa (e^{x_t} - 1 - x) \end{bmatrix}, \quad (\text{e.29})$$

Clearly, as in Appendix B.5 which studies our baseline model, the matrix A^* in (e.29) has trace 0, which means that the two eigenvalues of the matrix add up to 0. Furthermore, as long as we impose $\phi_\pi > \varphi(\Theta)$, the condition required for local determinacy of the targeted steady state, the determinant of A^* is positive:

$$\text{Det}(A^*) = \kappa(\phi_\pi - 1) - \rho^2 > 0 \quad \because \quad \phi_\pi > \varphi(\Theta^*) = 1 + \frac{\rho^2}{\kappa}$$

A positive determinant means that the eigenvalues must be purely imaginary, because if the roots were real and canceled each other out, then the determinant would be negative. Thus, as in our baseline model, the economy undergoes a Hopf bifurcation at $\Theta = \Theta^*$ (see Theorem 1). In fact, at $\Theta = \Theta^*$, the

two eigenvalues can be written as $\pm\omega i$, where $\omega = \sqrt{\kappa(\phi_\pi - 1) - \rho^2}$. Next, as in Appendix B.5, we can diagonalize A^* as $A^* = PDP^{-1}$, where

$$D = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} \rho & \omega \\ \kappa & 0 \end{bmatrix}$$

Next, we can pre-multiply both sides of (e.29) by P^{-1} to express the system in normal form:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = D \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix},$$

where

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \frac{\pi}{\kappa} \\ \frac{x}{\omega} - \frac{\rho}{\omega\kappa} \end{bmatrix} \Leftrightarrow \begin{bmatrix} x \\ \pi \end{bmatrix} = \begin{bmatrix} \rho u + \omega v \\ \kappa u \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} = \begin{bmatrix} -e^{\rho u + \omega v} + 1 + \rho u + \omega v \\ \frac{\rho e^{\rho u + \omega v} + \sigma \Lambda(\gamma \rho u + \gamma \omega v) - (\rho + \sigma)}{\omega} \end{bmatrix}$$

We can use the normal form to compute the first-Lyapunov coefficient, which can be written as:

$$\begin{aligned} \ell_1(0) &= f_{uuu}(0,0) + f_{uvv}(0,0) + g_{uuv}(0,0) + g_{vvv}(0,0) \\ &\quad + \frac{1}{\omega} \left[f_{uv}(0,0) (f_{uu}(0,0) + f_{vv}(0,0)) - g_{uv}(0,0) (g_{uu}(0,0) + g_{vv}(0,0)) - f_{uu}(0,0) g_{uu}(0,0) \right. \\ &\quad \left. + f_{vv}(0,0) g_{vv}(0,0) \right] \\ &= -\frac{\sigma \gamma^2 \kappa (\phi_\pi - 1)}{\omega^2} \left[\kappa (\phi_\pi - 1) \Lambda_0'' + \rho \sigma \gamma^2 (\Lambda_0'')^2 - \gamma \omega^2 \Lambda_0''' \right], \end{aligned}$$

where

$$\Lambda_0'' = \left. \frac{d^2 \Lambda(x)}{dx^2} \right|_{x=0} \quad \text{and} \quad \Lambda_0''' = \left. \frac{d^3 \Lambda(x)}{dx^3} \right|_{x=0}$$

are the second and third derivatives of the function $\Lambda(x)$ evaluated at $x = 0$. Finally, as in Appendix B.5, the Hopf bifurcation is supercritical as long as $\ell_1(0) < 0$, which requires that

$$\kappa (\phi_\pi - 1) \Lambda_0'' + \rho \sigma \gamma^2 (\Lambda_0'')^2 > \gamma \omega^2 \Lambda_0''' \quad (\text{e.30})$$

As long as (e.30) is satisfied, our HANK economy with a general $\Lambda(x)$ function would also feature global indeterminacy, even when the targeted equilibrium is locally determinate. This is because, if (e.30) is satisfied, Theorem 1 guarantees that, as in our baseline model, there exists a $\Theta^\diamond < \Theta^*$ such that for any $\Theta \in (\Theta^\diamond, \Theta^*)$, any trajectory (x_t, π_t) originating in the neighborhood of the targeted steady state $(x, \pi) = (0, 0)$ initially diverges away from $(0, 0)$, but coverage to a stable cycle and remain bounded asymptotically, thus implying that multiple bounded trajectories satisfy all equilibrium conditions no matter how large $\phi_\pi > \varphi(\Theta) > 1$ is.

Furthermore, the restriction in (e.30) is not very demanding and is satisfied as long as $\Lambda(\cdot)$ is sufficiently convex local to $x = 0$. This is easiest to see if we assume that $\Lambda_0''' < 0$, in which case (e.30) can be simplified to

$$\kappa (\phi_\pi - 1) \Lambda_0'' + \rho \sigma \gamma^2 (\Lambda_0'')^2 > 0,$$

A sufficient condition for which the above condition is satisfied is that $\Lambda_0'' > 0$, i.e., $\Lambda(\gamma x)$ is a decreasing and convex function local to $x = 0$. Note that the condition (e.30) is satisfied in our baseline specification since it is a decreasing concave function: $\Lambda_0'' = \Theta^2 > 0$ and $\Lambda_0''' = -\Theta^3 < 0$ for any $\Theta > 0$. Importantly, (e.30) only needs to be satisfied at $x = 0$, and does *not* require the function $\Lambda(x)$ to be concave over the entire domain. In other words, the existence of the stable cycle can be established without restricting the shape of $\Lambda(\gamma x)$ for x far away from its value in the steady state (other than the assumptions about continuity and differentiability); condition (e.30) only depends on how countercyclical risk is in the neighborhood of the targeted steady state ($x = \pi = 0$). A corollary of this is that the existence of the stable cycle *does not* require the existence of an untargeted steady state. It is also useful to note that even the restriction $\Lambda_0'' > 0$ is only a sufficient condition, (e.30) may even be satisfied in $\Lambda(\cdot)$ is concave, but not too concave.

Finally, it is worth pointing out that (e.30) is not satisfied if $\Lambda(\cdot)$ is linear in x since that implies that $\Lambda_0'' = \Lambda_0''' = 0$. However, the argument above makes clear that the case with $\Lambda(x)$ linear is non-generic and constitutes a rather restrictive assumption on the shape of $\Lambda(x)$. (e.30) is satisfied away from this knife-edge case under fairly nonrestrictive assumptions. In fact, in the main text, we argue that it is natural for $\Lambda(x)$ to be a decreasing convex function. Together, these arguments imply that global indeterminacy can easily arise in HANK economies with countercyclical risk as long as standard monetary policy rules are utilized.

However, even if we insist that $\Lambda(\cdot)$ be a linear function in x , then while the stable cycle cannot exist, global indeterminacy can easily arise under some mild conditions which ensure the existence of at least one untargeted steady state alongside the targeted steady state. We describe this scenario next.

E.4.2 Global Indeterminacy: Existence of multiple steady states

In this section, we provide some sufficient conditions on the properties of $\Lambda(\gamma x)$ under which at least one untargeted steady state exists alongside the targeted steady state. As discussed earlier, since our baseline feature no predetermined variables, the existence of two steady states already implies global indeterminacy even if a stable cycle does not exist. To characterize sufficient conditions under which the economy with a general $\Lambda(\cdot)$ function admits multiple steady states, impose $\dot{x} = \dot{\pi} = 0$ in (e.26) and (e.27) to get:

$$\begin{aligned} 0 &= (\phi_\pi - 1)\pi + \sigma(\Lambda(\gamma x) - 1) \\ 0 &= \rho\pi - \kappa(e^x - 1) \end{aligned}$$

Any combination of (x, π) which satisfy the two equations above constitute a steady state of the economy. As in Appendix B.3, we can combine these two equations into one equation, and any x which solves the following equation is steady state of the economy.

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho}(e^x - 1) + \sigma(\Lambda(\gamma x) - 1) = 0$$

Without further restrictions on the slope of $\Lambda(\cdot)$, it is not possible to ascertain the sign and magnitude of $F(x)$ for any x . However, since we assumed that $\Lambda(0) = 1$, we know that $x = 0$ satisfies $F(0) = 0$.

Thus, $(x, \pi) = (0, 0)$ is still a steady state of the economy. Furthermore, given the assumption (e.23) that $\Lambda'(0) = -\Theta < 0$, i.e., risk is countercyclical local to $x = 0$, we can write the derivative of $F(x)$ at $x = 0$ as:

$$F'(x)|_{x=0} = \frac{\kappa(\phi_\pi - 1)}{\rho} + \gamma\sigma\Lambda'(0) = \frac{\kappa}{\rho} \left(\phi_\pi - 1 - \frac{\rho\gamma\sigma\Theta}{\kappa} \right) = \frac{\kappa}{\rho} [\phi_\pi - \varphi(\Theta)],$$

where the second equality uses (e.23). Given our maintained assumption that $\phi_\pi > \varphi(\Theta)$, we have $F'(0) > 0$, which implies that as we lower x from 0, $F(x)$ declines below 0. This condition is sufficient to ensure that a untargeted steady state exists even when $\Lambda(\gamma x)$ is linear. To see this, given our assumption that $\Lambda'(0) = -\Theta$, when $\Lambda(\gamma x)$ is linear, it can be written as:

$$\Lambda(\gamma x) = \max\{0, 1 - \gamma\Theta x\},$$

where the max operator ensures that the function only takes non-negative values even when x is positive and large. In this case, for $x < 0$, we can rewrite $F(x)$ above as:

$$F(x) = \frac{\kappa(\phi_\pi - 1)}{\rho} (e^x - 1) - \sigma\Theta\gamma x$$

It is true by inspection that as $x \rightarrow -\infty$, we have $F(x) \rightarrow +\infty$. Since we know that $F(0) = 0$ and $F'(0) > 0$, it must be that for as we lower x starting from 0, $F(x)$ becomes negative. Since $F(x)$ tends to $+\infty$ as $x \rightarrow -\infty$, it must be the case that there exists $\underline{x} < 0$ for which $F(\underline{x}) = 0$. Thus, even when $\Lambda(\cdot)$ is linear, the untargeted steady state exists for any $\Theta > 0$, i.e., as long as risk is countercyclical. It follows immediately that even when $\Lambda(\gamma x)$ is non-linear, a sufficient condition for the existence of the untargeted steady state is that $\Lambda(\ln y)$ is convex. Furthermore, under this condition, the existence of this steady state can be guaranteed without making further assumptions about the behavior of $\Lambda(\gamma x)$ away from $x = 0$: convexity of $\Lambda(\cdot)$ is sufficient to ensure the existence of an untargeted steady state with lower economic activity as long as risk is countercyclical *local* to the targeted steady state $\Lambda'(0) = -\Theta < 0$. Of course, the exact value of output in the untargeted steady state depends on the exact functional form of $\Lambda(\cdot)$, but we can guarantee existence of such a steady state based purely on the properties of $\Lambda(\cdot)$ *local* to the targeted steady state.

Thus, our characterization of global determinacy holds under fairly weak restrictions on the shape of the function $\Lambda(\cdot)$. Furthermore, establishing this only depends on the properties of $\Lambda(\cdot)$ local to the targeted steady state $x = 0$, and does not require us to take a stance on how cyclical risk behaves as the economy moves far from the steady state. $\square$

E.5 Non-constant fraction of HtM households

In our baseline model we assumed that $\lambda_{h,t}$ adjusts endogenously to ensure that $\lambda_{h,t}\eta = \lambda_{l,t}(1 - \eta)$. This meant that the fraction of ξ_h and ξ_l households stays constant over time. This assumption was made for simplicity. In this appendix, we relax this assumption and allow the fraction of ξ_l households (η_t) to change over time. In particular, if we assume that ξ_l households transition to ξ_h at a constant

rate $\bar{\lambda}_h > 0$, we can express the law of motion of the mass of ξ_l households as:

$$\dot{\eta}_t = \lambda_{l,t} (1 - \eta_t) - \bar{\lambda}_h \eta_t, \quad (\text{e.31})$$

where we maintain the assumption that $\lambda_{l,t} = \bar{\lambda}_l y_t^{-\Theta}$. Under this specification, more households transition to ξ_l state and become borrowing constrained when the economy is in a downturn, i.e., when output is below steady state level.

Despite this change, the rest of our model remains largely unchanged relative to the baseline. Even though η_t cannot change over time, at any date t , the consumption of a ξ_h household and ξ_l household can be written as:

$$c_{h,t} = \left(\frac{\xi_h}{\psi} \right)^\gamma w_t^\gamma \quad \text{and} \quad c_{l,t} = \left(\frac{\xi_l}{\psi} \right)^\gamma w_t^\gamma,$$

respectively. As in the baseline model, we normalize ψ to ensure that output and wages in the targeted steady state equal 1. This now requires:

$$\psi = [(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma]^\frac{1}{\gamma},$$

where $\bar{\eta} = \frac{\bar{\lambda}_l}{\bar{\lambda}_l + \bar{\lambda}_h}$ denotes the fraction of ξ_l households in the targeted steady state.

However, since η_t can change over time, the goods market clearing condition is now given by:

$$(1 - \eta_t) c_{h,t} + \eta_t c_{l,t} = y_t \quad \Rightarrow \quad \frac{(1 - \eta_t) \xi_h^\gamma}{(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma} w_t^\gamma + \frac{\eta_t \xi_l^\gamma}{(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma} w_t^\gamma = y_t \quad (\text{e.32})$$

We can rewrite the above equation to derive a relationship between aggregate output and wages:

$$y_t = \mu_t w_t^\gamma \quad \text{where} \quad \mu_t = \frac{(1 - \eta_t) \xi_h^\gamma + \eta_t \xi_l^\gamma}{(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma} \quad (\text{e.33})$$

Equation (e.33) confirms that in the targeted steady state with $\eta_t = \bar{\eta}$, the relationship between wages and output is the same as in our baseline model. However, away from the targeted steady state $\eta_t \neq \bar{\eta}$, μ_t mediates the relationship between wages and output, and captures the fact that ξ_h and ξ_l households work different number of (effective) hours. In fact, there is a one-to-one relationship between μ_t and η_t , which can be written as:

$$\mu_t = 1 - \Gamma(\Delta_c) \cdot \left(\frac{\eta_t - \bar{\eta}}{1 - \bar{\eta}} \right) \quad \text{where} \quad \Gamma(\Delta_c) = \frac{(1 - \bar{\eta})(\Delta_c - 1)}{1 + (1 - \bar{\eta})(\Delta_c - 1)} \in (0, 1), \quad (\text{e.34})$$

and $\Delta_c = \frac{c_h}{c_l} = \left(\frac{\xi_h}{\xi_l} \right)^\gamma > 1$ denotes the relative consumption of ξ_h and ξ_l households in the targeted steady state.

IS equation Next, we can derive the IS equation by taking the time-derivative of (e.33) (which describes $c_{h,t}$):

$$c_{h,t} = \frac{\xi_h^\gamma}{(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma} \left(\frac{y_t}{\mu_t} \right) \quad \Rightarrow \quad \frac{\dot{y}_t}{y_t} = \frac{\dot{c}_{h,t}}{c_{h,t}} + \frac{\dot{\mu}_t}{\mu_t}$$

Then using the Euler equation of ξ_h households (which remains the same as in the baseline and is given by (9)), we can derive the following IS equation:

$$\frac{\dot{y}_t}{y_t} = \gamma(i_t - \pi_t - r_t^*), \quad (\text{e.35})$$

where the natural rate is now given by:

$$r_t^* = \rho - \sigma y_t^{-\Theta} - \frac{1}{\gamma} \frac{\dot{\mu}_t}{\mu_t} \quad (\text{e.36})$$

Equation (e.36) shows that, as in our baseline model, r_t^* can still endogenously fluctuate in response to contemporaneous output fluctuations. However, now r^* also responds negatively to increases in the fraction of hand-to-mouth households. Using the interest rate rule (6) $i_t = \rho - \sigma + \phi_\pi \pi_t$, we can substitute out the nominal interest rate and write the IS equation (e.35) as:

$$\frac{1}{\gamma} \frac{\dot{y}_t}{y_t} = (\phi_\pi - 1)\pi_t + \sigma(y_t^{-\Theta} - 1) + \frac{1}{\gamma} \frac{\dot{\mu}_t}{\mu_t}, \quad (\text{e.37})$$

Phillips curve We use the same specification of the Phillips curve as in our baseline (given by (5)):

$$\dot{\pi}_t = \rho \pi_t - \kappa(w_t - 1)$$

We can write this in terms of output, rather than wages using (e.33) as:

$$\dot{\pi}_t = \rho \pi_t - \kappa \left[\left(\frac{y_t}{\mu_t} \right)^{\frac{1}{\gamma}} - 1 \right] \quad (\text{e.38})$$

Evolution of μ_t Next, differentiating μ_t in (e.33) with respect to time and using the definition of $\dot{\eta}_t$ in (e.31), we can derive the law of motion of $\dot{\mu}_t$:

$$\frac{\dot{\mu}_t}{\mu_t} = -\lambda_l \left[\Gamma \left(\frac{y_t^{-\Theta} - 1}{\mu_t} \right) - (y_t^{-\Theta} - 1) (1 - \mu_t^{-1}) - \frac{1}{\bar{\eta}} (1 - \mu_t^{-1}) \right]. \quad (\text{e.39})$$

Overall, the three ordinary differential equations (e.37), (e.38) and (e.39) jointly describe the aggregate dynamics of y_t , π_t and μ_t . However, rather than working in terms of y_t and μ_t , it is more convenient to express the dynamics in terms of transformed variables $x_t = \frac{1}{\gamma} \ln y_t$ and $m_t = \frac{1}{\gamma} \ln \mu_t$. As in our baseline, x_t can be interpreted as a scaled version the output-gap and measures the (scaled) percentage deviation of output from its value in the targeted steady state. Similarly, $\mu = 1$ in the targeted steady state, and thus m_t can be interpreted as the (scaled) percentage deviation of μ_t from its value in the targeted steady state. Following the change of variables, we can rewrite (e.37), (e.38) and

(e.39) as:

$$\dot{x}_t = (\phi_\pi - 1) \pi_t + \sigma (e^{-\gamma \Theta x_t} - 1) + \dot{m}_t \quad (\text{e.40})$$

$$\dot{\pi}_t = \rho \pi_t - \kappa (e^{x_t - m_t} - 1) \quad (\text{e.41})$$

$$\dot{m}_t = -\gamma^{-1} \bar{\lambda}_l \left\{ (e^{-\gamma \Theta x_t} - 1) [\Gamma + (1 - \Gamma) (1 - e^{-\gamma m_t})] + \frac{1}{\bar{\eta}} (1 - e^{-\gamma m_t}) \right\} \quad (\text{e.42})$$

As in the baseline, it is useful to separate the terms which dominate aggregate dynamics local to the targeted steady state $x = \pi = m = 0$ and the non-linear terms which dominate dynamics further away from this steady state:

$$\begin{aligned} \begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \\ \dot{m}_t \end{bmatrix} &= \underbrace{A \begin{bmatrix} x_t \\ \pi_t \\ m_t \end{bmatrix}}_{\text{first-order terms}} \\ &+ \underbrace{\begin{bmatrix} \bar{\sigma} (e^{-\gamma \Theta x_t} - 1 + \gamma \Theta x_t) - \gamma^{-1} \bar{\lambda}_l (1 - \Gamma) (e^{-\gamma \Theta x_t} - 1) (1 - e^{-\gamma m_t}) + \frac{\gamma^{-1} \bar{\lambda}_l}{\bar{\eta}} (e^{-\gamma m_t} - 1 + \gamma m_t) \\ -\kappa [e^{x_t - m_t} - 1 - (x_t - m_t)] \\ -\gamma^{-1} \bar{\lambda}_l \left\{ \Gamma (e^{-\gamma \Theta x_t} - 1 + \gamma \Theta x_t) + (1 - \Gamma) (e^{-\gamma \Theta x_t} - 1) (e^{-\gamma m_t} - 1) - \frac{1}{\bar{\eta}} (e^{-\gamma m_t} - 1 + \gamma m_t) \right\} \end{bmatrix}}_{\text{higher-order terms}} \end{aligned} \quad (\text{e.43})$$

where the matrix A is the Jacobian of the system around the targeted steady state $(x, \pi, m) = (0, 0, 0)$ and is given by

$$A = \begin{bmatrix} -\gamma \bar{\sigma} \Theta & \phi_\pi - 1 & -\bar{\lambda}_l / \bar{\eta} \\ -\kappa & \rho & \kappa \\ \bar{\lambda}_l \Gamma \Theta & 0 & -\bar{\lambda}_l / \bar{\eta} \end{bmatrix} \quad \text{where} \quad \bar{\sigma} = \bar{\lambda}_l \left[(\Delta_c)^{\frac{1}{\gamma}} - 1 - \frac{1}{\gamma} \Gamma(\Delta_c) \right] \quad (\text{e.44})$$

As in our baseline, the first-term on the RHS of (e.43) describes the first-order dynamics of the economy and describes the dynamics of x_t, π_t, m_t local to the targeted steady state $(x, \pi, m) = (0, 0, 0)$, while the second term on the RHS denotes the higher-order terms which dominate the dynamics of the economy further away from the targeted steady state. Before proceeding, we establish that as long as $\Delta_c > 1, \bar{\sigma} > 0$ for any $\gamma > 0$.

Lemma E.1. *For any $\Delta_c > 1$, we have $\bar{\sigma} > 0$ for all $\gamma > 0$.*

Proof. For this proof, it is convenient to define $g = \gamma^{-1}$ and rewrite $\bar{\sigma}$ in terms of the coefficient of relative risk aversion g :

$$\bar{\sigma}(g) = \bar{\lambda}_l [(\Delta_c)^g - 1 - g \Gamma(\Delta_c)]$$

First notice that for $g = 0 (\gamma = \infty)$, we have:

$$\bar{\sigma}(0) = \bar{\lambda}_l [1 - 1 - 0] = 0$$

Next, notice that we can write the derivative of $\bar{\sigma}(g)$ with respect to g as:

$$\frac{\partial \bar{\sigma}(g)}{\partial g} = \bar{\lambda}_l [(\Delta_c)^g \ln \Delta_c - \Gamma(\Delta_c)]$$

Thus, to prove the claim that $\bar{\sigma}(g) > 0$ for all $g \geq 0$, because $\bar{\sigma}(0) = 0$, it is sufficient to show that this derivative is positive for all $g \geq 0$. Start with $g = 0$, in which case the derivative can be written as:

$$\frac{\partial \bar{\sigma}(0)}{\partial g} = \bar{\lambda}_l [(\Delta_c)^0 \ln \Delta_c - \Gamma(\Delta_c)] = \ln(\Delta_c) - \Gamma(\Delta_c)$$

Clearly this is positive for $\Delta_c >> 1$ since $\Gamma(\Delta_c)$ is bounded above by 1 while $\log(\Delta_c)$ grows unbounded as Δ_c increases. Thus, we need to show that the derivative is positive for Δ_c close to 1. To do so, consider Δ_c close to 1, which we can equivalently write as $\Delta_c = 1 + \delta$ for some positive δ close to 0. We know that for δ small we can write $\ln(1 + \delta)$ as:

$$\ln(1 + \delta) = \delta - \frac{\delta^2}{2} + \frac{\delta^3}{3} \dots$$

In a similar fashion, a power series expansion of $\Gamma(1 + \delta)$ around $\delta = 0$ can be written as:

$$\Gamma(1 + \delta) = \frac{(1 - \bar{\eta}) \delta}{1 + (1 - \bar{\eta}) \delta} = (1 - \bar{\eta}) \delta [1 + (1 - \bar{\eta}) \delta + (1 - \bar{\eta})^2 \delta^2 + \dots] = (1 - \bar{\eta}) \delta - (1 - \bar{\eta})^2 \delta^2 + (1 - \bar{\eta})^3 \delta^3,$$

and so the leading term of $\ln(1 + \delta) - \Gamma(1 + \delta)$ is given by:

$$\ln(1 + \delta) - \Gamma(1 + \delta) = \bar{\eta} \delta > 0,$$

which implies that even for Δ_c close to 1, we have $\ln(\Delta_c) - \Gamma(\Delta_c) > 0$ and so for any $\Delta_c > 1$, we have $\frac{\partial \bar{\sigma}(0)}{\partial g} > 0$. Then, since $(\Delta_c)^g > 1$ for all $g > 0$, it must be true for any $\Delta_c > 1$ and $g > 0$ that:

$$\frac{\partial \bar{\sigma}(g)}{\partial g} = \bar{\lambda}_l [(\Delta_c)^g \ln \Delta_c - \Gamma(\Delta_c)] > 0$$

So, we have $\bar{\sigma} > 0$. □

E.5.1 Local and Global determinacy of the targeted equilibrium

Next, we can derive the conditions for local determinacy of the targeted equilibrium. Since we now have two jump variables x_t, π_t , and one predetermined variable m_t , local determinacy of the targeted equilibrium requires that the matrix A in (e.44) have two eigenvalues with positive real parts and one negative real eigenvalue, which implies that a necessary condition for local determinacy is that the determinant of A must be negative. Before we evaluate the determinant, notice that the trace of A can be written as:

$$\text{tr}(A) = \rho - \frac{\bar{\lambda}_l}{\bar{\eta}} - \gamma \bar{\sigma} \Theta$$

In what follows, we will assume that the fraction of ξ_l households in the targeted steady state (give by $\bar{\eta}$ is not too large. This is formalized in Assumption E.1 below.

Assumption E.1. *In the targeted steady state with, the fraction of ξ_l households is not too large:*

$$\bar{\eta} < \frac{\bar{\lambda}_l}{\rho} \quad (\text{e.45})$$

Assumption E.1 ensures that the trace of A is negative even if risk is acyclical, and ensures that at least one of the roots of A is always negative.

With assumption E.1 in place, the determinant of A can be written as:

$$\det(A) = -\frac{\kappa \bar{\lambda}_l}{\bar{\eta}} (1 - \bar{\eta} \Gamma \Theta) [\phi_\pi - \varphi_\eta(\Theta)], \quad \text{where} \quad \varphi_\eta(\Theta) = 1 + \frac{\rho \sigma \gamma \Theta}{\kappa (1 - \bar{\eta} \Gamma \Theta)}$$

The expression above shows that a necessary condition for local determinacy of the target equilibrium is risk not be too countercyclical and that monetary policy be sufficiently aggressive in response to changes in inflation:

$$\phi_\pi > \varphi_\eta(\Theta) \quad \text{and} \quad 0 < \Theta < (\bar{\eta} \Gamma)^{-1} \quad (\text{e.46})$$

Thus, the condition above is analogous to (18) in the baseline model and shows that if risk is acyclical $\Theta = 0$, then $\varphi_\eta(0) = 1$, and the standard Taylor principle $\phi_\pi > 1$ suffices for local determinacy. However, as we increase Θ starting from 0 (which makes risk more countercyclical), the larger ϕ_π must be to ensure local determinacy. This can be seen from the fact that the derivative of $\varphi_\eta(\Theta)$ is an increasing function of Θ .

As in our baseline model (e.46) is only a necessary condition for local determinacy of the target equilibrium. In fact, as we show next, there exists $\Theta^* \in (0, (\bar{\eta} \Gamma)^{-1})$ such that for $\Theta > \Theta^*$, the target equilibrium is locally *indeterminate*. In particular, for $0 < \Theta < \Theta^*$, two of the eigenvalues of A are a pair of complex conjugates with positive real parts, while the third is a negative real root, which implies local determinacy (since we have one predetermined variable and two jump variables). At $\Theta = \Theta^*$, the pair of complex roots pass become purely imaginary (they have 0 real parts), while the third root stays negative. In other words, the system of ODEs describing aggregate dynamics undergoes a Hopf bifurcation at $\Theta = \Theta^*$ (we discuss this in detail in the following paragraph). Increasing Θ above Θ^* (but keeping it still below $(\bar{\eta} \Gamma)^{-1}$ corresponds to the case in which A now has a negative determinant via three roots with negative real parts, which implies that the targeted equilibrium is locally indeterminate as for a given m , a trajectory starting from any (x, π) close enough to $(0, 0)$, now converges to the targeted steady state. Finally, when risk is even more countercyclical, $\Theta > (\bar{\eta} \Gamma)^{-1}$, no matter how large ϕ_π is relative to $\varphi_\eta(\Theta)$, we still have local indeterminacy. This is because for $\Theta > (\bar{\eta} \Gamma)^{-1}$, the determinant is positive in this case, no matter how large ϕ_π is. A positive determinant implies that we cannot have two roots with positive real parts and one negative root and as in our baseline model, even the augmented Taylor principle cannot deliver local determinacy. Thus, the local determinacy properties are similar to those in our baseline model and are qualitatively unaffected by the fact that the fraction of ξ_l households can change over time in this extension.

Hopf Bifurcation and convergence to a stable cycle As mentioned above, the system of ODEs undergoes a Hopf bifurcation at $\Theta = \Theta^*$. As in our baseline model, this Hopf bifurcation means that even when the targeted steady state is locally determinate, there is global indeterminacy because there are multiple bounded trajectories which start off the stable manifold of the targeted steady state, initially diverge away from the targeted steady state, but then converge to a stable cycle surrounding the targeted steady state and remain bounded asymptotically. Next, we show that as long as Assumption E.1 holds, and if the targeted steady state is locally determinate, i.e., $\phi_\pi > \phi_\eta(\Theta)$, then there exists a $\Theta^* \in \left(0, \frac{1}{\bar{\eta}\Gamma}\right)$ such that evaluating the matrix A in (e.44) at $\Theta = \Theta^*$ ensures that two out of its three eigenvalues (denotes by z_1^*, z_2^*, z_3^*) are purely imaginary and one is real and negative. As a result, z_1^*, z_2^*, z_3^* satisfy Orlando's formula, which is given by:

$$-S(A(\Theta^*)) + \frac{\det(A(\Theta^*))}{\text{tr}(A(\Theta^*))} = 0,$$

where $S(A(\Theta^*)) = z_1^* z_2^* + z_2^* z_3^* + z_3^* z_1^*$, $\det(A(\Theta^*)) = z_1^* z_2^* z_3^*$ and $\text{tr}(A(\Theta^*)) = z_1^* + z_2^* + z_3^*$. Since the trace of A is negative, satisfying Orlando's formula is a sufficient condition for two of the roots to be purely imaginary with the third root being negative since the trace and determinant are negative. We prove this claim next.

Lemma E.2. Suppose that $\bar{\eta} \leq \bar{\lambda}_l / \rho$, $\phi_\pi > \phi_\eta(\Theta)$ and $0 \leq \Theta < (\bar{\eta}\Gamma)^{-1}$. Then, $\exists \Theta^* \in \left(0, \frac{1}{\bar{\eta}\Gamma}\right)$, such that

$$H(\Theta^*) = -S(\Theta^*) + \frac{\det(\Theta^*)}{\text{tr}(\Theta^*)} = 0$$

Proof. Let's start by calculating $H(\Theta)$ for $0 \leq \Theta < (\bar{\eta}\Gamma)^{-1}$. For convenience, we replicate $A(\Theta)$ here:

$$A(\Theta) = \begin{bmatrix} -\gamma\bar{\sigma}\Theta & \phi_\pi - 1 & -\bar{\lambda}_l / \bar{\eta} \\ -\kappa & \rho & \kappa \\ \bar{\lambda}_l \Gamma \Theta & 0 & -\bar{\lambda}_l / \bar{\eta} \end{bmatrix}$$

First, note that following basic linear algebra, we can calculate $S(\Theta)$ as the sum of the principal minors of A :

$$S(\Theta) = \begin{vmatrix} -\gamma\bar{\sigma}\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{vmatrix} + \begin{vmatrix} \rho & \kappa \\ 0 & -\bar{\lambda}_l / \bar{\eta} \end{vmatrix} + \begin{vmatrix} -\gamma\bar{\sigma}\Theta & -\bar{\lambda}_l / \bar{\eta} \\ \bar{\lambda}_l \Gamma \Theta & -\bar{\lambda}_l / \bar{\eta} \end{vmatrix},$$

which can be simplified to:

$$S(\Theta) = \kappa(\phi_\pi - 1) - \rho \frac{\bar{\lambda}_l}{\bar{\eta}} + \rho \left(\bar{\lambda}_l \Gamma - \gamma\sigma + \frac{\bar{\lambda}_l}{\rho \bar{\eta}} \gamma\sigma \right) \Theta$$

Next, as before, the simplified determinant of A can be written as:

$$\det(\Theta) = \frac{\bar{\lambda}_l}{\bar{\eta}} [\rho \gamma \sigma \Theta - \kappa(\phi_\pi - 1)(1 - \bar{\eta}\Gamma\Theta)],$$

Since we are imposing conditions so that the targeted steady state is locally determinate ($\phi_\pi > \phi_\eta(\Theta)$ and $0 \leq \Theta < (\bar{\eta}\Gamma)^{-1}$), we know that the determinant is negative for all Θ in the range we are considering. Finally, we can express the trace of A as:

$$\text{tr}(\Theta) = \rho - \frac{\bar{\lambda}_l}{\bar{\eta}} - \gamma\bar{\sigma}\Theta$$

Given our assumption that $\bar{\eta} \leq \bar{\lambda}_l/\rho$, the trace is negative even when risk is acyclical ($\Theta = 0$). Furthermore, the more countercyclical the risk (larger Θ), the more negative is the trace. Using these expressions, we can write $H(\Theta)$ as:

$$H(\Theta) = \frac{\mathbb{N}(\Theta)}{\rho - \frac{\bar{\lambda}_l}{\bar{\eta}} - \gamma\bar{\sigma}\Theta},$$

where

$$\mathbb{N}(\Theta) = a_2\Theta^2 + a_1\Theta + a_0,$$

with

$$\begin{aligned} a_2 &= \rho\gamma\bar{\sigma} \left[\bar{\lambda}_l\Gamma + \left(\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho \right) \frac{\gamma\sigma}{\rho} \right] > 0 \\ a_1 &= \kappa(\phi_\pi - 1)\gamma\sigma + \gamma\sigma \left(\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho \right)^2 + \left(2\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho \right) \rho\bar{\lambda}_l\Gamma > 0 \\ a_0 &= -\rho \left[\frac{\bar{\lambda}_l}{\bar{\eta}} \left(\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho \right) + \kappa(\phi_\pi - 1) \right] < 0 \end{aligned}$$

The signs of a_2 and a_1 are guaranteed by our assumption that $\bar{\eta} \leq \bar{\lambda}_l/\rho$, while the sign of a_0 also relies on the fact that we are imposing $\phi_\pi > \phi_\eta(\Theta) > 1$.

Since the denominator of $H(\Theta)$ is always negative for any $\Theta \geq 0$, in order to show that $H(\cdot)$ has a zero in the interval $0 < \Theta < (\bar{\eta}\Gamma)^{-1}$, we need to show that $\mathbb{N}(\Theta)$ has a zero in the interval $0 < \Theta < (\bar{\eta}\Gamma)^{-1}$. To see that this is in fact the case, notice that

$$\mathbb{N}(0) = a_0 < 0$$

and for any $0 \leq \Theta < (\bar{\eta}\Gamma)^{-1}$, we know that

$$\mathbb{N}'(\Theta) = 2a_2\Theta + a_1 > 0$$

Since $\mathbb{N}(0) < 0$ and because $\mathbb{N}(\Theta)$ is an increasing function of Θ in the relevant interval, to prove that $\mathbb{N}(\Theta)$ has a zero in the interval $0 < \Theta < (\bar{\eta}\Gamma)^{-1}$, it is sufficient to show that $\mathbb{N}\left((\bar{\eta}\Gamma)^{-1}\right) > 0$. We can

write $\mathbb{N}\left(\frac{1}{\bar{\eta}\Gamma}\right)$ as:

$$\begin{aligned}\mathbb{N}\left(\frac{1}{\bar{\eta}\Gamma}\right) &= \frac{\gamma\bar{\sigma}\rho}{\bar{\eta}^2\Gamma^2}\left[\bar{\lambda}_l\Gamma + \frac{\gamma\sigma}{\rho}\left(\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho\right)\right] + \frac{1}{\bar{\eta}^2\Gamma}\left\{\gamma\sigma(\bar{\lambda}_l - \bar{\eta}\rho)^2 + (2\bar{\lambda}_l - \bar{\eta}\rho)\rho\bar{\lambda}_l\Gamma\right\} - \rho\frac{\bar{\lambda}_l}{\bar{\eta}}\left(\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho\right) \\ &\quad + \left(\frac{\gamma\sigma}{\bar{\eta}\Gamma} - \rho\right)\kappa[\phi_\pi - 1] \\ &> \frac{\gamma\bar{\sigma}\rho}{\bar{\eta}^2\Gamma^2}\left[\bar{\lambda}_l\Gamma + \frac{\gamma\sigma}{\rho}\left(\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho\right)\right] + \frac{1}{\bar{\eta}^2\Gamma}\left\{\gamma\sigma(\bar{\lambda}_l - \bar{\eta}\rho)^2 + (2\bar{\lambda}_l - \bar{\eta}\rho)\rho\bar{\lambda}_l\Gamma\right\} - \rho\frac{\bar{\lambda}_l}{\bar{\eta}}\left(\frac{\bar{\lambda}_l}{\bar{\eta}} - \rho\right) \\ &\quad + \left(\frac{\gamma\sigma}{\bar{\eta}\Gamma} - \rho\right)\kappa\left[\varphi_\eta\left(\frac{1}{\bar{\eta}\Gamma}\right) - 1\right]\end{aligned}$$

Since $\varphi_\eta(\Theta) = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa(1-\bar{\eta}\Gamma\Theta)}$, we know that

$$\lim_{\Theta \rightarrow \frac{1}{\bar{\eta}\Gamma}} \varphi_\eta(\Theta) = \infty$$

So, as long as $\frac{\gamma\sigma}{\bar{\eta}\Gamma} - \rho > 0$, we know that $\mathbb{N}\left((\bar{\eta}\Gamma)^{-1}\right) > 0$. To see that $\frac{\gamma\sigma}{\bar{\eta}\Gamma} - \rho = \frac{\gamma\sigma - \rho\bar{\eta}\Gamma}{\bar{\eta}\Gamma} > 0$, notice that:

$$\begin{aligned}\gamma\sigma - \rho\bar{\eta}\Gamma &= \gamma\bar{\lambda}_l\left\{\left(\Delta_c^{\frac{1}{\gamma}} - 1\right) - \frac{\rho\bar{\eta}}{\bar{\lambda}_l}\frac{1}{\gamma}\frac{(1-\bar{\eta})(\Delta_c-1)}{1+(1-\bar{\eta})(\Delta_c-1)}\right\} \\ &\geq \gamma\bar{\lambda}_l\left\{\Delta_c^{\frac{1}{\gamma}} - 1 - \frac{1}{\gamma}\frac{(1-\bar{\eta})(\Delta_c-1)}{1+(1-\bar{\eta})(\Delta_c-1)}\right\} \quad \because \quad \bar{\eta} < \bar{\lambda}_l/\rho \\ &= \gamma\bar{\lambda}_l\bar{\sigma} \\ &> 0,\end{aligned}$$

where, we know that the last term is positive from Lemma E.1. Thus, there exists exactly one $\Theta^* \in (0, (\bar{\eta}\Gamma)^{-1})$ for which $H(\Theta) = \mathbb{N}(\Theta) = 0$. Thus, a Hopf bifurcation occurs at $\Theta = \Theta^*$. $\square$

Since a Hopf bifurcation occurs at $\Theta = \Theta^*$, Theorem 3 ensures that there exists $\Theta^\diamond \in (0, \Theta^*)$ such that for $\Theta \in (\Theta^\diamond, \Theta^*)$ trajectories which start off the stable manifold around the targeted steady state initially diverge but then converge to a stable cycle and remain bounded. Thus, for $\Theta \in (\Theta^\diamond, \Theta^*)$, even though a large enough ϕ_π ensures that the targeted equilibrium is locally determinate, it cannot deliver global determinacy on account of the multiple bounded trajectories which converge to a stable cycle around the targeted steady state and satisfy all equilibrium conditions. In contrast, for $\Theta \in (\Theta^*, (\bar{\eta}\Gamma)^{-1})$, the targeted equilibrium is locally indeterminate, and so we also have global indeterminacy if risk is too countercyclical $\Theta > \Theta^*$.

Multiple Steady States As in our baseline model, in addition to the global indeterminacy manifesting as the possibility of the economy converging to a stable cycle, global indeterminacy also manifests in the form of an untargeted steady state which exists alongside the targeted steady state. We discuss this next.

As in our baseline model, as long as risk is countercyclical ($\Theta > 0$), there are multiple steady states in this extension in which η_t is allowed to vary over time. The targeted steady state with $x = \pi =$

$m = 0$ always exists, but alongside this, there also exists another untargeted steady state in which $x = \underline{x} < 0$, $\pi = \underline{\pi} < 0$ and $m = \underline{m} < 0$ as long as $\phi_\pi > \varphi_\eta(\Theta)$. In other words, as long as risk is not too countercyclical $\Theta < (\bar{\eta}\Gamma)^{-1}$ and the targeted equilibrium is locally determinate, there always exists an untargeted steady state in which the output-gap is negative, inflation is below target and the fraction of ξ_l households is larger than in the targeted steady state $\eta = \underline{\eta} = \frac{\bar{\lambda}_l}{\bar{\lambda}_l + \bar{\lambda}_h e^{\gamma\Theta x}} > \frac{\bar{\lambda}_l}{\bar{\lambda}_l + \bar{\lambda}_h} = \bar{\eta}$. This higher η in the untargeted steady state implies that:

$$\underline{m} = \frac{1}{\gamma} \ln \left[1 - \Gamma \left(\frac{\eta - \bar{\eta}}{1 - \bar{\eta}} \right) \right] < 0$$

To see that this untargeted steady state exists, we can set $\dot{x}_t = \dot{\pi}_t = \dot{m}_t = 0$ in (e.40), (e.38) and (e.42) to describe the combinations of x, π, m which constitute a steady state:

$$0 = (\phi_\pi - 1) \pi + \sigma (e^{-\gamma\Theta x} - 1) \quad (\text{e.47})$$

$$\pi = \frac{\kappa}{\rho} (e^{x-m} - 1) \quad (\text{e.48})$$

$$0 = (e^{-\gamma\Theta x} - 1) [\Gamma + (1 - \Gamma) (1 - e^{-\gamma m})] + \frac{1}{\bar{\eta}} (1 - e^{-\gamma m}) \quad (\text{e.49})$$

Combining (e.47) with (e.48) to eliminate π , we can reduce the problem of solving for three equations in 3 unknowns to 2 equations in 2 unknowns x, m . Thus, in any steady state (x, m) must satisfy:

$$\bar{\eta} (e^{-\gamma\Theta x} - 1) [\Gamma + (1 - \Gamma) (1 - e^{-\gamma m})] + (1 - e^{-\gamma m}) = 0 \quad (\text{e.50})$$

$$\kappa (\phi_\pi - 1) (e^{x-m} - 1) + \rho \sigma (e^{-\gamma\Theta x} - 1) = 0 \quad (\text{e.51})$$

Next, we can rearrange (e.50) to express x as a function of m :

$$x = \Omega(m) = \ln \left[1 - \frac{1}{\bar{\eta}} \frac{(1 - e^{-\gamma m})}{\Gamma + (1 - \Gamma) (1 - e^{-\gamma m})} \right]^{-\frac{1}{\gamma\Theta}}$$

Plug this into (e.51) to get one equation in one unknown m :

$$G(m) = \kappa (\phi_\pi - 1) \left[e^{\Omega(m)-m} - 1 \right] - \frac{\rho\sigma}{\bar{\eta}} \frac{(1 - e^{-\gamma m})}{\Gamma + (1 - \Gamma) (1 - e^{-\gamma m})}$$

Clearly, by inspection, $m = 0$ satisfies $G(0) = 0$. Also, since $\Omega(0) = 0$, $x = \pi = m = 0$ solves the system of equations. Consequently, the targeted steady state always exists.

Next, notice that the derivative of $G(m)$ evaluated at $m = 0$ can be written as:

$$G'(0) = \kappa \left[\frac{(\Gamma\bar{\eta})^{-1} - \Theta}{\gamma\rho\Theta} \right] \left(\phi_\pi - 1 - \frac{\rho\sigma\gamma\Theta}{\kappa(1 - \bar{\eta}\Gamma\Theta)} \right),$$

which is positive as risk is not too countercyclical: $0 < \Theta < (\Gamma\bar{\eta})^{-1}$, and if monetary policy is aggressive enough to ensure that the targeted steady state is locally determinate $\phi_\pi > \varphi_\eta(\Theta) = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa(1 - \bar{\eta}\Gamma\Theta)}$. Next, note that as $m \rightarrow -\infty$, we have $G(m) \rightarrow \infty$. Consequently, it must be the case that as we lower m from

0 towards $-\infty$, $G(m)$ crosses 0 again at least once at some point $\underline{m} < 0$. So, there exists an untargeted steady state $(\underline{x}, \underline{\pi}, \underline{m})$ where $\underline{m} < 0$. Since $\underline{m} < 0$, we have $\underline{x} = \Omega(\underline{m}) < 0$ implying that $\underline{x} < 0$. It remains to show that $\underline{\pi} < 0$.

Recall that $\underline{\pi} = \frac{\kappa}{\rho}(e^{\underline{x}-\underline{m}} - 1)$. To show that $\underline{\pi} < 0$, we need to show that $\underline{x} - \underline{m} < 0$, so that $e^{\underline{x}-\underline{m}} - 1 < 0$. To see that this is in fact the case, we can evaluate (e.51) at $x = \underline{x} < 0$ and $m = \underline{m} < 0$, which we replicate for convenience below:

$$\frac{\kappa(\phi_{\pi} - 1)}{\rho}(e^{\underline{x}-\underline{m}} - 1) = \sigma(1 - e^{-\gamma\Theta\underline{x}})$$

We know that since $\underline{x} < 0$, the RHS of this expression is negative since for $\underline{x} < 0$, we have $e^{-\gamma\Theta\underline{x}} > 1$. For a given $\Theta > 0$, under the assumption that the targeted steady state is locally determinate, i.e., $\phi_{\pi} > \phi_{\eta}(\Theta) > 1$, we know that $\phi_{\pi} - 1 > 0$. So, the only way that the LHS can also be negative is if $e^{\underline{x}-\underline{m}} < 1$, which requires that $\underline{x} < \underline{m}$. Thus, we have $\underline{\pi} < 0$ in the untargeted steady state.

However, since this extension has a predetermined variable (the fraction of ζ_l households), simply the existence of a second steady state does not imply global indeterminacy. However, we next show that for a given Θ , as long as risk is not too countercyclical $0 < \Theta < (\bar{\eta}\Gamma)^{-1}$ and monetary policy is aggressive enough, $\phi_{\pi} > \phi_{\eta}(\Theta)$ (which is necessary for local determinacy of the targeted equilibrium), then the untargeted steady state $(\underline{x}, \underline{\pi}, \underline{m}) < 0$ is locally *indeterminate*. This implies that there exists a neighborhood around $\underline{m}$ around the untargeted steady state such that for each m_0 in this neighborhood, there exist multiple combinations of (x_0, π_0) such that trajectories originating at (x_0, π_0, m_0) remain bounded. This in turn implies that we have global indeterminacy.

To see that the untargeted steady state is locally indeterminate, note that the Jacobian of (e.40)-(e.42) evaluated at the untargeted steady state $(\underline{x}, \underline{\pi}, \underline{m})$ can be written as:

$$\underline{A} = \begin{bmatrix} -\gamma\bar{\sigma}\Theta' & \phi_{\pi} - 1 & -\bar{\lambda}_l/\eta' \\ -\kappa' & \rho & \kappa' \\ \bar{\lambda}_l\Gamma\Theta' & 0 & -\bar{\lambda}_l/\eta' \end{bmatrix}, \quad (\text{e.52})$$

where

$$\begin{aligned} \Theta' &= \Theta e^{-\gamma\Theta\underline{x}} > \Theta \\ \eta' &= \bar{\eta}e^{\gamma\underline{m}} < \bar{\eta} \\ \kappa' &= \kappa e^{\underline{x}-\underline{m}} < \kappa \end{aligned}$$

We know that for the untargeted steady state to be locally-determinate, $\underline{A}$ must have two roots with positive real parts and one negative root. This requires that the determinant of $\underline{A}$ be negative. However, as we show next, as long as $\phi_{\pi} > \phi_{\eta}(\Theta)$ (which is a necessary condition for the targeted steady state to be locally determinate), the determinant of $\underline{A}$ is positive, which implies that the untargeted steady state is *locally indeterminate*.

Before we evaluate the determinant of $\underline{A}$, notice that the trace of $\underline{A}$ can be written as:

$$\text{tr}(\underline{A}) = \rho - \frac{\bar{\lambda}_l}{\eta'} - \gamma\sigma\Theta'$$

Since $\eta' < \bar{\eta}$, Assumption E.1 ensures that the trace of $\bar{A}$ is always negative as long as $\Theta \geq 0$. This ensures that at least one of the eigenvalues of A is negative. Next, note that we can express the determinant of $\underline{A}$ as:

$$\det(\underline{A}) = \frac{\bar{\lambda}_l}{\eta'} \kappa' \left[\frac{\rho\gamma\sigma\Theta'}{\kappa'} - (\phi_\pi - 1)(1 - \eta'\Gamma\Theta') \right] \quad (\text{e.53})$$

To show that this is positive for any Θ as long as $\Theta > \phi_\eta(\Theta)$, we need to consider two cases: (i) $\eta'\Gamma\Theta' = \bar{\eta}\Gamma\Theta e^{\gamma(\underline{m} - \gamma\Theta\bar{x})} \geq 1$ and (ii) $\eta'\Gamma\Theta' = \bar{\eta}\Gamma\Theta e^{\gamma(\underline{m} - \gamma\Theta\bar{x})} < 1$. First consider the case in which $\eta'\Gamma\Theta' \geq 1$, which implies that $1 - \eta'\Gamma\Theta' \leq 0$. Then, since we are assuming that $\phi_\pi > \phi_\eta(\Theta) > 1$ (which is a necessary condition for the targeted steady state to be locally determinate), the determinant (e.53) of $\underline{A}$ must be positive for any $\Theta > 0$. Next, consider the case with $\eta'\Gamma\Theta' < 1$. In this case, it is convenient to factorize (e.53) as:

$$\begin{aligned} \det(\underline{A}) &= \frac{\bar{\lambda}_l}{\eta'} (1 - \eta'\Gamma\Theta') \kappa' \left[\frac{\rho\gamma\sigma\Theta'}{(1 - \eta'\Gamma\Theta') \kappa'} - (\phi_\pi - 1) \right] \\ &> \frac{\bar{\lambda}_l}{\eta'} (1 - \eta'\Gamma\Theta') \kappa' \left[\frac{\rho\gamma\sigma\Theta'}{(1 - \eta'\Gamma\Theta') \kappa'} - \frac{\rho\gamma\sigma\Theta}{(1 - \bar{\eta}\Gamma\Theta) \kappa} \right] \quad \because \quad \phi_\pi - 1 > \frac{\rho\gamma\sigma\Theta}{(1 - \bar{\eta}\Gamma\Theta) \kappa} \\ &= \frac{\bar{\lambda}_l}{\eta'} (1 - \eta'\Gamma\Theta') \rho\gamma\sigma \left[\frac{\Theta'}{1 - \eta'\Gamma\Theta'} - \frac{\Theta}{1 - \bar{\eta}\Gamma\Theta} e^{\bar{x} - \underline{m}} \right] \\ &> \frac{\bar{\lambda}_l}{\eta'} (1 - \eta'\Gamma\Theta') \rho\gamma\sigma \left[\frac{\Theta'}{1 - \eta'\Gamma\Theta'} - \frac{\Theta}{1 - \bar{\eta}\Gamma\Theta} \right] \quad \because \quad \bar{x} - \underline{m} < 0 \\ &= \frac{\bar{\lambda}_l}{\eta'} (1 - \eta'\Gamma\Theta') \rho\gamma\sigma \left[\frac{\Theta}{1 - \eta'\Gamma\Theta'} e^{-\gamma\Theta\bar{x}} - \frac{\Theta}{1 - \bar{\eta}\Gamma\Theta} \right] \\ &> \frac{\bar{\lambda}_l}{\eta'} (1 - \eta'\Gamma\Theta') \rho\gamma\sigma \left[\frac{\Theta}{1 - \eta'\Gamma\Theta'} - \frac{\Theta}{1 - \bar{\eta}\Gamma\Theta} \right] \quad \because \quad \bar{x} < 0 \\ &= \frac{\bar{\lambda}_l}{\eta'} (1 - \eta'\Gamma\Theta') \rho\gamma\sigma\Theta \left[\frac{1}{1 - \eta'\Gamma\Theta'} - \frac{1}{1 - \bar{\eta}\Gamma\Theta} \right] \end{aligned}$$

The last expression is positive if $1 - \eta'\Gamma\Theta' = 1 - \bar{\eta}\Gamma\Theta e^{\gamma(\underline{m} - \Theta\bar{x})} < 1 - \bar{\eta}\Gamma\Theta$, which in turn is true if $\underline{m} - \Theta\bar{x} > 0$. In order to prove that $\underline{m} - \Theta\bar{x} > 0$ for any $\Theta > 0$ and $\phi_\pi > \phi_\eta(\Theta)$, we have to use (e.49) evaluated at the untargeted steady state in which $m = \underline{m} < 0$ and $x = \bar{x} < 0$. We replicate this equation below for convenience:

$$(e^{-\gamma\Theta\bar{x}} - 1) \bar{\eta} [\Gamma + (1 - \Gamma)(1 - e^{-\gamma\bar{m}})] = e^{-\gamma\bar{m}} - 1 \quad (\text{e.54})$$

We proceed to prove that $\underline{m} - \Theta\bar{x} > 0$ by contradiction. Suppose, instead that $\underline{m} - \Theta\bar{x} \leq 0$, which in turn implies that:

$$e^{-\gamma\bar{m}} - 1 \geq e^{-\gamma\Theta\bar{x}} - 1$$

Using this, we can rewrite (e.54) as:

$$\left(e^{-\gamma\Theta\underline{x}} - 1\right) \bar{\eta} \left[\Gamma + (1 - \Gamma) (1 - e^{-\gamma\underline{m}})\right] = e^{-\gamma\underline{m}} - 1 \geq e^{-\gamma\Theta\underline{x}} - 1$$

Since, $\underline{x} < 0$, we can simplify the above expression by dividing both sides by $e^{-\gamma\Theta\underline{x}} - 1$ to get:

$$\bar{\eta} \left[1 - (1 - \Gamma) e^{-\gamma\underline{m}}\right] \geq 1,$$

which is a contradiction since $\bar{\eta} \in (0, 1)$ and $1 - (1 - \Gamma) e^{-\gamma\underline{m}} < 1$ because $\underline{m} < 0$ and $\Gamma \in (0, 1)$. Consequently, the LHS is the product of two numbers which are smaller than 1 and their product cannot be larger than 1. Thus, it must be that $\underline{m} > \Theta\underline{x}$, and so even in case (ii), we have $\det(\underline{A}) > 0$.

Thus, we have shown that for any $\Theta > 0$ and $\phi_\pi > \phi_\eta(\Theta)$, the untargeted steady state is locally indeterminate: a positive determinant and a negative trace imply that the Jacobian of $\underline{A}$ must have two negative roots and one positive root. This means that a 2 dimensional stable manifold (locally) surrounds the untargeted steady state $(\underline{x}, \underline{\pi}, \underline{m})$: for a given m_0 , there exist multiple combinations of x_0, π_0 starting from which (x_t, π_t, m_t) converges to the untargeted steady state $(\underline{x}, \underline{\pi}, \underline{m})$ and remain bounded. Consequently, we have *global indeterminacy* because there exists a subset of $m_0 \in (-\infty, \infty)$ for which there exist multiple combinations of (x_0, π_0, m) starting from which the economy remains bounded.

Overall, the determinacy properties of the economy in which we don't adjust $\lambda_{h,t}$ to keep η_t constant over time are qualitatively similar to those of our baseline model. Thus, even in this extension, the conclusion is that as long as risk is countercyclical, the equilibrium is globally indeterminate as long as monetary policy follows a standard Taylor rule, no matter how aggressively monetary policy responds to changes in inflation.

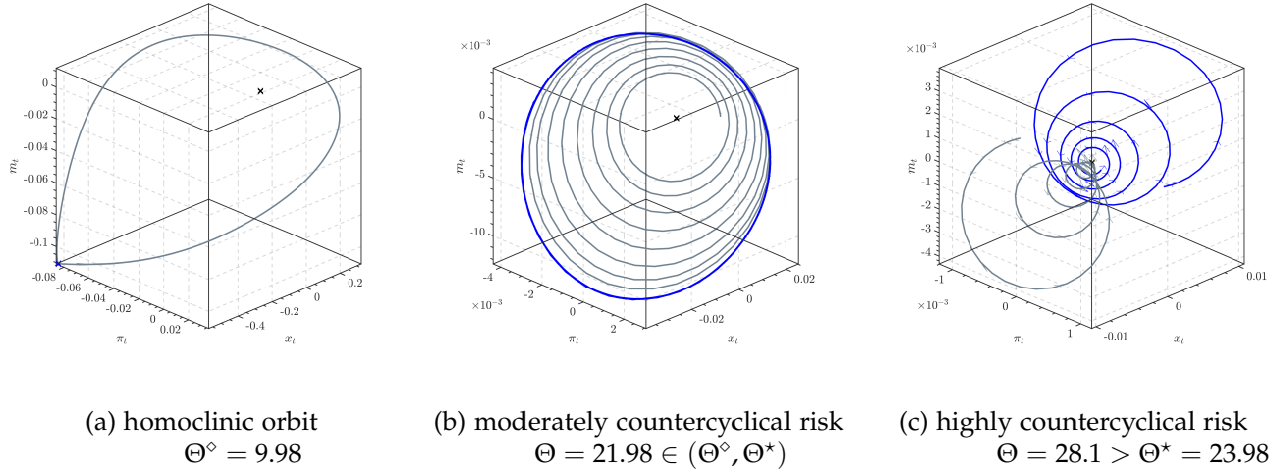


Figure E.4: Global dynamics as a function of Θ

E.5.2 Calibrated model

In this section, we numerically depict the global dynamics of the economy studied in this section as a function of Θ . In doing so, we maintain the baseline calibration. However, relative to the baseline model, we also need to calibrate $\bar{\eta}$, the fraction of households that are borrowing constrained (or hand-to-mouth) in the targeted steady state.¹⁴ So, following Kaplan et al. (2014), we set the fraction of borrowing constrained households in the targeted steady state $\bar{\eta} = 30\%$.

Given this calibration, the Hopf bifurcation occurs at $\Theta^* = 23.98$, which is a lower value of Θ^* than in our baseline model. More importantly, $\Theta^* = 23.98$ is lower than the median estimate of $\Theta = 28.1$ from Bilbiie, Primiceri and Tambalotti (2023). However, $\Theta^* = 23.98$ still lies within the posterior range of $\Theta \in (21.98, 29.9)$. Thus, at Bilbiie, Primiceri and Tambalotti (2023) model estimate of $\Theta = 28.1$, the above analysis implies that the targeted equilibrium is already locally indeterminate and hence we have both local and global indeterminacy. However, in a subset of the posterior range $\Theta \in (21.98, 23.98)$, the targeted equilibrium is locally determinate but we still have global indeterminacy as in our baseline model. Figure E.4b considers the case with $\Theta = 21.98$, and depicts (in gray) a sample trajectory which starts in the neighborhood of the targeted steady state $(0, 0, 0)$ (depicted by the black x), diverges away from it, only to converge to the stable cycle (depicted in blue). These dynamics are qualitatively similar for the case in which risk is moderately countercyclical, i.e., when Θ is in the interval $(\Theta^\diamond, \Theta^*)$, where at $\Theta = \Theta^\diamond$, the stable cycles get absorbed in to a homoclinic orbit (see Figure E.4a). Given our calibration, $\Theta^\diamond \approx 9.98$. Finally, Figure E.4c depicts local and global indeterminacy when risk is highly countercyclical Θ is between Θ^* and $(\bar{\eta}\Gamma)^{-1}$ and shows that for $m_0 = 0$, there are multiple combinations of (x_0, π_0) starting from which the economy converges to the targeted steady state. While we do not plot dynamics in the mildly countercyclical risk case $\Theta \in (0, \Theta^\diamond)$, we have proved analytically that the equilibrium is also globally indeterminate in this range. This is because we showed that as long as the targeted steady state is locally determinate, an untargeted steady state always exists which has $(\underline{x}, \underline{\pi}, \underline{m}) < 0$ and is locally indeterminate. In addition, one can invoke Theorem 4 to establish the existence of a saddle-connection along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state. However, numerically finding this saddle connection is quite hard in the context of a 3-dimensional system, so we refrain from doing that. $\square$

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¹⁴Given our simplifying assumption in the baseline model which kept the fraction of ξ_t households constant over time, the precise value of $\bar{\eta}$ did not matter for aggregate dynamics, provided that $\bar{\eta} > 0$.

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